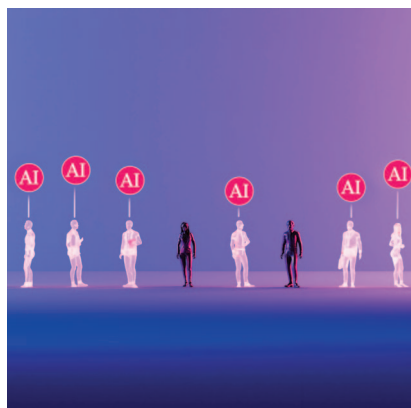


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AN AI JOB APOCALYPSE?



Rapid improvements in AI capabilities and growing corporate adoption have led to predictions that the technology could lead to large-scale job losses before the end of the decade. So, just how concerned should we be about an AI “job apocalypse”? MIT’s Daron Acemoglu and Neil Thompson and GS’ Joseph Briggs generally agree that these predictions likely won’t come to pass, though they differ on the scale of disruption ahead. Briggs expects significant labor displacement but only temporarily as new jobs eventually emerge. Thompson is less convinced about large-scale job displacement and takes comfort in the ability to anticipate changes, seeing AI as a rising tide rather than a crashing wave for labor. And Acemoglu expects a small net negative impact on labor over the next five years, with possible larger losses over

the longer term if investment remains focused on replacing rather than complementing workers. Amid this debate, we find that AI’s impact on corporate labor needs—and earnings—remains too uncertain for markets to reward (for now).

“

No general law of economics says that job creation must match job destruction. Job growth over the last 80+ years resulting from a combination of changing tasks, new tasks, and a shifting structure of occupations has not occurred at an even pace.

- Daron Acemoglu

Previous automation waves have resulted in many new jobs, which offers some reassurance. But AI also processes information in ways more similar to humans than previous technologies, so more worry may be warranted.

- Neil Thompson

Despite our expectation that AI-related job losses will lead to a meaningful amount of labor displacement, we continue to expect that labor market headwinds will be temporary. Key to this view is our expectation that over the long run AI will create many new jobs even as it destroys existing ones.

- Joseph Briggs

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The Goldman Sachs Group, Inc.

Macro news and views

We provide a brief snapshot on the most important economies for the global markets

US

Latest GS proprietary datapoints/major changes in views

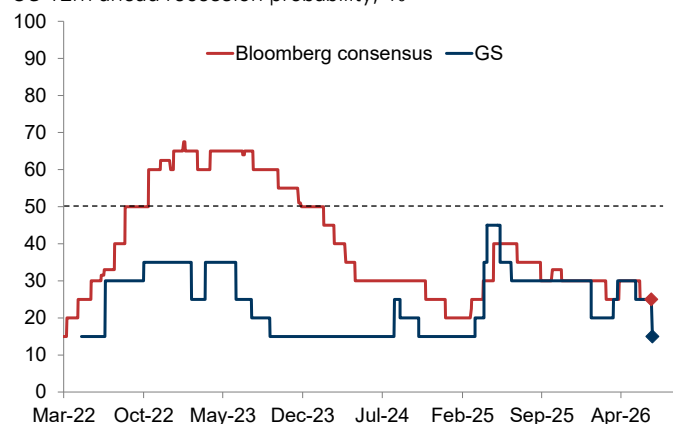
- We raised our sequential 2H26 GDP growth forecast to 2% (from 1.9%) and lowered our 12m recession odds to the 15% long-term norm (from 25%) following the US-Iran deal.

Datapoints/trends we're focused on

- Fed policy; the hawkish June meeting raises the risk of hikes later this year, but we expect the Fed to remain on hold in '26 as most FOMC voters likely still lean toward unchanged policy rates and the latest projections may be stale given the deal.
- Core PCE inflation, which we expect to remain close to 3% through YE before declining to close to 2% in 2027 (yoy).
- Real income, which will likely slow in 2H26, keeping real consumer spending growth at just 1.5%.

US recession risk: back to its long-term norm

US 12m ahead recession probability, %



Source: Bloomberg, Goldman Sachs GIR.

Europe

Latest GS proprietary datapoints/major changes in views

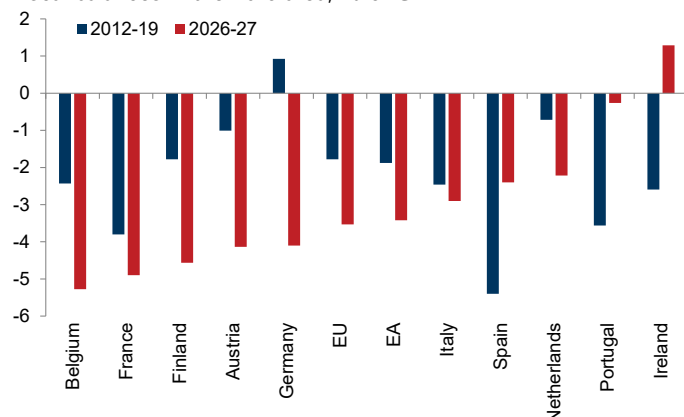
- We raised our 2026/27 EA GDP growth forecasts to 0.5%/1.2% (from 0.3%/0.9%, yoy) and lowered our EA/UK peak core inflation forecasts to 2.6%/2.9% (from 2.7%/3.1%, yoy) partly to reflect the lower energy prices we now expect.

Datapoints/trends we're focused on

- UK politics; Burnham is expected to succeed Starmer as PM, which could lead to slower fiscal consolidation.
- BoE policy; recent weak activity data and a continued deceleration in underlying inflation measures have raised our confidence that the BoE will remain on hold this year before cutting rates in 2027.
- EA fiscal deficits, which have widened on average vs pre-covid.

Euro area: deteriorating fiscal balances (for most)

Fiscal balances in the Euro area, % of GDP



Source: Haver Analytics, European Commission, Goldman Sachs GIR.

Japan

Latest GS proprietary datapoints/major changes in views

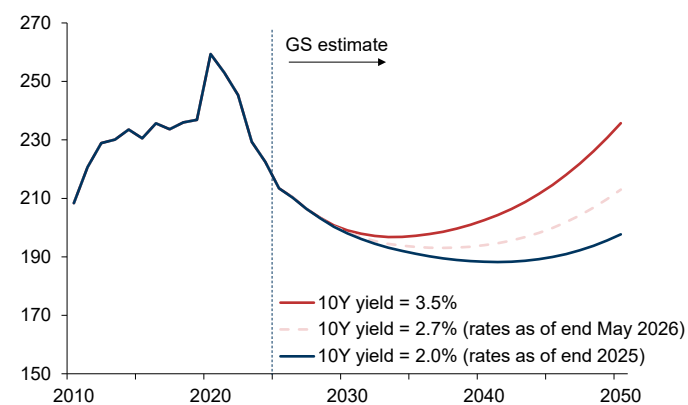
- No major changes in views.

Datapoints/trends we're focused on

- BoJ policy; we expect the BoJ to continue hiking rates roughly every six months, with the next hike likely in Jan 2027, to a policy rate of 1.5% next summer, though uncertainty around the timing of the next hike remains high.
- Japan fiscal outlook, which could worsen if PM Takaichi implements her expansionary plans and long-term interest rates rise further toward the levels of other G10 economies.
- Japanese real GDP growth, which we expect to decline to 0.5% in 2026 (from 1.1% last year, both yoy).

Japan: a potentially less sustainable fiscal outlook

Debt/GDP ratio by interest rate scenario, % of GDP



Source: Cabinet Office, Goldman Sachs GIR.

Emerging Markets (EM)

Latest GS proprietary datapoints/major changes in views

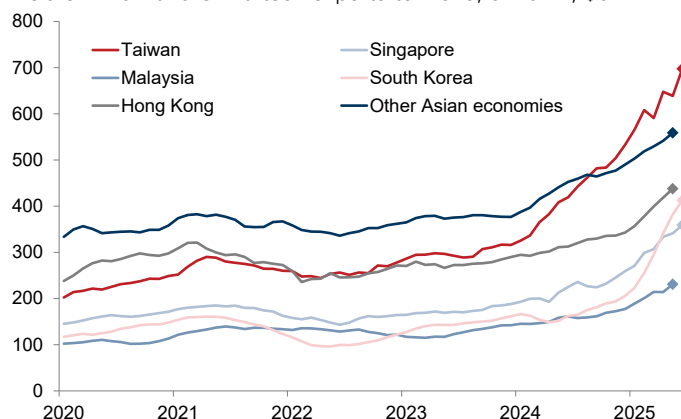
- We lowered our 2Q26 China real GDP growth forecast to 3.5% (from 4.0%, both qoq ann.) given weak April and May activity data, though we expect growth to rebound to 5.0% in 3Q26 (qoq ann.), for full-year real GDP growth of 4.7%.

Datapoints/trends we're focused on

- AI investment boom, which will likely continue boosting growth in several Asian economies involved in the tech and AI supply chain, especially Taiwan and South Korea.
- China's current account surplus, which continues to grow and could weigh on trading partners' export growth.
- RMB internationalization, which could accelerate ahead.

Asia: surging tech exports

Asia ex. Mainland China tech exports to world, 3m ann., \$bn



Source: Haver Analytics, Goldman Sachs GIR.

An AI job apocalypse?

Rapid improvements in AI capabilities and growing corporate efforts to integrate the technology into enterprise workflows have raised concerns that AI could trigger a “job apocalypse”, especially as some companies have cited AI as a factor in recent layoffs (see pg. 16), entry-level hiring [appears](#) to be softening in some AI-exposed occupations, and unemployment among college graduates has risen above the pre-pandemic average. So, just how concerned should we be about the disruptive potential of AI for the US labor market?

Some [prominent technologists](#) predict that AI could ultimately [eliminate](#) a massive number of jobs. They argue that, unlike previous technological shifts that primarily automated routine manual labor and clerical tasks, increasingly capable AI models will soon be able to handle much of the cognitive work humans perform, leaving many workers with less to do. They also point to the unprecedented speed of AI advances, which could outpace the labor market’s natural ability to adapt, leading to substantial displacement in as soon as the next few years. And even if the technology eventually creates new roles, they warn of a skills mismatch, which could mean that these new roles will be filled by AI itself rather than displaced human workers.

GS Senior Global Economist Joseph Briggs also expects significant labor displacement, estimating more than 9% of the labor force—around 15mn workers—could lose their jobs during the 10-year AI transition he assumes. But he believes these losses will prove temporary owing to his expectation that AI will create many new jobs over the long term even as it destroys existing ones, taking comfort from the historical precedent of technological innovations creating much new work and the dynamism of the US economy (which GS CEO David Solomon also [cites](#) as a key reason why he believes the narrative of an AI job apocalypse is overblown). And the 10-year AI transition period that Briggs assumes should provide the labor market time to adapt.

Neil Thompson, Director of the FutureTech research project at MIT’s Computer Science and AI lab, is less convinced that AI will drive large-scale job displacement. He explains that “capability is only one step in the process that leads to labor market change”—AI must also perform reliably, have access to the right information, and be cost effective to meaningfully impact jobs. And because most jobs consist of many tasks, only some of which can be automated, he expects a slower and more uneven labor market adjustment than AI’s current capabilities suggest. Thompson also takes some—though not complete—comfort from previous automation waves that created many new jobs. So, he describes AI’s impact on the labor market more as a “rising tide” than a “crashing wave”: real change and churn, especially if adoption accelerates quickly, but not sudden, widespread job loss.

We then spoke again with Daron Acemoglu, Institute Professor at MIT and winner of the Nobel Prize in Economics. He expects AI to have only a small net negative impact on labor over the next five years—far less than some technologists predict—as current AI models are more likely to replace than complement workers (GS US Economist Elsie Peng indeed finds that AI is currently substituting more than augmenting, resulting in a small net drag on the labor market). But Acemoglu warns that

job losses could be larger over the longer term if AI investment continues to focus more on replacing than complementing workers, and especially so if agentic advances, improvements in AI models’ social capabilities, or AI-robotics integration expand the set of jobs that the technology could impact (GS equity analysts James Schneider and Luya You also argue that falling token costs could make more enterprise workflows economically viable to automate). And Acemoglu is less comforted than Briggs and Thompson by history, pointing out that “no general law of economics says that job creation must match job destruction.”

So, which workers will be most impacted by these labor market shifts? While much of the concern about AI-driven job losses has centered on new graduates and white-collar workers more broadly, GS US Economists Jessica Rindels and Pierfrancesco Mei find that AI has had little impact on college graduates’ job prospects so far. But they argue that could change over the near term as college graduates are disproportionately employed in industries with high AI adoption rates and occupations with more potentially automatable tasks. Even so, Rindels and Mei remain relatively optimistic about college graduates’ longer-term prospects, finding that they have historically adjusted more nimbly than other workers to technological disruption.

As for white-collar workers more broadly, Acemoglu argues that those performing cognitive, routine tasks under predictable conditions—such as customer service and back-office work—will bear the brunt of AI displacement. As these tasks are disproportionately performed by lower-paid white-collar workers, he expects this to lead to a rise in income inequality over the coming years. However, Acemoglu notes that if workers in higher-paid managerial roles are also displaced, as [some think they could be](#), then inequality could actually decline.

Thompson agrees that much will depend on what types of work AI ultimately automates, with the key being whether it automates the most or least expert parts of a job. If AI automates the least expert tasks, employment would fall while wages rise as the work becomes more specialized; if it instead automates the most expert tasks, employment would rise while wages fall as more workers can do the job. That framework suggests the impact will cut across the occupation and wage distribution rather than hitting one worker group in particular.

Amid this debate, GS Senior US Equity Strategist Ryan Hammond finds that the technology’s impact on corporate labor needs—and, crucially, earnings—remains too uncertain for markets to reward today. As a result, he says, investors continue to favor the AI infrastructure complex, where earnings benefits are more tangible. But Hammond expects this to change, with the AI trade eventually broadening to include labor productivity beneficiaries as clearer evidence that AI’s much-touted productivity benefits can generate a durable earnings uplift emerges.

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Interview with Daron Acemoglu

Daron Acemoglu is an Institute Professor at MIT and author of several books. He received the Nobel Memorial Prize in Economic Sciences in 2024 (jointly with Simon Johnson and James A. Robinson). Below, he argues that AI is likely to have a small net negative impact on labor over the next five years and a larger negative impact over the longer term unless the industry directs more investment toward advances that complement, rather than replace, workers.

The views stated herein are those of the interviewee and do not necessarily reflect those of Goldman Sachs.



Allison Nathan: The consensus among economists seems to be that aggregate labor market data is not yet indicating significant labor market impacts from AI technology despite headlines of large AI-related layoffs. Do you agree?

Daron Acemoglu: Yes. Well-informed people like my colleague Erik

Brynjolfsson [have observed](#) an impact on the entry-level job market. But I broadly agree with the consensus that no huge effect is notable in the aggregate data yet. Some of the layoff headlines may be AI whitewashing, and some of them are genuine attempts by a handful of brave companies to experiment with completing more tasks with AI. But these efforts are not large enough to show up in the aggregate data, except possibly in some slowdown in hiring in AI-exposed entry-level positions.

Allison Nathan: Do you expect a larger impact to be visible anytime soon?

Daron Acemoglu: Yes. While it's very difficult to make predictions with any degree of certainty, as these company efforts intensify, I suspect these impacts will become more visible in 2027 in the form of layoffs or a hiring slowdown in jobs where AI can at least have a chance of replacing some tasks. AI is more likely to replace than augment jobs in the near term simply because current AI models are not good at complementing workers beyond simple tasks like reference checks, except in a few fields such as biological and chemical research. So, I expect a net negative impact on the number of jobs in coming years.

But it is important to emphasize that, in my estimation, the scale of job losses won't be anywhere close to the very large layoffs [some](#) are predicting. I estimate such losses will be less than, say, 2-4% over the next five years. My caution about AI's spread and, in turn, its economic impact, reflects the relative lack of user-friendly applications today that would enable large employers to leverage the foundation models.

Right now, workers are largely engaged in prompt engineering, which is difficult, inconsistent, and time-consuming. Coding/software engineering remains the notable exception because many aspects of coding are now routinized and software engineers today are AI experts that know how to leverage, troubleshoot, and check the models. Agentic advances may open the door to developing applications that enable broader AI use, but that's unlikely if every company must develop its own applications. What's required to move the needle is the

development of applications that all companies can leverage, like the Microsoft Office version of AI, so to speak.

So, I expect only a small net negative impact on labor over the next five years or so, concentrated in white-collar workers whose tasks are cognitive and routine and involve performing similar functions under predictable conditions rather than typically confronting new and unusual things or constant innovation, creation, and intense social interaction or judgment. These exposed occupations include customer service representatives or back-office work. In the US, these workers number roughly 8-9 million, which is not a small number but accounts for only roughly 5% of the workforce. And the impact could be even smaller and slower if the early movers that are already experimenting with AI labor replacement get their mouths burned.



AI is more likely to replace than augment jobs in the near term simply because current AI models are not good at complementing workers beyond simple tasks like reference checks... So, I expect a net negative impact on the number of jobs in coming years. But... the scale of job losses won't be anywhere close to the very large layoffs some are predicting."

Allison Nathan: Does your calculus about AI's labor market impact shift over the medium-to-longer term?

Daron Acemoglu: Much more uncertainty exists over a 10- to 15-year horizon. How this evolves depends on where investments are directed. I have [argued](#) for over 10 years that the complementary path—whereby AI technology complements versus replaces human labor—could be quite productive. But such a path requires different investments in terms of pre-training and domain-specific applications and data. And those investments are not happening on a sufficient scale today to support a human-complementary path. A shift in the industry's priorities and incentives that leads to greater investment in this direction could result in a much more positive outcome for labor. But should things continue as they are, I would expect bigger net job losses within the next 10 to 15 years.

Many wild cards could also worsen these losses. If agentic advances lead to widely accessible, user-friendly applications, then job losses could extend beyond back-office white-collar workers to middle-manager roles that involve more judgement and even to more senior managerial roles should the

technology's cognitive abilities vastly improve from here. The large chunk of jobs that involve social interactions could also come under threat if the social dimensions of models continue to improve and human consumers become more accepting of interacting with bots, which younger generations seem more open to.

And perhaps one of the biggest wildcards is the potential integration of AI and robotics. Significant efforts are being made on this front, especially outside of the US, but several challenges exist to such integration on both the robotics and AI sides. Robots need to be flexible and gentle while still being able to perform physically challenging tasks, and AI models currently don't do very well on spatial causal relations, which combine spatial reasoning with causal inference. AI models also lack immediate feedback controls, which require much greater awareness of the environment than what large language models (LLMs) can achieve by simply reading text and understanding words. So, many advances would be required, which suggests progress will likely be slow. But should amazing breakthroughs occur, that would greatly increase the number of jobs that AI could impact given that roughly 50% of jobs in the US economy require physical labor.

“A shift in the industry's priorities and incentives that leads to greater investment in [the complementary path] could result in a much more positive outcome for labor. But should things continue as they are, I would expect bigger net job losses within the next 10 to 15 years.”

Allison Nathan: The vast majority of job growth since 1940 has been driven by technology and the creative destruction process. So, do you think this time will be different?

Daron Acemoglu: Not necessarily. But no general law of economics says that job creation must match job destruction. Job growth over the last 80+ years resulting from a combination of changing tasks, new tasks, and a shifting structure of occupations has not occurred at an even pace. For example, fewer jobs for workers without a college degree have been created since 1980 than before 1980, leading to a decline in this group's employment-to-population ratio and wages, especially among men. During the 1950s, 1960s, and early 1970s, job creation exceeded job destruction, which led to increased demand for labor and a faster rise in wages than productivity. But since the late 1970s, job creation has mostly fallen short of job destruction, with the destruction centered in manufacturing jobs. The job losses I expect in the coming years amount to a similar scenario, but with the job losses instead centered around cognitive office jobs. So, nothing is completely different this time around, but every period is different in terms of the balance between job creation and destruction.

Allison Nathan: What implications will AI likely have for wages and income inequality?

Daron Acemoglu: To answer that, it's important to understand that inequality and employment are more tightly linked than is sometimes implied because declining or stagnant wages for some groups typically reduce these groups' employment-to-population ratio or participation rate as they become less motivated to work. So, a decline in wages and employment will likely go hand-in-hand for specific groups. And, assuming lower-paid white-collar workers will be hit hardest by AI, at least in the coming years labor income inequality will likely rise.

Of course, if job losses also encompass high-paid managerial roles, as some people seem to expect, then income inequality could actually decline. But, as we've discussed, I don't believe that's the more likely scenario. And even if well-paid managers did find themselves in the crosshairs, they would likely have more success at finding new jobs than the next layer of displaced workers. So, they'd still be unlikely to bear the brunt of AI displacement.

Allison Nathan: What about the implications for capital versus labor income? Which is more likely to reap the benefits of AI technology?

Daron Acemoglu: Labor-capital inequality—the gap between income earned from work and income generated from assets—will more certainly increase. LLMs are increasing the capital intensity of production. The huge capital investment they require is a telltale sign that capital income will increase, even if some of those investments aren't fully profitable or don't break even. They nevertheless generate substantial revenue, almost all of which is capital-based; while AI companies presumably pay their employees well, the main input is data and compute. So, labor-capital inequality will rise. Moreover, since many of these companies have concentrated ownership, inequality will increase even within capital income earners.

Allison Nathan: What role, if any, will policy have in shaping labor market outcomes?

Daron Acemoglu: Policy has a vital role to play. But before we even involve policy, we must first arrive at a societal consensus about what we want from AI and what AI can deliver. In my view, complementary AI could result in productivity *and* labor benefits and prove enormously useful for the economy and society at large. But what's missing today is a clear conversation about what it would take to purposefully build human-complementary AI tools. Some people argue that the best approach would be to double down on LLMs and then use them in human-complementary ways. I disagree. In my view, an approach that shifts the focus from “machine intelligence” to “machine usefulness”—or from machines designed to mimic and replicate human abilities to those designed to augment and empower human capabilities—would be better.

Once the way forward is agreed, we can then start to think seriously about the right policies to implement, spanning everything from the structure of the tax system, to what the government can do in terms of creating lodestars and directions for the AI industry, and broader competitive measures to ensure that the industry doesn't become too concentrated. As I have long said, technology is not destiny—we have the power to shape what it does and who it benefits based on the choices we make.

Interview with Neil Thompson

Neil Thompson is the Director of the FutureTech research project at MIT's Computer Science and Artificial Intelligence Lab and a Principal Investigator at MIT's Initiative on the Digital Economy. Below, he argues that AI's impact on the labor market is more likely to resemble a rising tide than a crashing wave, driving uneven churn that workers, firms, and policymakers can anticipate and manage rather than sudden, widespread job loss.

The views stated herein are those of the interviewee and do not necessarily reflect those of Goldman Sachs.



Jenny Grimberg: What is the right way to think about AI's potential impact on the labor market?

Neil Thompson: We should think of AI as a transformative technology that is already highly capable and quickly becoming even more so but recognize that capability is only one step in the process that leads to labor market

change. For AI to meaningfully affect jobs, it must be able to perform a task when given the right information, have access to that information, and be cost effective. That's often a tall order. Even with perfect information, AI doesn't have a 100% task success rate. It may be able to automate part of a process, like check-in at a doctor's office, but it can't give reliable medical advice without access to privacy records or other sensitive data that isn't easily obtained. And [we've found](#) that, in many cases, an AI system is too expensive to run at the level of precision required.

In practice, that means that while AI's capabilities will continue to improve rapidly, which will undoubtedly lead many people to conclude that it will have a very large impact on labor and the economy more broadly, the adoption process will be lengthy. Large businesses will likely automate before small ones, and the highest-value use cases will be prioritized before a long tail of tasks that will take much longer to automate, or never be automated at all. In that sense, AI will likely resemble previous automation waves: the set of tasks the technology can theoretically automate will be much bigger than the set it actually automates. So, the impact AI ultimately has on the labor market may not be nearly as large as its impressive capabilities suggest.

“The impact AI ultimately has on the labor market may not be nearly as large as its impressive capabilities suggest.”

Jenny Grimberg: You often make the distinction between “expert” and “inexpert” tasks. Why is that distinction so important for understanding how AI will impact labor?

Neil Thompson: We have found that, for most jobs, only a fraction of the tasks in that job can be automated. What happens to the worker whose job is partially automated is a key question that my colleague [David Autor and I have explored](#). People often think about this through the demand lens: if 30% of a job is automated, demand for that job may fall by 30%, resulting in lower wages and fewer workers. But thinking about

this through the supply lens is more useful. If AI automates the least expert part of a job, workers can spend more time on valuable tasks. In that scenario, employment falls while wages rise because the remaining work is more specialized. But if AI instead automates the most expert part of a job, workers are left doing the lower-value tasks, leading employment to rise while wages fall as more people can do the work and it becomes less valuable.

Think about the fate of taxi drivers vs. proofreaders over the last several decades. GPS technology automated the most expert part of a taxi driver's job—knowing all the streets and routes around a city. So, competition in the space has increased, with many more taxi drivers now than ever before, which has driven down wages. Conversely, word processing technologies automated the least expert part of a proofreader's job—checking spelling and grammar—and left them with much more complex tasks like figuring out how to structure an argument and present evidence. So, fewer people are now employed as proofreaders, and those who remain are more highly compensated.

AI technology will likely do both, automating expert tasks in some jobs and inexpert tasks in others. So, the labor market impacts will be diverse, and businesses and governments must plan for much more uneven outcomes than the headline debate often suggests.

Jenny Grimberg: What impact is AI having on expertise so far, and what does that suggest for how it will ultimately impact the aggregate level of jobs?

Neil Thompson: We find that AI is expertise-increasing in some jobs and expertise-degrading in others, with these effects appearing evenly across the job distribution rather than being concentrated in high- or low-wage jobs. That matters because it implies that when AI partially automates jobs, it doesn't push the labor market in a single direction. In some cases, the remaining work becomes more valuable, in others less so, and the employment effects run both ways, as we've discussed.

But at the aggregate level, what really matters is that some jobs will likely be largely automated and disappear, while new jobs are created. The balance between those two forces is uncertain. Previous automation waves have resulted in many new jobs, which offers some reassurance. But AI also processes information in ways more similar to humans than previous technologies, so more worry may be warranted. Still, it's too early to know whether AI will result in substantial unemployment or whether the new tasks it creates, along with the extra leverage it gives workers, will ensure that plenty of meaningful work continues to exist.

Jenny Grimberg: But do AI's broad capabilities make significant job replacement more likely than in the past?

Neil Thompson: Not necessarily. Capability is not the same as reliability. Traditional technologies like spreadsheets or databases have a limited scope, but within that scope they perform perfectly—we never worry about Excel multiplying numbers incorrectly or a database missing records so long as we query them correctly. AI tools have a much broader scope, but they can hallucinate or even go off the rails, and those problems haven't yet been solved. So, AI is a powerful assistive tool, but it can't be slotted into a process and just forgotten about. The technology's broad scope also means it will likely create new tasks and increase output at the individual and firm levels, potentially leading organizations to grow both in terms of headcount and their bottom lines. So, we should be skeptical of the notion that AI will destroy work without creating new work along the way just because it's so broad and powerful in its capabilities.

“We should be skeptical of the notion that AI will destroy work without creating new work along the way just because it's so broad and powerful in its capabilities.”

Jenny Grimberg: If AI inference costs continue to fall, would that translate into much wider adoption and larger labor market impacts?

Neil Thompson: Not as much as you might think. Sure, if smaller and cheaper “[meek models](#)”—which operate with limited computational resources—can eventually perform tasks that today require the most advanced frontier models, that would ease adoption somewhat by, for example, making it possible to run powerful AI on a phone or a single GPU. But for most AI systems, the marginal cost of running them is not the main barrier to adoption. The cost of training the model, which includes infrastructure and data costs, also factors into the price of a model, whether it's an in-house model or third-party subscription, and often matters more. Even if AI models become cheaper and easier to deploy, a lack of access to private data, rapidly changing decision criteria, or a need for significant back-and-forth interaction between a human and the AI system may still impede adoption, as could the fact that AI performance is known to degrade fairly quickly as tasks become more complex. So, a further decline in inference costs could increase adoption, but it wouldn't on its own determine the scale of AI's labor market impact.

Jenny Grimberg: Who will capture the economic value that AI creates? Does labor have any shot of doing so?

Neil Thompson: The value will accrue to the owner of the scarce resource. If an AI system and a human jointly perform a task and the necessary AI expertise is rare or hard to access, the system provider can charge a premium. But if that expertise is commoditized—which we're seeing in many cases today given intense competition among providers—the value is more likely to accrue to the worker, especially if AI automates the least expert parts of the job. The same logic applies to

businesses. Even if the provision of the AI tool itself is generating the economic value, it may not accrue to the model builder. In some cases, proprietary data is the scarcer asset, so the data provider captures the value. And firms with unique datasets that are important for running a manufacturing process, managing client relationships, or supporting other key operations will find those assets becoming even more valuable as AI tools improve.

Jenny Grimberg: Ultimately, how important will the speed of AI adoption be in shaping labor market outcomes?

Neil Thompson: Speed is critical. Consumer AI adoption has been very rapid but shallow, with consumers mainly using AI for simple queries. And enterprise adoption remains limited. We find that only around [10-20% of S&P 500 firms](#) are currently using AI in ways that are generating revenue and meaningfully benefitting them. If that number were to rise sharply, the labor market impact would be more significant. Fast, wide-scale adoption would cause more disruption, not necessarily because fewer jobs would exist overall, but because churn would occur more quickly: some tasks and jobs would disappear, new ones would be created, and workers would have less time to transition into new roles. Workers, businesses, and the government will need to manage that process, and the faster it occurs, the more challenging it will be.

Jenny Grimberg: So, are the worries that AI will lead to a job apocalypse warranted, or overblown?

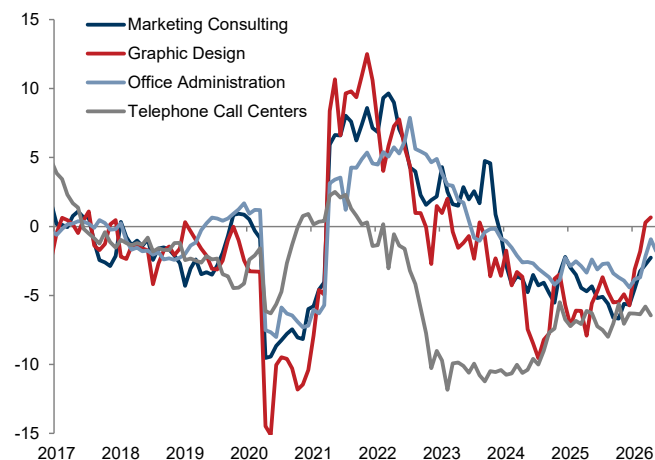
Neil Thompson: I am neither complacent nor alarmist. People are right to focus on AI's rapidly evolving capabilities as a potential challenge to labor. But it's still too early to say that we are headed for large-scale job destruction, and I am skeptical that widespread AI layoffs are already occurring despite how some firms are characterizing recent headcount reductions. The more likely near-term outcome is partial rather than complete automation of jobs, which will push some workers to become more expert and others less expert. That points to change and churn, but not necessarily widespread job losses. Over the longer term, the key question is how quickly AI will create new tasks and how effectively we leverage the technology to generate an “AI dividend” in the form of higher productivity and prosperity. So, I am worried about the churn, optimistic about the dividend, and uncertain about the long-term impact on jobs.

I think about this as the difference between crashing waves and rising tides. In a crashing wave scenario, workers are walking along the seashore on a beautiful day and, out of nowhere, a wave hits and washes many of them away. But [our research](#) suggests that is not the dominant pattern. Instead, the more common pattern is that of a rising tide. In such a scenario, some workers are on the sand, some are ankle-deep in water, and some are up to their knees as the tide comes in. AI is coming just as the rising tide is, but not without warning. That doesn't mean the adjustment will be easy, especially if the tide rises fast. But it does mean workers and businesses can anticipate it and have a better shot at managing the transition. Policy also has an important role to play, both in helping workers navigate the churn through retraining and wage insurance and in thinking more broadly about how to ensure that the gains from AI accrue to labor. So, the risks to the labor market are real, but likely not apocalyptic.

AI & the US labor market: a visible...

Employment growth has fallen below its 2015-2019 trend in many non-tech...

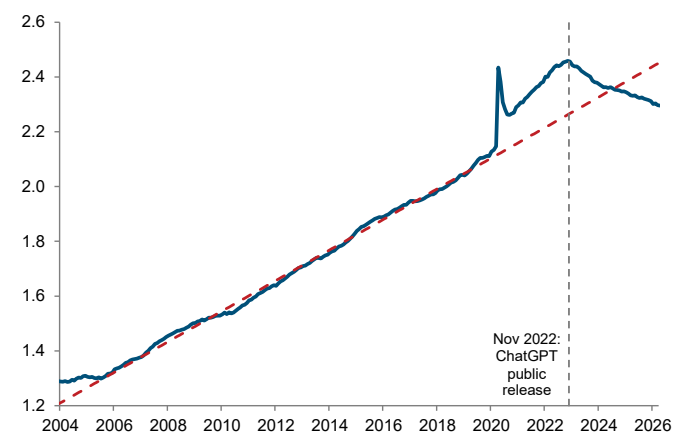
Year-over-year payrolls growth deviation from 2015-19 trend, non-tech service sectors, pp



Source: US Bureau of Labor Statistics, Haver Analytics, Goldman Sachs GIR.

...with tech's share of total employment continuing to decline relative to its long-run trend

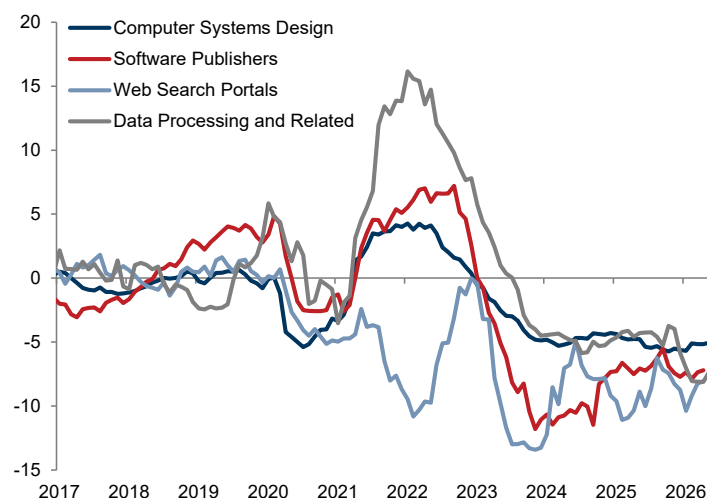
Tech sector* share of total employment, %



*Tech refers to the software publishers, data processing & related, web search & related, and computer systems design subsectors. Red line is 2007-19 trend. Source: US Bureau of Labor Statistics, Haver Analytics, Goldman Sachs GIR.

...and tech service sectors where AI use cases have been established...

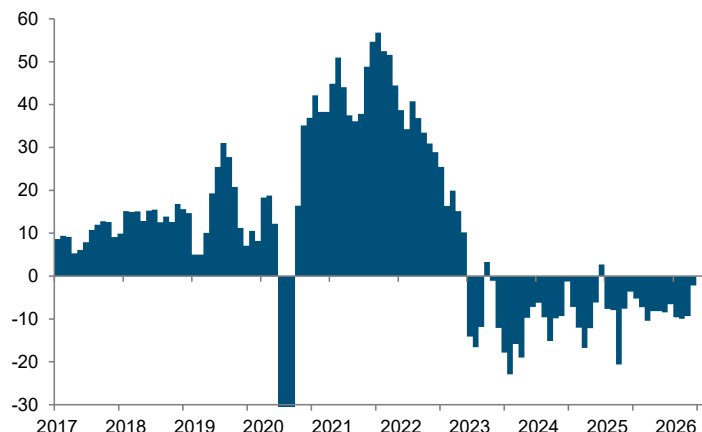
Year-over-year payrolls growth deviation from 2015-19 trend, tech service sectors, pp



Source: US Bureau of Labor Statistics, Haver Analytics, Goldman Sachs GIR.

In aggregate, industries with established AI use cases are seeing employment contract by around 11k per month...

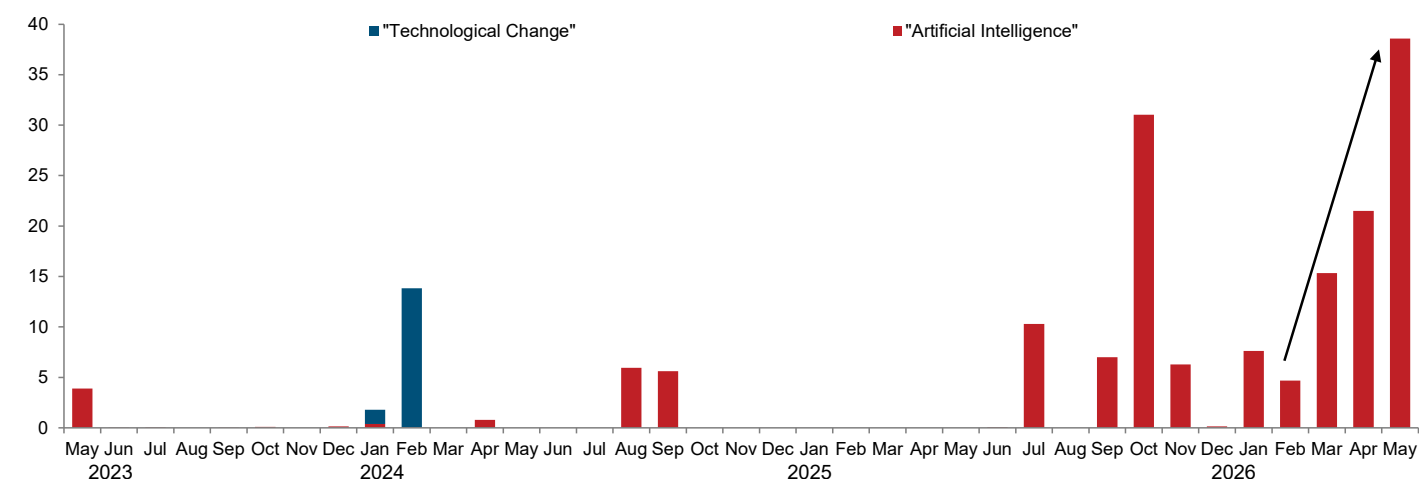
Avg monthly job growth in AI-affected sectors*, last 3m, thousands



*Management consulting, graphic design, office administration, telephone call centers, computer systems, software publishers, web search, data processing, movie production, broadcasting, publishing, and document preparation services. Source: US Bureau of Labor Statistics, Goldman Sachs GIR.

...and layoffs attributed to AI have accelerated in recent months as many high-profile tech companies have announced layoff plans

Layoffs attributed to each factor in corporate announcements, thousands

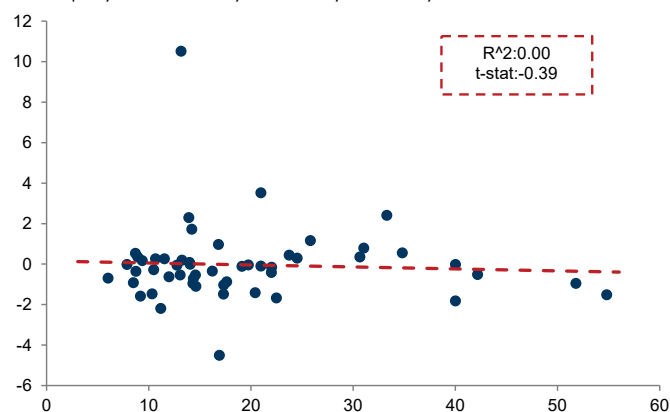


Source: Challenger, Gray & Christmas, Goldman Sachs GIR.

...but overall limited impact so far

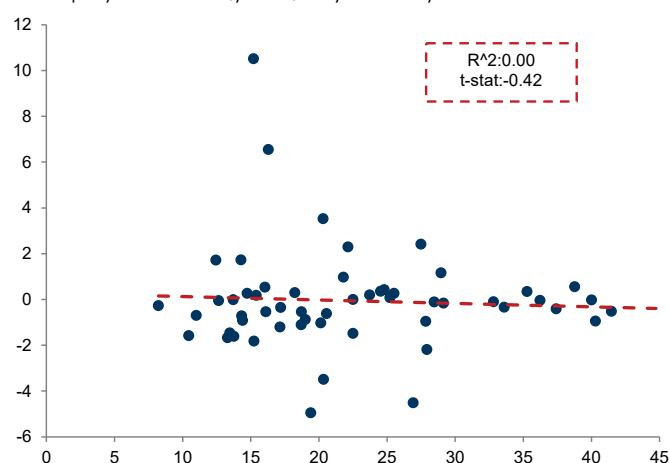
Despite this, AI's overall impact on the labor market remains limited, with no statistically significant relationship between AI adoption and unemployment...

Relationship between AI adoption (x-axis) and the unemployment rate (y-axis)* by industry



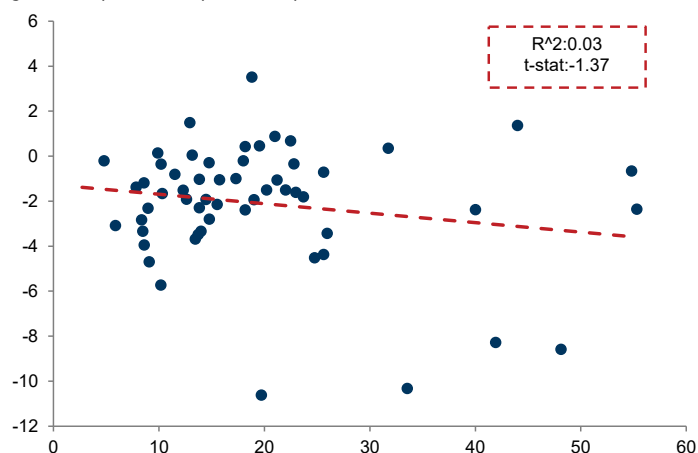
...AI exposure and unemployment...

Relationship between AI exposure (x-axis)** and the unemployment rate (y-axis)* by industry



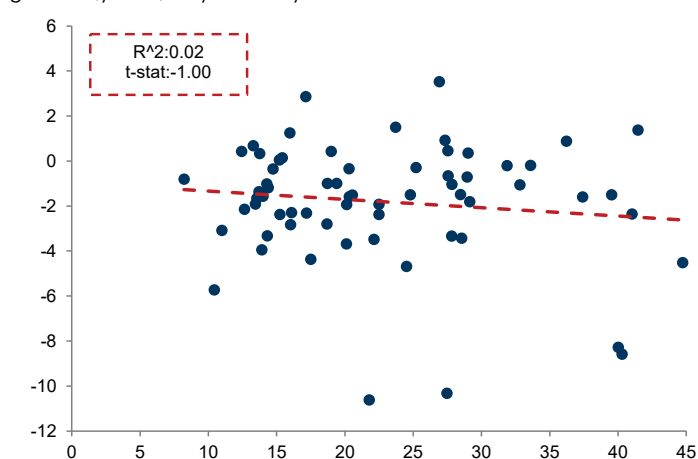
...AI adoption and employment growth...

Relationship between AI adoption (x-axis) and employment growth (y-axis)* by industry



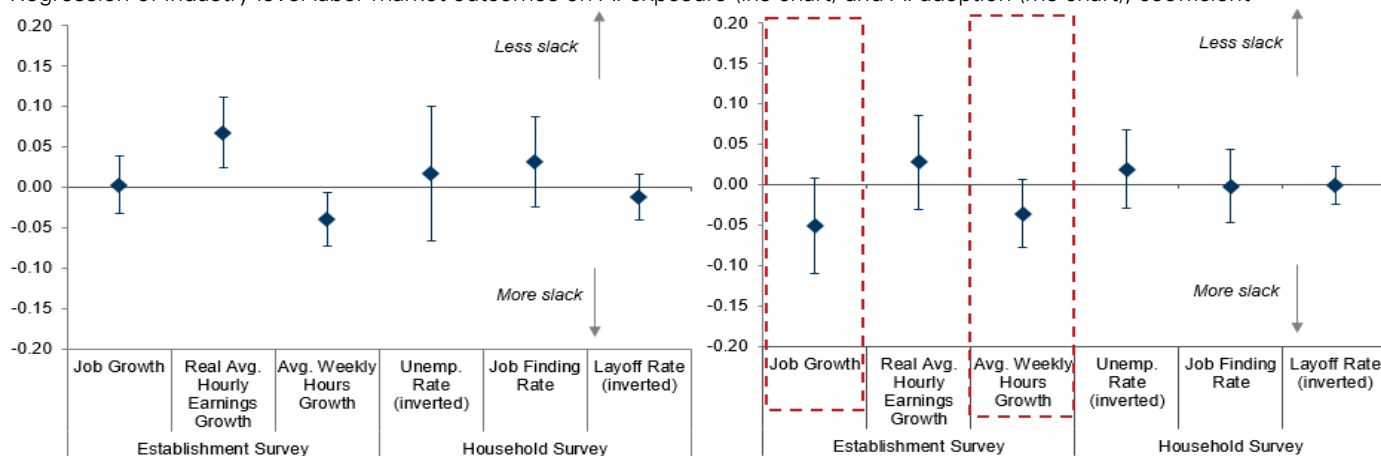
...or AI exposure and employment growth

Relationship between AI exposure (x-axis)** and employment growth (y-axis)* by industry



The correlation between AI exposure/adoption and broad measures of labor market slack also remains limited, though early signs of a negative relationship between AI adoption and job growth and hours worked growth have emerged

Regression of industry-level labor market outcomes on AI exposure (lhs chart) and AI adoption (rhs chart), coefficient***



*Latest 3m average minus the 2015-2019 average.

**See this note for more detail on how we calculate AI exposure scores.

***All dependent variables are taken as the deviation from 2015-2019 averages. Average coefficient over the past three months is plotted.

Source for all charts: US Bureau of Labor Statistics, IPUMS, US Census Bureau, Goldman Sachs GIR.

Special thanks to GS global economists Joseph Briggs and Sarah Dong for charts, which were originally featured in the AI Adoption Tracker.

How much AI labor displacement?

Joseph Briggs argues that while the AI transition could displace more than 9% of workers, the labor market hit should be temporary as new job creation absorbs these losses over the long term

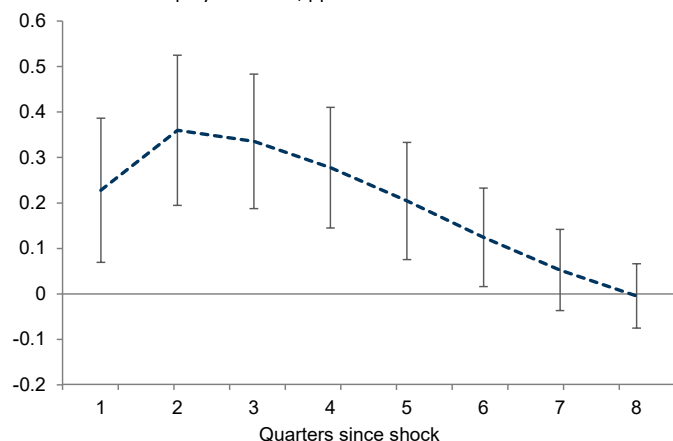
We continue to expect that AI will significantly boost economic activity, with our baseline forecasts assuming a 15% uplift to economy-wide productivity and GDP following full adoption. While the AI transition will undoubtedly lead to some labor displacement, we remain positive on the long-run benefits of AI and expect the new jobs it creates to more than offset any near-term labor market headwinds (which remain modest today, see pgs. 8-9) over the long run.

Moderate near-term displacement

Last year, we estimated that each 1pp increase in technology-driven productivity growth raises the unemployment rate by 0.3-0.4pp in the near term—as AI-displaced workers experience an extended period of joblessness before finding new work—but that this “frictional” unemployment fades back to zero after two years as workers find new jobs.

Each 1% increase in technology-driven productivity raises the unemployment rate by 0.3-0.4pp

Effect of a 1pp increase in labor productivity due to a positive technology shock on the unemployment rate, pp



Source: Gali (1999), Goldman Sachs GIR.

Combining this historical pattern with our baseline forecast for a 15% productivity boost from AI implies that around 6% of workers—equivalent to around 10-11mn workers—would experience an unemployment spell due to AI over its adoption cycle. This estimate is consistent with our bottom-up, task-based analysis of AI displacement risks, that led us to conclude that AI will likely displace 6-7% of workers (with plausible estimates ranging from 3-14%). If adoption occurs over a 10-year period, then the peak impact on the unemployment rate would be a bit more than 0.5pp.

Our prior estimates of the likely rise in the unemployment rate (i.e., the *stock* of unemployed workers at any given time) likely understate the total amount of displacement (i.e., the *flow* out of existing jobs) if some workers find new positions quickly after being displaced. Furthermore, some displaced workers

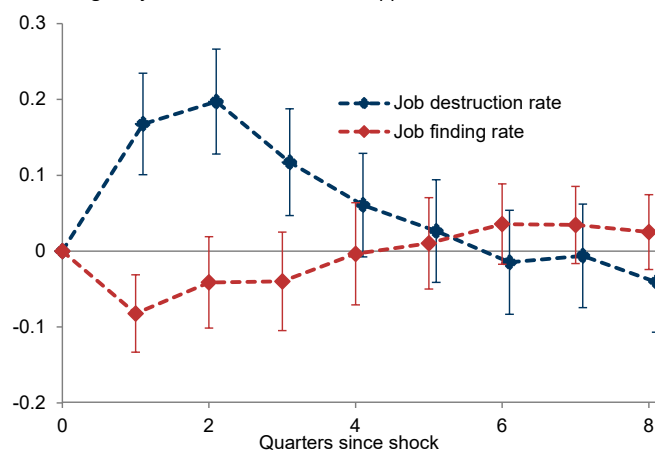
may exit the labor force rather than look for a new job, thereby further dampening the unemployment rate increase.

To finetune our estimates of total displacement, we examine how technology-driven productivity gains impact gross job formation and destruction rates (both measured as the share of overall employment). These updated estimates imply that technology-driven productivity gains modestly lower new job creation (as some businesses pause hiring) and moderately increase job destruction.

Most importantly, our new estimates suggest that each 1% increase in productivity due to technological change cumulatively raises the job destruction rate by around 0.5-0.6pp in the two years following the shock—roughly 50% higher than our prior displacement estimate—implying that up to 9% of workers could be displaced over the entire 10-year AI adoption cycle under our baseline productivity forecast.

Each 1% increase in technology-driven productivity cumulatively raises the job destruction rate by 0.5-0.6pp over two years

Effect of a 1pp increase in labor productivity due to a positive technology shock on gross job formation/destruction, pp



Source: BLS, Goldman Sachs GIR.

And historical patterns suggest some upside risk to even this larger estimate. Leveraging industry-level, utilization-adjusted total factor productivity shocks (TFP) to help disentangle factors that could amplify job displacement reveals two patterns that suggest further displacement upside.

First, larger TFP shocks tend to lead to greater displacement. For TFP shocks above the 75th percentile of their historical distribution, we find that each 1% boost to productivity raises the cumulative job loss rate by 0.6-0.7% over a two-year window. Given that we anticipate the AI boost to productivity will be larger than most historical episodes, this pattern suggests more scope for displacement during the AI transition.

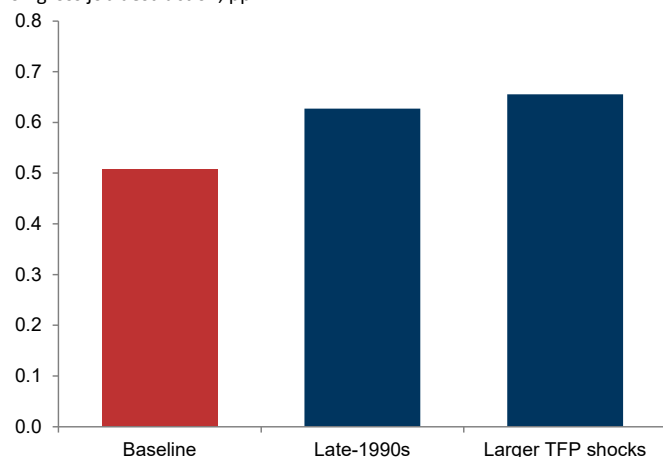
Second, the effects of productivity shocks were larger in the late 1990s and early 2000s, with each 1% technology-driven boost to productivity cumulatively raising the gross job destruction rate by 0.6-0.7%. Given that the information and communication technology (ICT) productivity boom is the closest historical analogue to the AI productivity boom that we anticipate, this also suggests more scope for displacement.

In light of this new evidence, it seems reasonable to expect that the AI transition could lead to the displacement of over 9% of the labor force—or around 15mn workers. Assuming these

job losses are distributed across a 10-year adoption period and that most workers find new work within a year, the unemployment rate would likely still rise by less than 1pp.

We see more scope for displacement given that we expect a large productivity uplift, especially if job losses follow patterns seen in the late 1990s or the goods sector

Cumulative effect of a 1% increase in TFP due to a positive technology shock on gross job destruction, pp



Source: Fernald (2014), BLS, Goldman Sachs GIR.

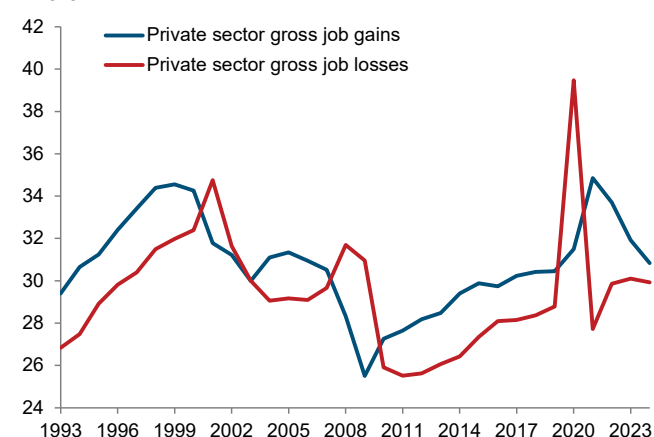
New job creation is a powerful offset to labor disruption

Despite our expectation that AI-related job losses will lead to a meaningful amount of labor displacement, we continue to expect that labor market headwinds will be temporary. Key to this view is our expectation that over the long run AI will create many new jobs even as it destroys existing ones.

Historical evidence provides support for this view. As we've noted, technology has been a key driver of long-run job growth. About 60% of workers today are in occupations that didn't exist in 1940, which account for 85% of the job growth since then.

The US economy creates 25-35mn gross new jobs in a typical year

Millions



Source: BLS, Haver Analytics, Goldman Sachs GIR.

Most importantly, the US labor market is incredibly dynamic. Although the US economy adds a few million net jobs per year at most—well below the annual AI-related job losses that we anticipate during the transition—this number is the difference between much larger gross job gains (defined as net job gains at expanding and opening establishments) and gross job losses (defined as net job losses at expanding and closing establishments). These gross measures have historically

fluctuated between 25 and 35mn per year, implying that the economy is constantly creating and destroying jobs, even during normal times.

In a dynamic AI-enabled economy, it is quite plausible that a moderate step-up in the pace of job creation will boost gross job gains enough to absorb most AI-related job losses and mitigate the labor market hit from AI disruption.

Where will these new jobs emerge? It is hard to say exactly, but history again provides a guide.

First, technology can directly create new occupations, as was the case for the nearly 15mn jobs in occupations enabled by the emergence of the digital economy.

Second, it can create new occupations by enabling increased specialization, a dynamic that has been a key driver of the increase in healthcare employment from roughly 2mn to over 18mn over the last 60 years.

Third, technology can indirectly boost discretionary demand in occupations that emerge due to income gains, even if these roles were already technologically feasible. For example, 1mn workers today are employed in industries like pet care, nail salons, educational support/tutoring, and athletic coaching that have grown rapidly in recent decades.

In our view, the combination of these dynamics should create enough new jobs to reabsorb AI-displaced workers, thereby limiting the risks of persistent job loss over the long run.

Risks around displacement

We see two key risks around our relatively benign AI labor market outlook.

First, even if our updated estimate that the AI transition will displace over 9% of workers is correct, the job losses could be concentrated over a shorter period, resulting in a notably larger rise in the unemployment rate. Indeed, the long-run decline in the "routine occupation" share of the labor force has historically mostly occurred in economic recessions, when the economy is less able to create new positions. The labor market transition would likely be more painful if this dynamic repeats itself, but even in this scenario displacement-related job losses would still likely be temporary.

The second more existential risk is that "this time is different" and AI is fundamentally a more labor-replacing technology than those that inform our historical analysis. While tech sector commentators extrapolating from their own experience tend to lean into this view, we believe that past technology shocks provide a useful guide to the likely labor displacement from AI automation (although effects could be larger if the eventual emergence of AGI eventually makes human input in knowledge-based work tasks redundant).

That said, the answer to "is this time different?" will be hard to prove or disprove until the labor market holds up or doesn't in the coming years. So, the debate around this question will remain lively for the foreseeable future.

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AI & labor: substitution vs. augmentation

Elsie Peng investigates AI's impact on the US labor market through two key channels of impact: substitution and augmentation

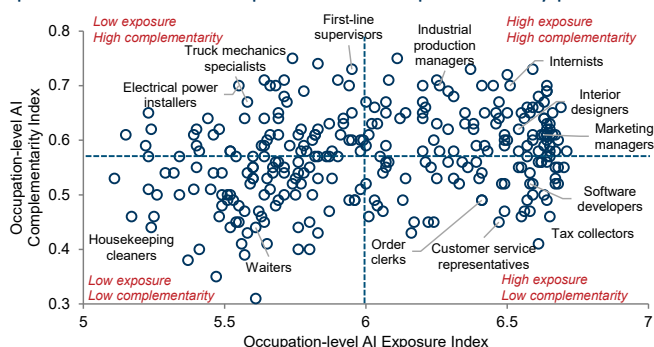
AI's impact on the US labor market is more nuanced than the headline debate—namely, that AI will either lead to significant job losses or usher in a historic era of productivity growth—suggests. We find that AI is both *substituting* for labor by performing tasks humans once did and *augmenting* labor by making workers more productive.

This nuance matters because the aggregate labor market impacts from substitution versus augmentation of labor differ. When AI substitutes for labor, the employment effect is straightforwardly negative—firms automate and hire fewer people. When AI augments labor, the impact is less clear-cut—a more productive worker means that fewer workers produce the same output, but lower production costs may also increase demand sufficiently to create more jobs overall—a dynamic known as the Jevons paradox. In practice, we find that AI augmentation has created jobs, but not enough to fully offset the job losses from AI substitution, resulting in a small net drag on the labor market.

Complementarity is key

Most AI exposure indices simply measure the overlap between AI capabilities and job requirements. But two occupations with similar AI exposure may differ in their degree of AI “complementarity.” For example, customer service representatives and interior designers face similar levels of AI exposure, but AI is far more complementary to interior designers because their work involves more unstructured tasks and requires a physical presence at worksites that AI cannot yet replicate. Combining an exposure score with a complementarity score¹ therefore provides a better gauge of AI's occupation-level impacts. Occupations with high exposure but low complementarity—e.g. telephone operators, insurance claims clerks, and bill collectors—face the highest substitution risk, while occupations with high exposure and high complementarity—e.g. education workers, construction managers, and judges—have the highest augmentation potential.

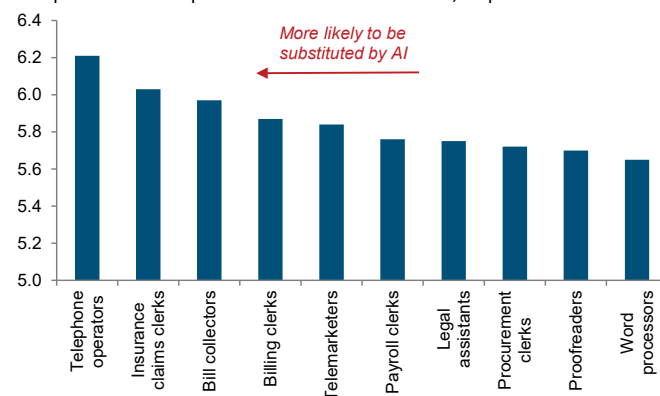
Whether AI is more likely to augment or substitute for labor depends on its level of exposure and complementarity profile...



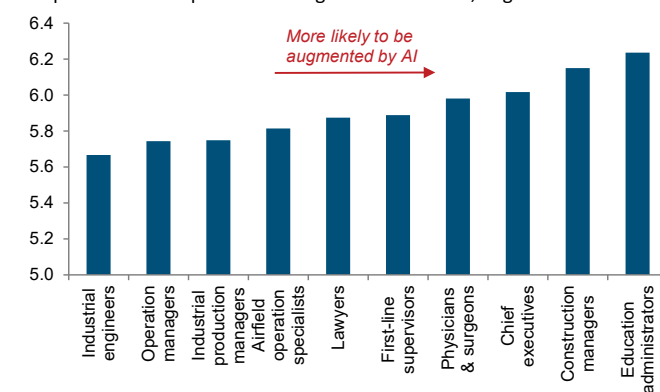
Source: Goldman Sachs GIR.

...with telephone operators, insurance claims clerks, and bill collectors facing the highest substitution risk and education workers, judges, and construction managers having the highest augmentation potential

Occupations most exposed to AI substitution effect, displacement score



Occupations most exposed to AI augmentation effect, augmentation score



Source: Goldman Sachs GIR.

Negative substitution effects, positive augmentation effects

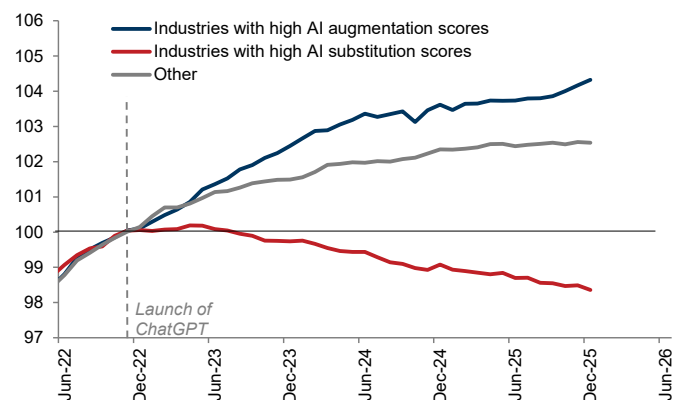
Since ChatGPT's launch in November 2022, this gauge suggests that industries with high AI substitution risk have experienced a notable decline in payroll employment while industries with high augmentation potential have experienced somewhat larger employment gains than those less exposed to AI, especially in recent months. Consistent with this pattern, occupations with high substitution risk have also experienced a larger rise in unemployment rates than other occupations, while those with high augmentation potential have experienced a slightly smaller increase, with much of this divergence emerging since mid-2025 when enterprise AI adoption started to rise (we exclude healthcare, education, and government from this analysis because post-pandemic job growth in these sectors largely reflected catch-up hiring rather than AI effects).

Occupations more exposed to AI substitution have also experienced a sharper decline in job openings, with the effect most visible in computer and information services, sales, and management occupations. By contrast, occupations more likely to be augmented by AI, as well as those with limited AI applicability, have experienced a more gradual normalization toward pre-pandemic baselines.

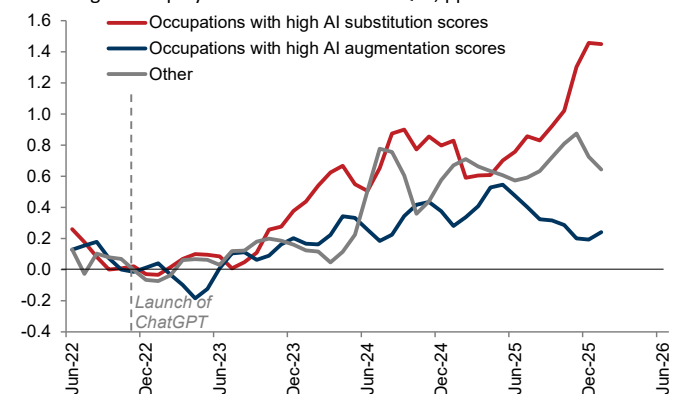
¹ We draw on an occupation-level AI complementarity index developed by [IMF economists Pizzinelli et al. \(2023\)](#). The index is constructed based on six work abilities: interpersonal interaction, accountability, physical proximity, human judgment, the share of unstructured tasks, and the level of expertise required to perform work tasks effectively. Jobs that rely less on these attributes receive a lower complementarity score, meaning AI is more likely to replace rather than assist the worker.

Since ChatGPT's launch, industries and occupations with high AI substitution risk have experienced larger declines in employment and increases in unemployment on average

Payroll employment by industry exposure to AI, index, 4Q22=100



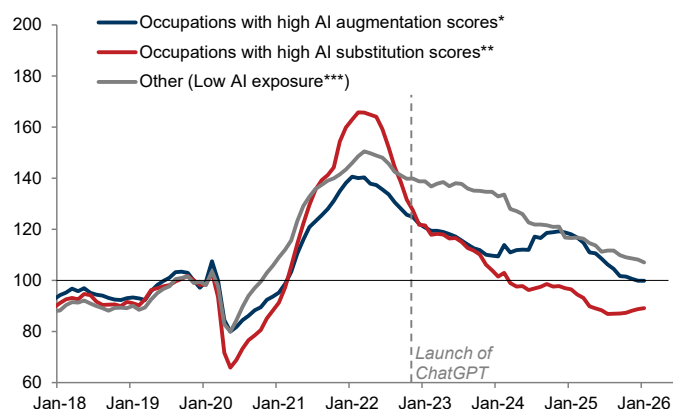
3m average unemployment rate relative to 4Q22, pp



Source: Department of Labor, Goldman Sachs GIR.

Occupations with high AI substitution risk have experienced larger declines in job openings over the past couple of years

Job openings by AI exposure, index, 4Q19=100



*We define an occupation as having a high AI augmentation score if only some of its core tasks are automatable by AI, with others still requiring human judgement, creativity, or interpersonal skills.

**We define an occupation as having a high AI substitution score if most of its core tasks can be readily automated.

***We define an occupation as having a low AI exposure if only a small share of its core tasks can be automated by AI.

See this note for more details on the methodology.

Source: LinkUp, Goldman Sachs GIR.

A small labor market drag

This analysis suggests that AI substitution has reduced monthly payroll growth by roughly 25k and raised the unemployment

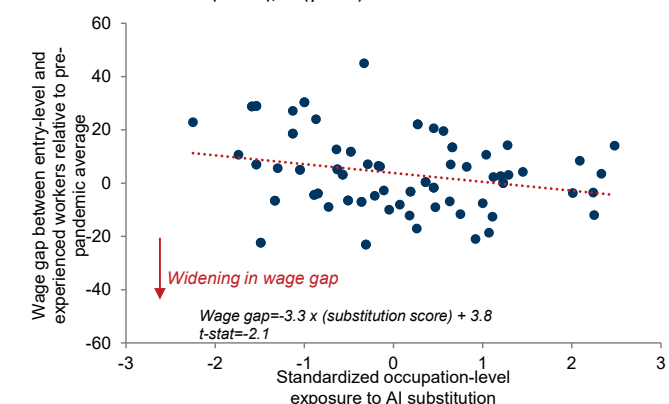
rate by 0.16pp over the past year. Augmentation has partially offset this shift, adding around 9k to monthly payroll growth and lowering the unemployment rate by 0.06pp. This implies a still-small net AI drag of 16k on monthly payroll growth and a 0.1pp increase in the unemployment rate. Younger and less experienced workers have been hit harder by AI substitution effects, bearing the brunt of the impact, with the wage and unemployment rate gaps between entry-level and experienced workers widening by 1.3% and 0.6pp, respectively, from pre-pandemic averages.

The unemployment rate and wage gaps between entry-level and experienced workers have widened more sharply in occupations more exposed to AI substitution

Standardized measure (x-axis), pp (y-axis)



Standardized measure (x-axis), % (y-axis)



Note: We use age to approximate for work experience. Entry-level workers are defined as workers under the age of 30, and experienced workers are defined as workers aged 31-50. The top chart plots the unemployment rate gap as of January 2026 using the monthly CPS sample, and the bottom chart plots the wage gap as of 2025 using the annual ASEC sample.

Source: Department of Labor, Goldman Sachs GIR.

That said, augmentation measures alone don't fully capture the boost to construction hiring from the AI infrastructure investment boom or the additional labor demand that AI-driven productivity and income gains have generated. So, the true AI drag on the labor market is likely smaller than our estimates imply.

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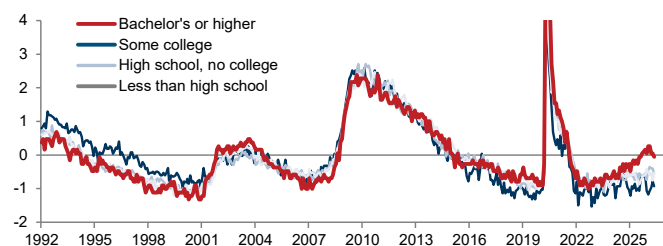
Is AI hurting new graduates?

Jessica Rindels and Pierfrancesco Mei explore how AI is and will likely impact college graduates' employment prospects

The unemployment rate for US college graduates stood at 2.7% last month, well above the 2019 average of 2.1% and much higher than the unemployment rates for other education levels on a standardized basis (relative to its normal level and volatility over the business cycle). Amid accelerating AI adoption and reports of AI automating routine tasks traditionally done by entry-level white-collar employees, this has raised the question of whether AI adoption is especially tough on new graduates. While [some studies](#) have found that AI has had a significant negative impact on early-career workers—including new college graduates—we find little impact on graduates' employment prospects so far. We think this dynamic could change over the near term but are more optimistic about college graduates' prospects over the longer term.

College graduates are currently unemployed at a higher rate than other education levels

Unemployment rates by education level, normalized score



Source: Department of Labor, Goldman Sachs GIR.

Little impact so far...

We are less convinced than some others that AI adoption has significantly impacted college graduates' job prospects so far. AI-related employment headwinds have so far been modest and limited to a few specific AI-exposed subindustries, such as software publishing, data processing, and call centers. Notably, the unemployment rate for young tech workers—which rose markedly in 2024—has normalized and is now in line with the unemployment rate for all tech workers. We also find only a slight positive correlation between industry-level AI adoption and unemployment for workers aged 20-30 and college grads.

...but some disruption ahead...

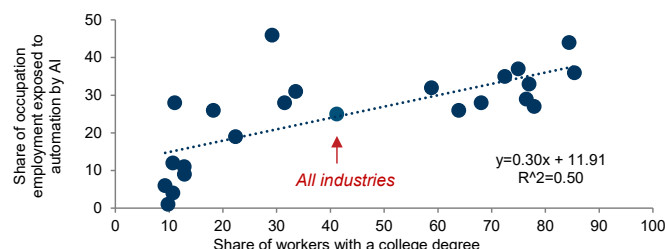
That said, we believe AI could impact college graduates more meaningfully ahead, for two reasons.

First, we find a strong positive correlation between the share of workers with a college degree and AI adoption at the industry level, with adoption highest in the management, information services, professional and business services, and finance sectors—where college graduates account for over 60% of the workforce—and lowest in the food services, construction, and transportation sectors—where they account for less than 25%.

Second, we find that college graduates are disproportionately employed in occupations with greater shares of work tasks that could be automated following the full adoption of generative AI. Roughly 40% of employment in legal, architecture and

engineering, and physical, life, and social science occupations—all areas where college graduates account for over 75% of the workforce—is exposed to AI automation, while the comparable figure for construction, repair and maintenance, and grounds cleaning occupations—where college graduates account for only around 10% of the workforce—is less than 10%.

College graduates are disproportionately employed in occupations with greater exposure to AI automation



Source: Department of Labor, Goldman Sachs GIR.

...that should be offset in the longer run

While these facts suggest that AI could eventually impact college graduates more meaningfully, several factors should help offset the potential burden over the longer term.

First, higher AI adoption does not necessarily imply greater risk of job displacement because AI can also augment labor and boost employment growth in many occupations, not just substitute for it. Second, we expect AI-related job gains in other areas to partly offset AI-related job displacement, as technological change has historically been a main driver of employment growth via the creation of new occupations.

Third, we find that college graduates and young workers in particular tend to fare better than other education and age cohorts when displaced by new technologies. Relative to these other cohorts, college graduates have historically experienced smaller earnings declines after being displaced from jobs that technology disrupted, and displaced workers under the age of 30 were around 10pp more likely to find a new job in a different occupational category and move up the occupational ladder into positions requiring more advanced skills.

Lastly, we have found that college students are beginning to respond to AI-driven labor market shifts. Over the past year, enrollment in majors such as computer and information sciences—which feed into occupations with high displacement risk and weak recent job growth—has declined while majors such as healthcare and engineering—whose related occupations are less exposed to AI and have experienced stronger job growth—have seen enrollment rise.

Taken together, these patterns suggest that while early-career college graduates may be disproportionately exposed to AI-related employment headwinds in the near term, they will likely adjust more flexibly to potential disruptions through occupational mobility and skill upgrading over the long term.

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AI's uneven toll on workers

Pierfrancesco Mei argues that AI will likely have large but uneven impacts on workers

AI is set to significantly reshape the labor market, bringing large productivity gains but uneven consequences for workers, much like past technological shifts. Using history as a guide¹, we find that while most workers will benefit from AI-driven gains in productivity and real income, the transition could impose real costs on displaced workers, who face earnings losses, slower career progression, and higher unemployment risk. However, younger, more mobile workers tend to adjust more easily, and retraining can help improve outcomes over time.

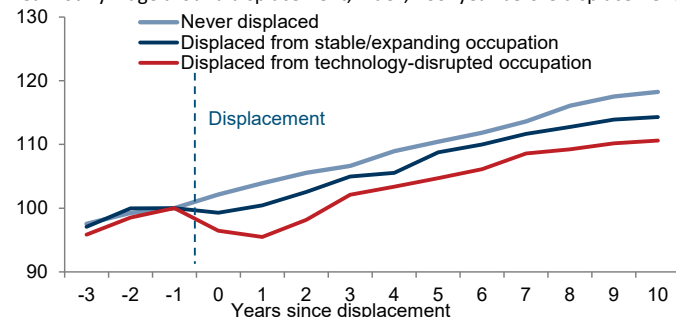
Technological displacement shapes workers' trajectories

History offers four main insights into the impact of technological displacement on workers:

- Workers displaced from technology-disrupted jobs tend to face a somewhat more challenging return to employment than those leaving more stable roles.** On average, they take about one month longer to find a new job and incur real earnings losses of over 3% upon reemployment, compared with negligible losses for workers displaced from more stable occupations. This largely owes to occupational downgrading: technology-displaced workers are more likely to move into roles with higher routine content and lower analytical and interpersonal demands, likely reflecting a decline in the market value of their existing skills.

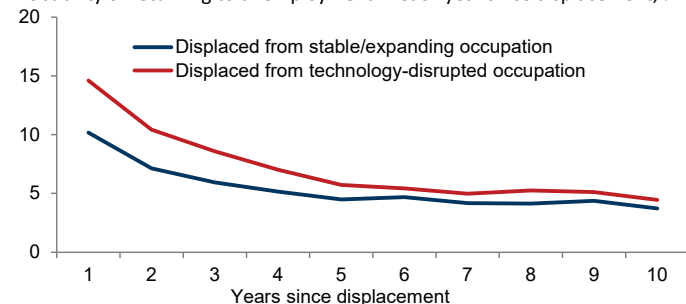
Technology-displaced workers face persistently lower earnings...

Real hourly wage around displacement, index, 100=year before displacement



...and greater job instability in the years following displacement

Probability of returning to unemployment in each year since displacement, %



Source: Census Bureau, US Bureau of Labor Statistics, Goldman Sachs GIR.

- The effects of technological displacement extend beyond initial job loss.** Over the decade following displacement, technology-displaced workers see real earnings grow nearly 10pp less than workers who were never displaced, and roughly 5pp less than workers displaced from non-

technology-impacted occupations. The likelihood of subsequent unemployment also remains higher for technology-displaced workers, particularly in the first five years. And those displaced early in their careers tend to accumulate wealth more slowly—largely due to delayed homeownership—and are slightly less likely to form a household relative to their never-displaced peers.

3. The impact of technological displacement varies widely across workers.

Among technology-displaced workers, younger, college-educated, and urban workers tend to experience roughly half the cumulative earnings losses of other technology-displaced workers as greater occupational mobility allows them to move to less routine roles that are less exposed to further displacement risk. Workers with shorter job tenure also tend to fare better, likely because their skills are more readily transferable.

Retraining can also help cushion these impacts. Workers who participate in retraining programs within three years of a job loss achieve ~2pp stronger cumulative real wage growth over the next decade, and their probability of future unemployment declines by ~10pp. This likely reflects skill upgrading as retrained workers tend to move into roles with higher abstract content and greater complementarity with technology, reducing their exposure to further automation.

4. Recessions amplify the costs of technological displacement.

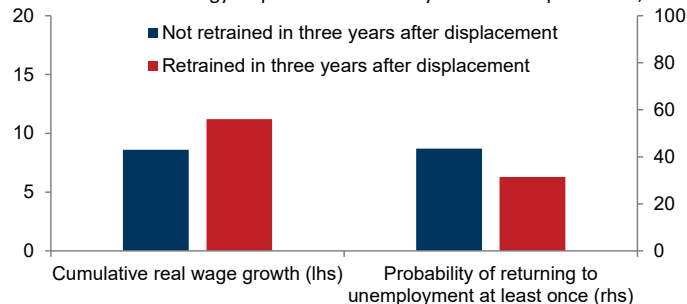
Firms tend to shed routine jobs during downturns, and workers displaced by technology in these periods face even wider gaps in outcomes relative to other displaced workers. On average, they spend about three more weeks unemployed and experience a roughly 5pp higher risk of future unemployment and labor force exit.

Mitigating the costs of technological displacement

All told, as with previous waves of technological change, AI-driven displacement will likely impose real but varied costs on workers—costs that could be larger in the event of a downturn. Encouragingly, however, younger workers most at risk from the erosion of entry-level positions have historically adjusted more quickly to technology-induced shifts in labor demand. And retraining programs can help mitigate these effects for workers across all age groups, helping displaced workers achieve more stable employment and modestly higher wages over time.

Retraining improves outcomes for technology-displaced workers

Outcomes for technology-displaced workers 10 years after displacement, %



Source: Census Bureau, US Bureau of Labor Statistics, Goldman Sachs GIR.

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¹ We analyze employment trends across over 300 occupations since the 1980s in the US, identifying in each decade those most exposed to technological change and tracking the individuals who worked in them through the National Longitudinal Surveys—a dataset following over 20k workers across their careers.

Corporate take on AI's labor impact

Company	Key message	Implied workforce impact	Recent layoff/workforce announcements
Alphabet (Google)	Nearly half of all code is generated by AI; agents used in finance teams.	Leveraging AI to "moderate the pace of headcount growth."	Cut an est. 1.5-3k jobs this year, largely in its cloud business, as part of its AI restructuring.
Amazon	Scaling robotics and automation is a many-year effort to improve productivity. AI will reinvent every customer experience.	Decelerating headcount growth (1% yoy in Q1 2026) as automation replaces manual roles.	In January 2026, Amazon said it would cut 16k additional corporate employees, on top of 14k white-collar layoffs in October 2025.
Block (Square)	Strategic shift to an "automated" company; production code changes per engineer increased 2.5x; 60% of code changes now made by non-engineers.	Workforce reduction plan involves an "increased reliance on automation and AI tools."	In February 2026, Block announced a workforce reduction of over 40%, or about 4k employees.
C.H. Robinson	Initiated a 2025 Restructuring Program leveraging AI to streamline operations.	Automating manual tasks to drive operating leverage; focus on decoupling headcount growth from volume growth.	Average headcount was down about 12% year-over-year in Q1 2026 and was down about 3% sequentially.
Cisco	"AgenticOps" is the new operating model for AI-driven IT; 90% of support cases use AI.	Shifting resources from legacy infrastructure and processes toward investments into silicon, security, and internal AI adoption.	In May 2026, Cisco laid off ~4k employees as part of its push toward AI adoption. It expects to pay \$1bn in termination costs.
Citigroup	AI deployment is transforming operational capabilities and freeing up ~100k hours of developer capacity every week.	Shifting focus to "next-gen execution" with a leaner, higher-performing team.	In 2024, Citi announced plans to reduce its workforce by 20k roles over the medium term as part of efforts to streamline operations.
Cloudflare	Adopting an agentic AI-first operating model, Cloudflare OS, where AI is a core part of the workforce.	Internal AI usage has risen; 97% of R&D workers use AI coding tools; AI agents review all production code before deployment.	In May 2026, Cloudflare announced it was reducing its current workforce by approximately 20% (1.1k).
Coinbase	Restructuring to manage operating expenses and "optimize the company for the AI era."	65% of customer support is automated; using LLM agents for compliance investigations.	In May 2026, Coinbase said it was reducing its workforce by 14%, citing the use of AI to improve operations.
Dow	Using AI to simplify end-to-end work processes and automate manual hand-offs.	Reduce workforce as part of cost-saving efforts that will use AI to boost productivity.	Announced a workforce reduction of 4.5k roles in January 2026 amid increasing use of AI.
IBM	\$4.5bn productivity gains achieved; uses "IBM Bob", AI coding agent, to boost software development productivity.	Transforming into a "hybrid model of people plus software."	Reported a \$0.3bn "workforce rebalancing" charge in its Q1 2026 earnings results.
Intuit	"System of intelligence" strategy leverages AI + HI (Human Intelligence) to complete complex jobs on behalf of customers.	Shifting from manual reporting to AI agents; coders delivering 39% more code while achieving 40% faster coding speeds.	Announced 17% reduction in full-time workforce in May 2026 to simplify structure and accelerate AI.
Meta Platforms	AI is enabling single individuals to do projects that used to require big teams. Agentic coding is driving a 30-80% increase in output per engineer.	Organization is "flattening"; saving resources to invest in "superintelligence."	In April 2026, Meta announced it will lay off about 10% of its workforce to streamline operations and help fund more investment in AI.
Microsoft	Focusing on building high-performing teams and evolving operations to increase pace and agility.	Continued investment toward AI, R&D compute capacity, and talent while overall headcount contracts.	FY2026 guidance expects decline in yoy headcount; voluntary retirement buyouts impacting up to 7% of its US workforce; Xbox division plans to announce significant layoffs in July 2026.
Nasdaq	Established a dedicated AI Integration Team to streamline implementation and accelerate adoption across functions.	Shift human roles from resource-intensive manual workflows to high-value investigation.	Established a \$100mn AI expense efficiency target; said areas in finance, marketing, legal and HR could see headcount reductions.
Oracle	AI "fusion agentic apps" autonomously execute routine work.	Optimizing labor costs and reducing time spent on manual coordination.	Some estimates suggest Oracle could cut up to 30k jobs this year as part of targeted moves toward operational efficiencies to offset the significant capacity buildup.
Pinterest	50% of code is AI-generated; reallocating resources to AI roles.	Strategic shift to "AI-powered" destination.	Recorded a roughly \$47mn restructuring charge in its Q1 2026 results.
Salesforce	Shift toward an "Agentic Enterprise" with AI agents driving greater productivity and operational efficiency.	Ongoing workforce and office space reduction to drive greater operational efficiencies; redeployment of roles to high growth areas.	In February 2026, Salesforce cut 1k roles with another round of layoffs in June 2026 across sales, admin, tech, and product roles.
ServiceNow	Introduction of "autonomous workforce" with AI specialists designed to execute jobs end-to-end.	AI agents handle ~90% of internal support and customer service requests; automating tasks in finance, marketing, legal, and HR.	In June 2026, ServiceNow announced layoffs to hundreds of staff, saying that "real AI efficiencies" are being generated.
Snap	40% of new code is AI-generated; AI enables "small squads" to do more.	Reduction of repetitive work; shift from AI R&D to deployment phase.	In April 2026, Snap said it would cut 1k jobs as it gains efficiency through deploying AI.
Verizon	Transformation into an "AI-first" company. Beginning to embed AI and automation into operations and customer interactions.	Shift toward hiring "AI-savvy" individuals to complement the workforce; focus on using AI to replace a large share of customer service jobs.	Cut 13k jobs in Q4 2025, though Verizon framed the move as a response to customer churn/earnings pressures. Set up a \$20mn fund to re-skill laid off workers with AI skills.
Walmart	AI and automation will drive labor productivity and lower marginal costs.	Growth at a lower marginal cost; focus on upskilling roles for "retail's next era."	In May 2026, Walmart said it could cut or relocate about 1k workers as it combines more of its technology and product teams.
Workday	Strategic pivot to agentic AI that automates workflows while maintaining human-in-the-loop guardrails.	>50% of committed code is AI generated; 22% growth in engineering output over last 6m; dev. timelines reduced to 3m driven by AI.	Eliminated about 400 jobs in February 2026 within Global Customer Operations which also includes customer support roles.

Note: Table does not constitute an exhaustive list of company commentary on AI and recent corporate layoffs.

Source: Company data, AlphaSense, WSJ, various news sources, compiled by Goldman Sachs GIR.

AI: limited labor productivity uplift so far

Ryan Hammond argues that AI's impact on labor and profits remains uncertain, keeping focus on infrastructure beneficiaries for now

While recent corporate layoff announcements (see pg. 16) have intensified concerns about AI-driven job losses, uncertainty remains high about how AI will ultimately impact companies' labor force needs and, in turn, corporate profits. The key question is whether AI will be revenue-enhancing (i.e., generate more revenues with the same labor force), margin-enhancing (i.e., generate similar revenues with a smaller labor force), or both. The earnings impact will also depend on the timeline of corporate AI adoption and how much of the labor productivity gains accrue to capital. Given this uncertainty, investors have continued to favor companies with tangible near-term earnings benefits, which are largely concentrated in the AI infrastructure complex. We expect this to eventually broaden to include labor productivity beneficiaries, but only as more definitive evidence of an earnings uplift emerges.

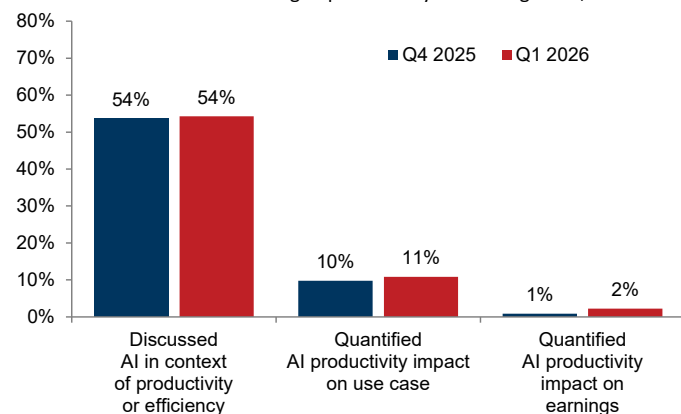
AI workforce impact remains uncertain

Corporate commentary during 1Q26 earnings season indicated that uncertainty around AI's impact is unlikely to be resolved in the near term. Enterprise AI adoption remains nascent, consistent with broader economy-wide adoption surveys. While 54% of S&P 500 companies discussed AI in the context of labor productivity in Q1 earnings calls, only 11% quantified AI labor productivity gains in specific use cases (vs. 10% in the prior quarter)—largely in coding, engineering, call centers, customer support, and marketing. And just 2% of firms tied such gains to earnings, with the majority of those firms focused on quantifying the cost savings related to AI labor productivity.

At the same time, rising concerns around token economics—which have improved (see pg. 18) but remain challenging, with several companies flagging the high costs associated with token usage in recent commentary and capping employee token budgets—suggest that the upfront costs associated with AI consumption could offset some of the potential earnings uplift from increased labor productivity.

Only 2% of companies tied AI productivity gains to earnings in Q1

Share of S&P 500 firms discussing AI productivity on earnings calls, %



Source: Goldman Sachs GIR.

Markets aren't buying it (yet)

Intra-market performance suggests investors share this uncertainty about the long-term impact of AI on companies' labor forces and earnings. Share price performance following Q1 earnings results did not meaningfully differ between companies that discussed AI productivity benefits and those that did not. And a basket of stocks that have mentioned specific plans to implement AI into their internal workflows (GSXUPROD¹) has traded sideways relative to the equal-weight S&P 500 during the past few years (+33% vs. +27%).

Our basket of stocks with the largest potential earnings upside from AI productivity gains (GSTHLTAI) sends a similar signal². The sector-neutral basket has lagged the equal-weight S&P 500 since the end of 2023 and has traded sideways on a relative basis in the past two months (+1% vs. +3%). This suggests investors are not yet confident that AI-driven productivity gains will translate into a meaningful earnings boost.

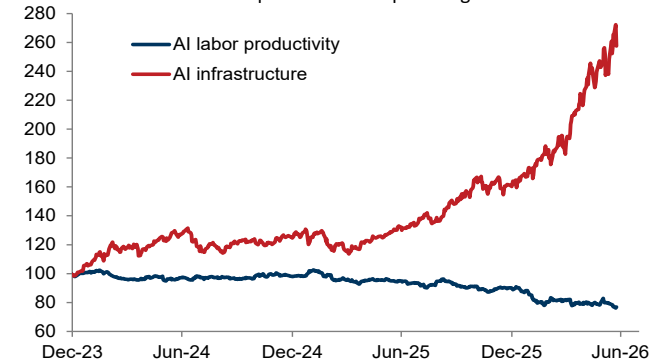
Infrastructure still leads the AI trade... but likely not forever

Investors have instead continued to reward companies seeing more meaningful near-term earnings benefits from AI, which are mostly concentrated in the infrastructure complex. A basket of AI infrastructure stocks (GSCBAIP2³) has risen by 74% YTD, outperforming the equal-weight S&P 500 by 65pp, with gains largely driven by positive earnings revisions. We expect this dynamic to persist in the near term. While consensus estimates imply hyperscaler capex of \$920bn in 2027—implying a sharp deceleration in growth from 84% in 2026 to 22% in 2027—we believe these estimates are too conservative. Continued robust capex growth should support further earnings and share price upside for AI infrastructure beneficiaries in the near term, albeit with more volatility.

Eventually, we expect investors to broaden their exposure beyond the AI infrastructure complex to subsequent phases of the AI trade, including labor productivity beneficiaries. However, this shift will likely require clear evidence of a meaningful and tangible earnings impact, which may take time to emerge.

The AI infrastructure complex has outperformed the equal-weight S&P 500 since the end of 2023

Indexed return of AI-related portfolios vs. equal-weight S&P 500



Source: Goldman Sachs GIR.

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¹ Basket was developed by the GS Global Banking & Markets division.

² Our basket is based on two factors: (1) Our economists' estimate of the share of each firm's wage bill that is exposed to AI automation and (2) Our estimate of each firm's labor costs as a share of revenue. Combining these, we calculate the potential earnings boost from AI adoption assuming stable margins or stable revenues.

³ Basket was developed by the GS Global Banking & Markets division.

AI agents vs. humans: who costs more?

As token unit costs fall, AI agents are becoming increasingly competitive with human labor across a wider range of enterprise workflows. Simulating agent workloads across coding, call center, and data-entry tasks, we find that even million-token daily workflows can compare favorably with human labor costs—especially as token unit costs continue to fall—with meaningful implications for margins and returns for both AI model makers and enterprise end-users.

What AI agents and tokens actually are

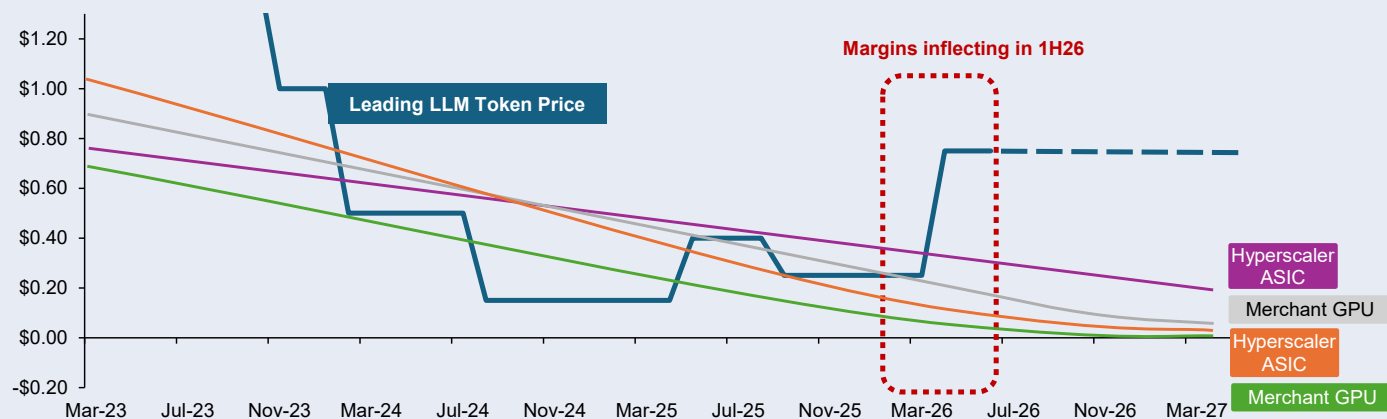
AI agents can do much more than traditional chatbots; they can plan, use tools, validate work, and take actions with limited human input. Tokens are the unit that powers this activity, and as agents become more capable, they consume vastly more tokens.

Token economics are turning more favorable

Agentic AI will likely continue driving token consumption, just as token unit economics are improving. We estimate that global token demand could rise by roughly 24x by 2030 versus 2026 levels, reflecting growing consumer and enterprise AI agent adoption. More importantly, all-in token production costs appear to be falling faster and more consistently than expected, even as leading token prices have begun to stabilize in some areas. This suggests that the industry may be entering a new phase where increased token usage can support not only revenue growth, but also incremental margin expansion for hyperscalers and model makers.

Token economics are beginning to inflect, as compute unit cost declines continue and token prices stabilize

Cost/price per one million tokens, \$



Note: Token costs are the total computing costs realized by hyperscalers and model makers; token prices are the prices charged to enterprises and end-users.

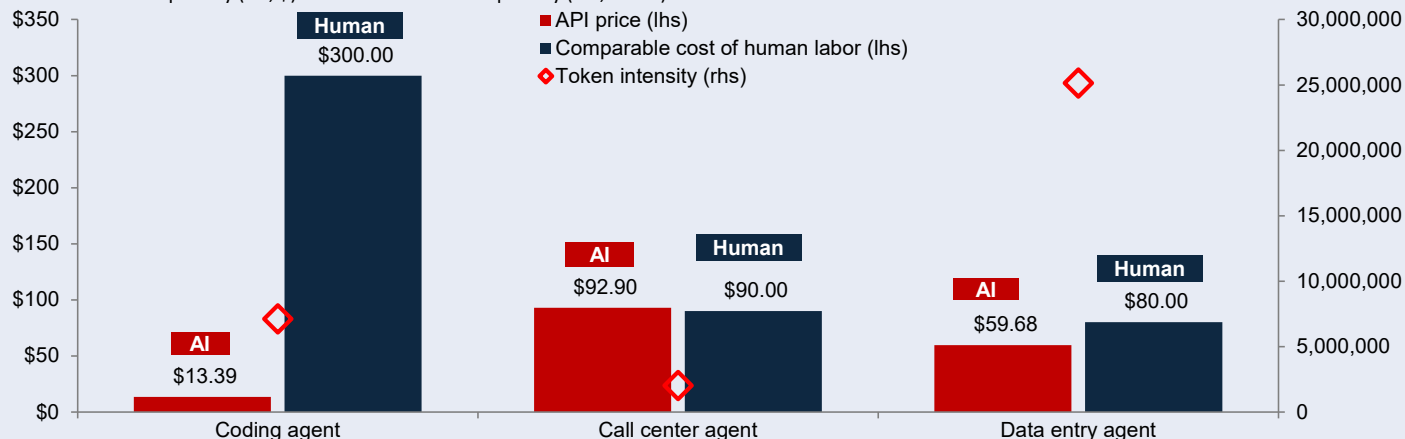
Source: Data compiled by Goldman Sachs GIR.

Falling token costs can expand the AI frontier

As token unit costs continue to decline, AI agents may become increasingly economical relative to human labor, expanding the range of enterprise workflows that firms can automate at an attractive return on investment (ROI). Coding is a compelling example: It is high value, largely text-based, and supported by a mature tooling stack, which allows even token-heavy agentic workflows to remain less expensive than equivalent human labor. Other categories that are operationally harder and more expensive to execute like real-time voice and call center automation are more constrained for now. However, the broader trend is clear: ongoing declines in token costs and, in turn, prices may lower the breakeven point for automation and render more AI use cases economically viable.

Coding agents are currently cost effective against human labor, while other categories remain more constrained for now

Human vs. AI cost per day (lhs, \$) and tokens consumed per day (rhs, count)



Source: Data compiled by Goldman Sachs GIR.

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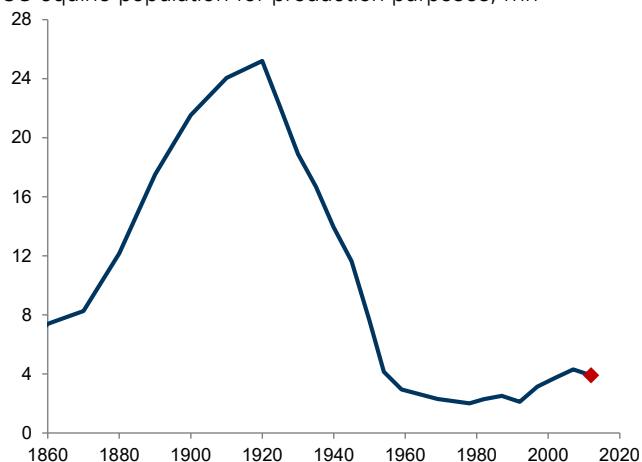
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Prior tech transitions & the labor market

Growing concerns about an AI job apocalypse have sparked fears that humans could “go the way of horses” as production inputs...

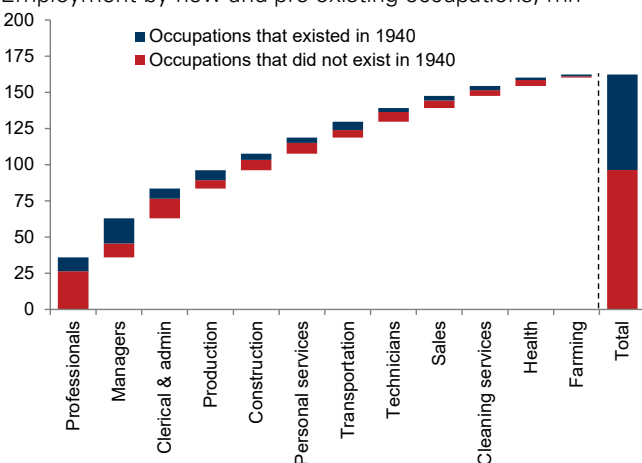
US equine population for production purposes, mn



Source: Goldman Sachs GIR.

In fact, technological change has been a key driver of employment growth via the creation of new occupations...

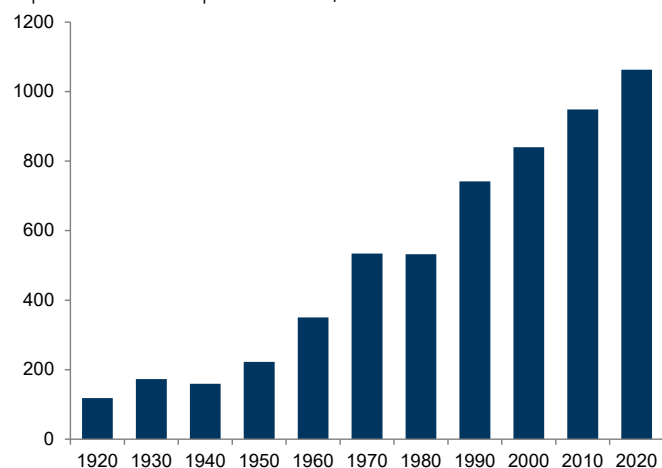
Employment by new and pre-existing occupations, mn



Source: Autor et al. (2022), Goldman Sachs GIR.

Technology can also create new and more specialized roles within an occupation, as has been the case for healthcare

Distinct healthcare occupations according to the Census Alphabetical Occupation Index, count

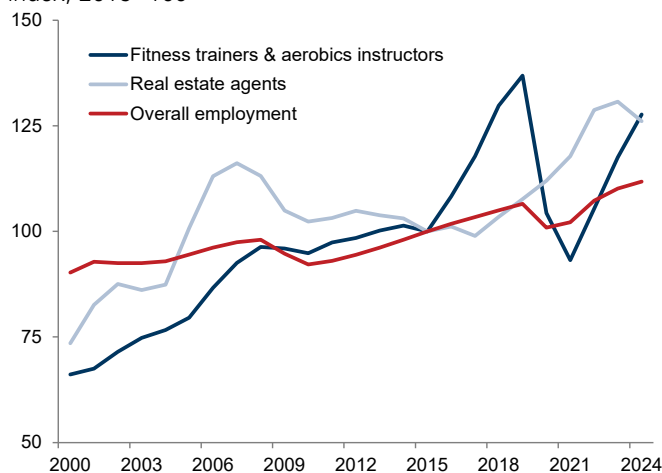


Source: Census Bureau, Goldman Sachs GIR.

Special thanks to GS global economists Joseph Briggs and Sarah Dong for charts.

...but technological replacement potential does not always lead to job losses, as evidenced by continued employment growth in roles such as fitness trainers and real estate agents

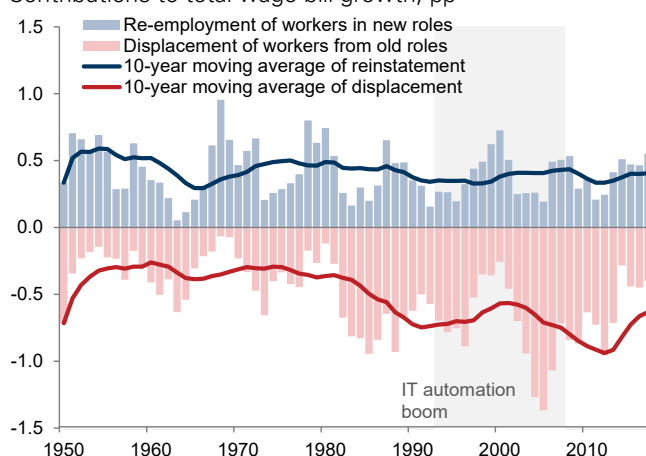
Employment in technology-disrupted discretionary services, index, 2015=100



Source: US Bureau of Labor Statistics, Haver Analytics, Goldman Sachs GIR.

...and, historically, the creation of new roles has roughly offset worker displacement from automation

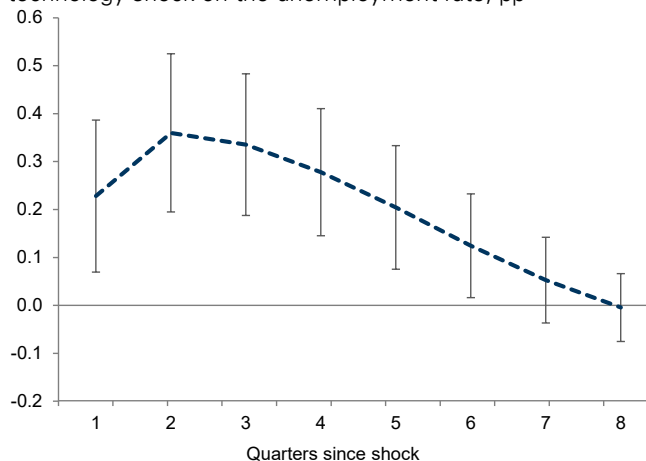
Contributions to total wage bill growth, pp



Source: Acemoglu and Restrepo (2019), Goldman Sachs GIR.

But even as technology creates new jobs, it also leads to a rise in frictional unemployment as workers transition to these jobs—a pattern we expect to persist under greater AI adoption

Effect of a 1pp increase in labor productivity due to a technology shock on the unemployment rate, pp



Source: Gali (1999), Goldman Sachs GIR.

Summary of our key forecasts

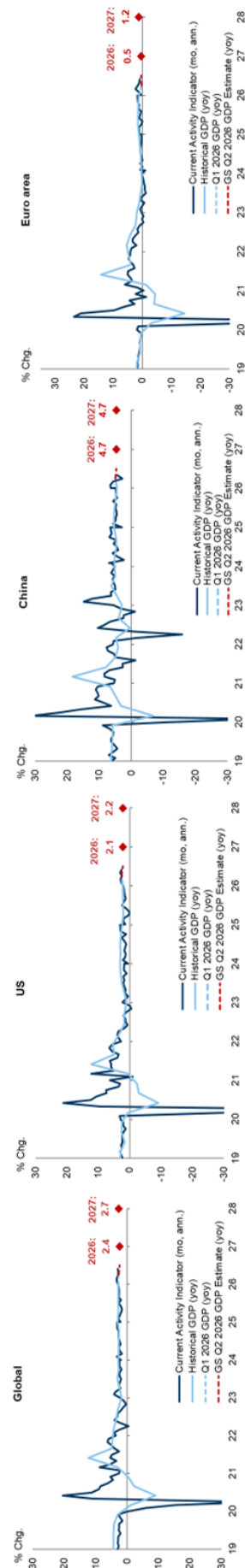
GS GIR: Macro at a glance

Watching

- **Globally**, we expect real GDP growth to slow to 2.4% yoy in 2026 amid lingering headwinds from the rise in energy prices from the Iran conflict. We expect global core inflation to end the year at 2.9%, reflecting a fading tariff boost and further normalization in shelter and wage inflation but a boost from energy price passthrough.
- **In the US**, we expect moderate real GDP growth of 2.0% on a Q4/Q4 basis in 2026, reflecting still-subdued consumer spending growth but a boost from the AI boom via higher equity wealth as well as strong capex. We expect core PCE inflation of 3.2% by December 2026 given the effects of tariffs, energy price passthrough, AI demand exaggerated by measurement issues, and higher financial services prices, with it falling closer to 2% in 2027. We expect the unemployment rate to end 2026 at 4.4%.
- **We expect the Fed** to leave the policy rate unchanged at 3.5-3.75% this year, though the hawkish June FOMC meeting raises the risk of interest rate hikes later this year.
- **In the Euro area**, we expect real GDP growth to improve in 2H26 given easing financial conditions and likely lower energy prices for full-year real GDP growth of 0.5% yoy. We expect core inflation to rise to a peak of 2.6% yoy in 4Q26 amid lingering energy price passthrough before declining to 2.0% by end-2028.
- **We expect the ECB** to deliver one more 25bp hike in September to a peak policy rate of 2.5% before cutting back to 2% in 2027.
- **In China**, we expect sequential growth to rebound in 3Q26 from 2Q26 as the oil shock reverses and fiscal spending increases for full-year real GDP growth of 4.7% yoy. We expect CPI/PPI inflation to rise to 1.0%/2.0% yoy this year largely owing to commodity price passthrough.
- **WATCH THE US-IRAN AGREEMENT.** The agreement between the US and Iran has reduced the downside risks to our economic outlook, though the situation remains uncertain and we will be closely watching how oil flows through the Strait of Hormuz evolve from here.

Goldman Sachs Global Investment Research.

Growth



Source: Haver Analytics, Goldman Sachs Global Investment Research.
Note: GS CAI is a measure of current growth. For more information on the methodology of the CAI please see "Technical Updates to Our Global CAIs," Global Economics Comment, Sep. 01, 2025.

Forecasts

Economics	Markets										Equities									
	GDP growth (%)		Interest rates 10Yr (%)		FX		Last		E2026		E2027		Returns (%)		12m		YTD		E2026 P/E	
Policy rates (%)	2026		2027		2026		2027		2026		2027		2026		2027		2026		2027	
	GS (Q4/Q4)	Cons. (Q4/Q4)	GS (CY)	Cons. (CY)	GS (CY)	Cons. (CY)	GS (CY)	Cons. (CY)	GS (CY)	Cons. (CY)	GS (CY)	Cons. (CY)	GS (CY)	Cons. (CY)	GS (CY)	Cons. (CY)	GS (CY)	Cons. (CY)	GS (CY)	Cons. (CY)
Global	2.3	2.0	2.4	2.5	2.7	2.6	2.7	2.6	4.41	4.40	4.25	EUR/USD	1.13	1.14	1.20	Price	8,000	8,000	8,000	8,000
US	2.0	2.0	2.1	2.2	2.2	2.1	2.2	2.1	2.87	3.00	3.00	GBP/USD	1.32	1.33	1.33	EPS	\$340	\$340	\$340	\$340
China	4.6	4.5	4.7	5.0	4.7	4.6	4.7	4.6	2.67	2.50	2.25	USD/JPY	162	160	155	Growth	24%	24%	24%	24%
Euro area	0.6	0.6	0.5	1.4	1.2	0.6	1.2	0.6	4.56	4.50	4.35	USD/CHF	6.79	6.80	6.50	Growth	24%	24%	24%	24%
Commodities																				
Credit (bp)																				
Commodities																				
Crude Oil, Brent (\$/bbl)																				
US	3.63	3.98	3.13	3.85	3.22	2.95	3.50	USD	74	98	80	IG	74	73	82	US	3.3	3.3	3.3	3.3
Euro area	2.50	2.47	2.00	2.42	40.95	47	40	HY	277	275	295	Euro area	2.9	6.3	6.2	China	1.0	1.0	1.0	1.0
China	1.40	1.29	1.30	--	13,030	13,400	13,700	EUR	90	88	88	China	1.0	--	--	China	1.0	--	--	--
Japan	1.00	1.21	1.50	1.73	4,024	4,583	4,836	HY	265	277	315	Japan	1.0	--	--	Japan	1.0	--	--	--

Market pricing as of June 24, 2026

Source: Bloomberg, Goldman Sachs Global Investment Research. For important disclosures, see the Disclosure Appendix or go to www.gs.com/research/hedge.html.

Glossary of GS proprietary indices

Current Activity Indicator (CAI)

GS CAIs measure the growth signal in a broad range of weekly and monthly indicators, offering an alternative to Gross Domestic Product (GDP). GDP is an imperfect guide to current activity: In most countries, it is only available quarterly and is released with a substantial delay, and its initial estimates are often heavily revised. GDP also ignores important measures of real activity, such as employment and the purchasing managers' indexes (PMIs). All of these problems reduce the effectiveness of GDP for investment and policy decisions. Our CAIs aim to address GDP's shortcomings and provide a timelier read on the pace of growth.

For more, see our CAI page and Global Economics Comment: Technical Updates to Our Global CAIs.

Dynamic Equilibrium Exchange Rates (DEER)

The GSDEER framework establishes an equilibrium (or "fair") value of the real exchange rate based on relative productivity and terms-of-trade differentials.

For more, see our GSDEER page, Global Economics Paper No. 227: Finding Fair Value in EM FX, 26 January 2016, and Global Markets Analyst: A Look at Valuation Across G10 FX, 29 June 2017.

Financial Conditions Index (FCI)

GS FCIs gauge the "looseness" or "tightness" of financial conditions across the world's major economies, incorporating variables that directly affect spending on domestically produced goods and services. FCIs can provide valuable information about the economic growth outlook and the direct and indirect effects of monetary policy on real economic activity.

FCIs for the G10 economies are calculated as a weighted average of a policy rate, a long-term risk-free bond yield, a corporate credit spread, an equity price variable, and a trade-weighted exchange rate; the Euro area FCI also includes a sovereign credit spread. The weights mirror the effects of the financial variables on real GDP growth in our models over a one-year horizon. FCIs for emerging markets are calculated as a weighted average of a short-term interest rate, a long-term swap rate, a CDS spread, an equity price variable, a trade-weighted exchange rate, and—in economies with large foreign-currency-denominated debt stocks—a debt-weighted exchange rate index.

For more, see our FCI page, Global Economics Analyst: Our New G10 Financial Conditions Indices, 20 April 2017, and Global Economics Analyst: Tracking EM Financial Conditions – Our New FCIs, 6 October 2017.

Goldman Sachs Analyst Index (GSAI)

The US GSAI is based on a monthly survey of GS equity analysts to obtain their assessments of business conditions in the industries they follow. The results provide timely "bottom-up" information about US economic activity to supplement and cross-check our analysis of "top-down" data. Based on analysts' responses, we create a diffusion index for economic activity comparable to the ISM's indexes for activity in the manufacturing and nonmanufacturing sectors.

Macro-Data Assessment Platform (MAP)

GS MAP scores facilitate rapid interpretation of new data releases for economic indicators worldwide. MAP summarizes the importance of a specific data release (i.e., its historical correlation with GDP) and the degree of surprise relative to the consensus forecast. The sign on the degree of surprise characterizes underperformance with a negative number and outperformance with a positive number. Each of these two components is ranked on a scale from 0 to 5, with the MAP score being the product of the two, i.e., from -25 to +25. For example, a MAP score of +20 (5;+4) would indicate that the data has a very high correlation to GDP (5) and that it came out well above consensus expectations (+4), for a total MAP value of +20.

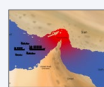
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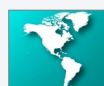
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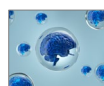
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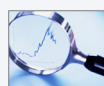
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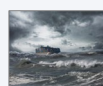
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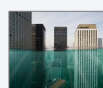
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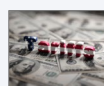
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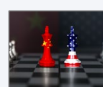
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High bond yields: here to stay?
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TOP^{of} MIND

CRACKS IN PRIVATE CREDIT



After over a decade of rapid growth and relative calm, private credit has come under pressure as high-profile defaults, concerns about inflated valuations, and substantial exposure to a software industry vulnerable to AI disruption have fueled a surge in redemption requests. Could these stresses trigger a systemic crisis? And even if not, could they dampen the outlook for private credit? Oaktree's Howard Marks, Ares' Michael Arougheti, Marathon's Bruce Richards, and GS' Amanda Lynam agree current private credit stresses don't pose systemic risk. But they somewhat disagree on private credit's near-to-medium term outlook. Arougheti and Lynam aren't particularly worried about software exposure and expect credit fundamentals to remain healthy as long as the economy holds up, while Richards is more concerned

about software risk and sees a painful default cycle ahead. But all three remain optimistic about private credit's long-term growth, while Marks expects more cautious investors but a healthier investment environment to ultimately emerge.



The structure of [direct lending investment] vehicles introduces serious concerns around pricing and liquidity... [but] I don't see the same systemic exposure here as was the case in the run-up to the GFC.

- Howard Marks

While defaults will inevitably rise as the cycle matures and some managers will underperform, that doesn't mean the entire credit market will tip over, despite a handful of attention-grabbing defaults.

- Michael Arougheti

3% of the HY bond market is software loans versus 13% of the broadly syndicated loan market and a whopping 23% of the private credit market... So, it's hard to argue that there isn't too much software risk in direct lending.

- Bruce Richards

A key feature of private credit is asset-liability matching. Most capital is locked up, and even in retail-focused structures, liquidity is limited by design. That significantly reduces the risk of a 'run on the bank.'

- Amanda Lynam



WHAT'S INSIDE

INTERVIEWS WITH:

Howard Marks, Co-Founder and Co-Chairman, Oaktree Capital Management

Michael Arougheti, Co-Founder and CEO, Ares Management

Bruce Richards, CEO, Chairman, and Co-Founder, Marathon Asset Management

Amanda Lynam, Chief Credit Strategist, Goldman Sachs

BUSINESS DEVELOPMENT COMPANIES: NAVS VS. PRICES

Shamshad Ali, GS Credit Strategy Research

PRIVATE CREDIT: LIMITED MACRO SPILLOVERS

Manuel Abecasis, GS US Economics Research

ALTS MANAGERS: PRIVATE CREDIT OPPORTUNITIES

Alexander Blostein and Anthony Corbin, GS Equity Research

BANKS AND PRIVATE CREDIT: THE INTERSECTION

Richard Ramsden, GS Equity Research

...AND MORE

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US

Latest GS proprietary datapoints/major changes in views

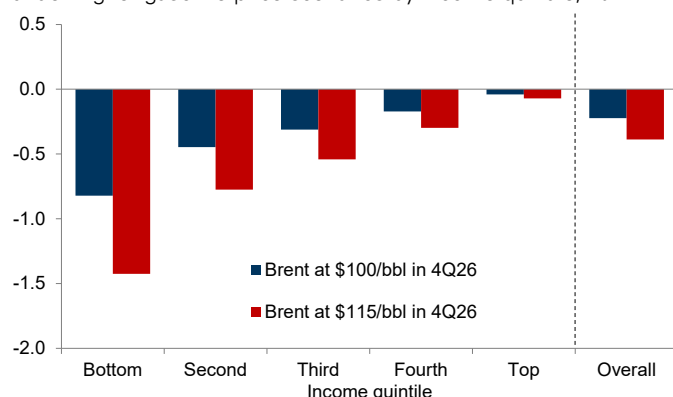
- We raised our Dec 2026 headline/core PCE inflation forecasts to 3.4%/2.6% yoy (from 3.1%/2.5%) given the even higher oil prices we now expect.

Datapoints/trends we're focused on

- Iran war; longer energy supply disruptions that result in even higher gasoline prices would likely lead to a larger drag on consumer spending.
- Fed Chair transition; we don't expect a major reduction in the Fed's balance sheet under Kevin Warsh as Fed Chair.
- US labor market, which we expect to soften this year due to weaker growth, higher energy costs, and modestly greater reluctance to hire for jobs that can be done by AI.

US: higher gasoline prices bite consumers

Effective change in real PCE spending vs. GS baseline in 2026 under higher gasoline price scenarios by income quintile, %



Source: Goldman Sachs GIR.

Europe

Latest GS proprietary datapoints/major changes in views

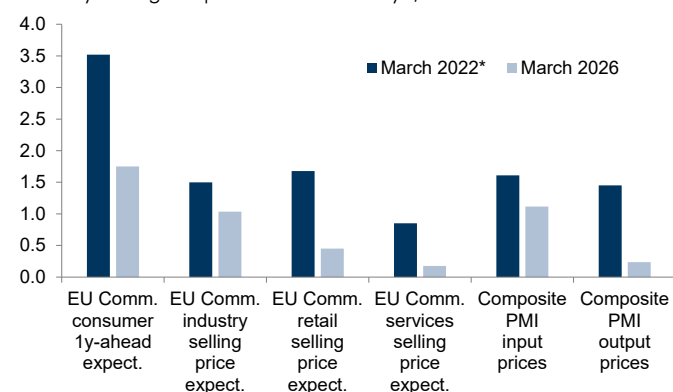
- We raised our 1Q26 UK GDP growth forecast after better-than-expected February GDP data, bringing our full-year 2026 growth forecast to 1.0% (Q4/Q4, vs. 0.6% before).

Datapoints/trends we're focused on

- Iran war; we think the impact of the current energy price shock on European manufacturing should be notably smaller and less persistent than the 2022/23 energy crisis.
- ECB policy; we expect the ECB to hike rates in June and September but view whether the ECB hikes twice or remains on hold this year as a close call.
- UK employment growth, which remains weak.

Euro area: a less painful energy shock than 2022

Monthly change in price-related surveys, std. dev.



*Normalized for size of the change in equally-weighted avg. of oil and gas prices.

Source: Haver Analytics, Goldman Sachs GIR.

Japan

Latest GS proprietary datapoints/major changes in views

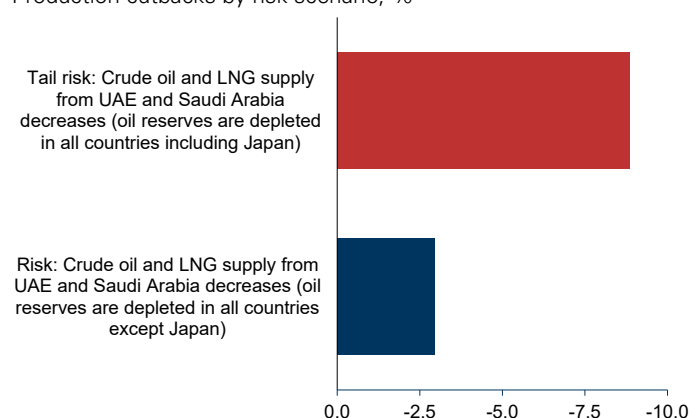
- We lowered our 2026 Japan real GDP growth forecast to 0.3% (from 0.4%) and raised our 2027 core CPI forecast to 2.3% (from 2.1%) given the higher oil prices we now expect.

Datapoints/trends we're focused on

- Iran war; even with sufficient oil reserves, we think Japan faces potentially significant indirect oil supply constraints as shortages abroad impact global supply chains.
- BoJ policy; we continue to expect the next rate hike in July, though uncertainty about the timing remains high.
- Shunto* spring wage negotiations show a marginally softer base pay rise from 2025, but still high in the mid-3% range.
- Japanese consumer confidence, which deteriorated sharply.

Japan: potential oil shortage impacts

Production cutbacks by risk scenario, %



Source: OECD, Goldman Sachs GIR.

Emerging Markets (EM)

Latest GS proprietary datapoints/major changes in views

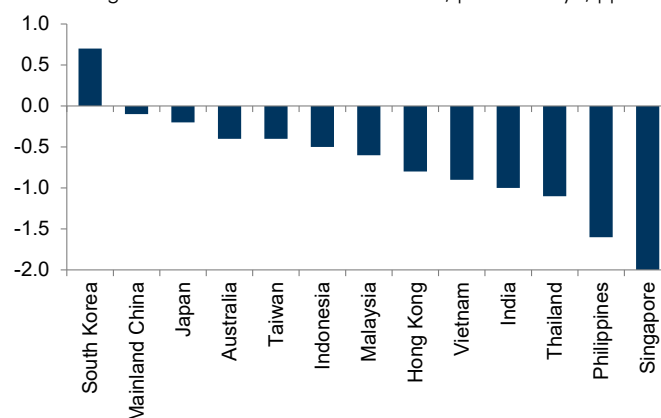
- We raised our 2026/27 China PPI forecasts to 1.2%/0.9% (from 1%/0.6%) given the higher oil prices we now expect.

Datapoints/trends we're focused on

- Iran war; we now largely expect lower growth and higher inflation across Asia-Pacific from the energy shock, but the impact on inflation should be less persistent than in 2022.
- China growth and activity, which has generally been better than expected this year, implying little urgency for large near-term policy stimulus and adding conviction to our low expectations for stimulus at the April Politburo meeting.
- Hungary elections; opposition party Tisza's decisive win in the parliamentary elections should strengthen EU-Hungary relations and could pave the way for Euro membership.

Asia-Pacific: energy shock weighs on growth

Net change in GS 2026 real GDP forecasts, past 60 days, pp



Source: Goldman Sachs GIR.

Cracks in private credit

After over a decade of rapid growth and relative calm, the private credit market has come under pressure as several high-profile defaults, concerns about potentially inflated valuations, and especially substantial exposure to a software industry vulnerable to AI disruption have fueled a surge in redemption requests (see pg. 16)—drawing [some comparisons](#) to the setup heading into the Global Financial Crisis (GFC). So, could these private credit stresses trigger a systemic crisis? And even if not, could they meaningfully impair the outlook for a market that has become a key source of financing for companies and an increasingly important allocation for investors?

We ask several long-time industry watchers for their perspectives: Howard Marks, Co-Founder and Co-Chairman of Oaktree Capital Management, Michael Arougheti, Co-Founder and CEO of Ares Management, Bruce Richards, CEO, Chairman, and Co-Founder of Marathon Asset Management, and Amanda Lynam, GS Chief Credit Strategist.

Arougheti and Lynam argue that the concerns around private credit stresses seem overdone as they are concentrated in non-traded business development companies (BDCs)—the primary direct lending investment vehicle for retail investors (see pg. 8)—which comprise only a modest portion of the private credit market. And they explain that despite the current worries about elevated redemption requests, the limited liquidity of non-traded BDCs (which Lynam describes as a “feature, not a bug”, though Marks finds some retail investors’ apparent disappointment in this feature problematic) substantially reduces the likelihood of crisis-inducing fire sales.

This, combined with the lack of leverage in private credit funds, well-secured and substantially-protected bank exposure to private credit (see pg. 19 for GS Head of the Financials Group Richard Ramsden’s deep-dive into the evolving intersection between banks and private credit), and GS US Economist Manuel Abecasis’ finding that even a sharp rise in private credit default rates would only moderately impact the flow of new lending to the private sector, all agree: the current private credit stresses don’t pose systemic risk, with comparisons between these stresses and the lead-up to the GFC unwarranted.

But even if private credit doesn’t pose meaningful risk to the financial system or the economy, will the industry itself still experience significant pain? Arougheti and Lynam don’t think so. Arougheti sees no indication of a turn in the credit cycle in private or bank credit portfolios or credit card charge-off data, and doesn’t view last fall’s attention-grabbing defaults as particular cause for worry because they were relatively small exposures and, importantly, mainly funded by bank-led asset-based facilities rather than private credit managers.

Lynam also points out that underlying credit fundamentals appear relatively healthy and expects them to remain so as long as economic growth holds up, which GS Senior Global Credit Strategist Shamshad Ali finds is consistent with the signals from the micro signposts he’s watching (namely, what tends to happen to loans flagged as “PIK-at-restructuring” and the size and profile of loans coming due).

But even if underlying fundamentals appear healthy today, will that remain the case given direct lending’s significant exposure to the software industry, which is ripe for AI disruption?

Arougheti, Lynam, and Marks aren’t particularly concerned, with Arougheti explaining that the software loans in private credit portfolios are senior secured, high up in the capital structure, and extended to large, cash-generative firms, Lynam noting that software exposures are not a uniquely private credit problem and that managers have long underwritten AI disruption risk into their investment criteria, and Marks arguing that even significant losses on software loans wouldn’t be “existential” for most senior lenders.

Richards, however, is very worried about software risk. He points out that 23% of the private credit market is software loans, which he says is an excessive figure by any standard, with leverage at the fund level and especially the company level further magnifying the risk from this exposure. And he doesn’t take comfort from private lenders’ high position in the capital structure given that a significant part of the cushion that sits below these lenders has already been eaten through. So, he expects “incredibly high software sector default rates in 2027, 2028, and 2029, with peak defaults of 15% and double-digit default rates in each of those years” and software loan loss rates in the 70-100% range. Lynam, for her part, takes a more nuanced view as she thinks software exposure is more likely to drive performance dispersion among managers than broad-based deterioration in the fundamental outlook, and, absent a sharp downturn in economic growth—which is not GS economists’ base case—doesn’t expect private credit realized losses to rise materially above the longer-term average.

So, the distribution of views on private credit’s near-to-medium outlook is wide. But perhaps the bigger question is whether the recent developments will dent the longer-term growth outlook for private credit, which has become a key financing tool for many firms and growing share of investors’ portfolios. Perhaps surprisingly, Richards doesn’t think so. While he expects the coming software correction to be painful, he expects direct lending—the largest segment of private credit (see pg. 9)—to emerge from it “bigger and stronger than ever” as the correction results in the return of manager discipline.

Arougheti and Lynam are also relatively optimistic, arguing that the current stresses will likely shift market share within private credit rather than slow the industry’s overall growth. In particular, they expect capital to shift from retail-focused strategies to institutional strategies like opportunistic credit, which GS US Financials analysts Alexander Blostein and Anthony Corbin argue should benefit alternative asset managers, though they expect dispersion among them depending on their exposure to the wealth channel.

But Marks is somewhat more cautious, expecting retail investors to be more deliberate and circumspect “now that the tide may have begun to go out on private asset vehicles sold to the public and some of their flaws have been exposed.” That said, he also believes that experiencing a full credit cycle may ultimately lead to the emergence of a healthier investment environment for direct lending and private credit more broadly.

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Interview with Howard Marks

Howard Marks is Co-Founder and Co-Chairman of Oaktree Capital Management. Below, he argues that while nothing is inherently wrong with direct lending, structural aspects of direct lending *investment vehicles* can be problematic for some investors, but don't pose systemic risk.

The views stated herein are those of the interviewee and do not necessarily reflect those of Goldman Sachs.



Allison Nathan: You've characterized the rapid expansion of direct lending as a "gold rush." What precipitated that, and is the rapid growth a cause for concern?

Howard Marks: Direct lending—private loans made by non-bank lenders for mid-size private equity deals—has become an integral aspect

of the private equity business, which runs on using borrowed money to lever up equity returns on the purchase of companies. As post-Global Financial Crisis (GFC) regulations curtailed banks' ability to lend for levered transactions in an effort to make banks less risky, private lenders stepped in to fill the void. And because only a few private lenders existed and many private equity firms wanted to borrow money, private lenders were able to demand high interest rates and safety through strong covenants.

But, as in any gold rush, this success led to a flood of new entrants. Available capital increased relative to demand for it, which meant that lenders couldn't demand as much interest or safety, so the lender's advantage was somewhat competed away. But because demand for private equity was growing, demand for financing from private lenders to lever up transactions was also increasing, allowing private lenders to substantially grow their assets under management and generate sizable fees. So, everyone was motivated to hang up a shingle as a direct lender. The result was rapid growth in the direct lending market, which didn't exist in its current form before 2011 and is approaching \$2tn today, with the vast majority of firms engaged in direct lending today not in business 20 years ago. And, as in every upsurge in financial activity that brings heated competition, some participants may have engaged in risky behavior to get in on the gold rush.

Allison Nathan: Is direct lending inherently problematic?

Howard Marks: There's nothing inherently wrong with lending money. But lending can be done well or poorly. Lending carefully to the right companies at an interest rate that appropriately compensates for the default risk can be a profitable and prudent activity. But the opposite is also true. In 1978, I was fortunate to participate in the start of the high yield (HY) bond market. Prior to 1977, a company rated below investment grade (IG) could not issue bonds. The innovation of the time was the concept that a non-IG company should be able to issue bonds if it offered enough interest to compensate for the risk of default. Most people considered this imprudent, calling them "junk" bonds. But, as with early direct lending, because few investors were eager to lend to these companies, those that did could demand high interest rates and strong safety. We have now been in the business of lending money to non-IG companies for 48 years and have produced good returns

with manageable risk. So, there's nothing inherently safe or risky about making a loan, even a non-IG loan.

Allison Nathan: But are there inherent flaws in direct lending investment vehicles?

Howard Marks: The structure of such vehicles introduces serious concerns around pricing and liquidity. On pricing, if investors choose to take their money out of a non-traded business development company (BDC)—the vehicle for retail participation in direct lending—they are paid at the fund's net asset value (NAV), which third parties estimate periodically. But is the NAV accurate? Is it fair? A perception exists that NAVs for private assets—both equity and credit—may be overvalued, which would align with most participants' incentive to show good performance. But since these loans don't trade, it's difficult to determine their fair price. This is problematic because if the NAV is too high, investors that leave are paid too much at the expense of the remaining investors. And if the NAV is too low, investors that leave are penalized unfairly.

It's important to recognize that the whole concept of determining a fair and accurate price is more complicated than many appreciate. For example, should a loan be valued at its "fundamental value"—which is hard to estimate—or at the price an investor could sell it for today, or at the price they could get a month from now after prospective buyers have had the opportunity to conduct due diligence, or at the price an investor would have to pay to buy an equivalent loan? Along the same lines, should a \$1bn loan be valued at the price an investor could get for the whole \$1bn, or for the first \$10mn? Those prices might differ. The point is that an asset does not "have a price", which is true even for traded assets; it depends on market conditions and the size of the trade.

So, valuing assets fairly is a hornet's nest, and especially so for private assets that have no reference point. Private asset investors must therefore accept that there is no correct answer, and no *one* answer, when it comes to a loan's fair value. The only way to avoid the inherent unfairness that the uncertainty around NAVs introduces would be to pay investors leaving the fund the proceeds the manager receives when they sell off the departing investor's share of the assets. But that's not how these vehicles are structured.

Allison Nathan: You also mentioned that liquidity of direct lending vehicles is a concern. Why?

Howard Marks: To avoid managers having to sell assets at unrealistically low prices that would be injurious to their investors and potentially to their business, private credit vehicles are designed either to prohibit investors from taking back their capital, as is the case with traded BDCs, or to allow them to take back only a certain percentage—typically 5% of NAV—per quarter, as is the case with non-traded BDCs. This has recently proven problematic because redemption requests for some non-traded BDC funds have exceeded the 5% cap.

You can imagine how it feels to invest in something and then not be able to get your money back. It can be a problem when a vehicle that invests in private assets promises liquidity.

Allison Nathan: But isn't the limited liquidity of non-traded BDCs a feature—not a flaw—given that such liquidity constraints are embedded in the vehicle's design?

Howard Marks: It's absolutely true that investors participating in a fund that limits redemptions shouldn't be surprised when full liquidity isn't available. But "should" is one of the most interesting words in the English language. If people shouldn't expect liquidity, why are they disappointed when they don't have it? The answer is either that they weren't adequately informed, or they were well informed but didn't understand or take cognizance of the terms. The risk of either of these is highest for retail investors, raising the question of whether these vehicles are suitable for the retail base. But there's nothing inherently wrong with retail investor participation if they know what they're buying.

Allison Nathan: A key concern around private lending is the potential for large redemption requests to force fire sales. But don't the liquidity constraints of non-traded BDCs—well understood or not—prevent that?

Howard Marks: Forced sales are indeed a common cause of crises; I've been around long enough to see several of them. But the redemption limits on non-traded BDCs should prevent such a crash in this market. When people say non-traded BDCs' liquidity limitations are doing their job, that's what they mean. But those limitations are still unpleasant for investors.

Allison Nathan: So, you don't see similarities between what's happening in direct lending today and the GFC?

Howard Marks: No, for three reasons. First, the GFC occurred largely because too much money was invested in highly levered residential mortgage-backed securities (RMBS) that were built out of subprime mortgages, which ultimately had no substance. A great deal of money went into such mortgages because structuring RMBS produced a lot of fees; thus, mortgage brokers were paid to make loans, but not necessarily good loans. Nobody seemed to wonder about the wisdom of lending money to unsuitable borrowers. Today, no inherent lack of substance exists in direct loans to creditworthy companies.

Second, another key contributor to the GFC was banks' high leverage. Structuring RMBS from subprime mortgages was so profitable that the institutions making and securitizing the loans had incentive to keep doing so. But that required somebody to take the riskiest tranche, which often ended up being the institution itself. So, highly levered financial institutions—namely, banks, which were levered up to 32x—held the riskiest tranches. And a 3% decline in the value of the portfolio at an institution that is 32x levered could render it insolvent. Conversely, private credit funds are not highly levered. Some are unlevered, and some are slightly levered—up to perhaps 2x at most. So, they don't face the same type of meltdown risk.

Third, private credit funds aren't interdependent. The years leading up to the GFC saw substantial use of derivatives and hedges contracted in with other institutions—one example is credit default swaps—that created what we called

"counterparty risk", setting up for cascading losses if one institution failed. That type of interconnectedness doesn't exist among private credit funds. So, I don't see the same systemic exposure here as was the case in the run-up to the GFC.

Allison Nathan: Even if there's no systemic risk, are concerns about direct lending's exposure to software firms that are vulnerable to AI disruption warranted, especially since many of these firms were levered up in buyouts?

Howard Marks: It's cause for concern for the funds that loaned money for those buyouts. But the downside is probably manageable in most cases because, if a first-lien loan is made to a software company, the value of the company must decline enough to wipe out the equity as well as the junior lenders before that loan is jeopardized. That type of value destruction is not impossible, but it shouldn't be assumed. And even if the first-lien lender lost half their money in such a case, that wouldn't be too bad in a diversified portfolio. If a quarter of the investments are in software and they all incur such losses, then the lender would lose half their money in a quarter of their investments, or 12.5% of the portfolio. And even if they're levered 1:1, that would amount to a 25% loss, which still isn't existential. There's an old saying that in times of crisis, all correlations go to one. In other words, everybody panics and sees all outcomes as equally probable and equally bad. But losing a quarter of your capital because the software industry loses most of its value isn't abysmal. It's just bad for the fund's investors. And it doesn't jeopardize the financial system.

Allison Nathan: But what could default rates look like?

Howard Marks: I don't make economic predictions. So, I can't say how high default rates could climb. And it's important to note that in private credit, what constitutes a default—or even whether lenders will report defaults—is unclear. Lenders have an incentive to grant the borrower an extension rather than report a default, and nobody may be enforcing the reporting of defaults. Particularly for that reason, I wouldn't put much stock in estimated default rates.

Allison Nathan: Given everything we've discussed, how do you expect the direct lending market to evolve from here?

Howard Marks: It's important to first note that the growth of the private equity industry, which is the main borrower, has slowed. So, the growth in demand for these loans will also likely slow. And, as Warren Buffett famously said, it's only when the tide goes out that you find out who's been swimming naked. Now that the tide may have begun to go out on private asset vehicles sold to the public, and some of their flaws have been exposed, I would expect retail investors to be a bit more deliberate and circumspect. In a recent memo, I ended with a quote from Bob O'Leary, Oaktree's co-CEO, who says direct lending will be okay, but it might take a credit cycle to get there. Only after investors have experienced a full cycle in which it swings from being too easy to borrow money to being too hard are they likely to really achieve the understanding of both the positives and the negatives that enables them to make better investment decisions the next time around. That's ultimately a healthier investment environment, and it may be what lies ahead for direct lending and private credit more broadly. But learning that lesson isn't always pleasant.

Interview with Michael Arougheti

Michael Arougheti is Co-Founder and CEO of Ares Management. Below, he discusses the evolution of private credit and his perspective on the recent concerns around it.

The views stated herein are those of the interviewee and do not necessarily reflect those of Goldman Sachs.



Allison Nathan: First, what is private credit, and how has it evolved?

Michael Arougheti: Private credit is self-originated loans that institutional money managers make to private companies, asset owners, and portfolios of assets and cash flows.

The largest segment of private credit is direct lending to small- and medium-sized businesses largely owned by private equity sponsors. It also includes asset-based credit, real estate and infrastructure lending, opportunistic credit, and credit secondaries, where investors trade existing private credit interests. These strategies are generally funded in the sub-investment grade (IG) market. Over the past decade, IG private credit has also grown as insurers have merged with institutional asset managers. So, private credit is a diverse market spanning many strategies, sectors, and geographies.

While there is a perception that the private credit market developed largely in response to post-Global Financial Crisis (GFC) regulation, its origins actually trace back to the 1980s, when the advent of the high yield (HY) market revealed a whole swath of sub-IG borrowers whose funding needs banks and public markets were not meeting. Since then, several structural and capital innovations have expanded the market, including the growth of business development companies (BDCs)—closed-end investment vehicles that lend to small- and mid-sized private companies—increased insurer participation post the GFC, and, most recently, the rise of retail investment. So, private credit has experienced a decades-long evolution.

Allison Nathan: The growth in private credit in recent years seems to be a source of worry. Should it be?

Michael Arougheti: No. Private markets have grown because banks and public markets have shrunk, leaving them increasingly unable to meet firms' capital needs. The number of US banks has fallen by roughly half over the last 25 years. The share of bank balance sheet assets that represent commercial and industrial (C&I) loans—a proxy for loans made in private credit portfolios—has also declined from ~30% in the 1980s to <15% today. At the same time, the US has half as many publicly traded companies today as 25 years ago, with only ~2,100 firms listed on the NYSE and 3,400 on the NASDAQ, a tiny fraction of the 18 million companies in the US today.

Private credit growth also hasn't been extraordinary. Private market AUM has grown at a 13% compound annual growth rate (CAGR) over the past decade, comparable to private credit's 14% CAGR, with private credit growing only slightly faster than the overall private market over the last three years as the Silicon Valley Bank (SVB) crisis disrupted private equity flows and capital shifted toward opportunistic credit. Capital raised in institutional private credit has also declined for the last four years. And direct lending dry power remains modest relative to private equity dry powder, at roughly \$300-400bn vs.

over \$1tn. More broadly, leveraged credit—non-IG private credit, bank C&I loans, and HY and leveraged loans—as a share of US nominal GDP has remained steady at around 23% since 2015. So, private credit growth hasn't been excessive, and many parts of the market still don't have enough capital to meet demand.

Allison Nathan: Even if the growth doesn't warrant worry, aren't private credit markets taking risks that banks won't?

Michael Arougheti: No. Private markets are taking credit risks that banks *can't*. It's underappreciated that banks are 10-15:1 leveraged while private credit funds are only 0-1:1 leveraged. Leverage amplifies returns but also risks. Banks' inherent asset-liability mismatch and regulatory capital frameworks mean that whole swaths of loans don't make sense for banks to extend. Private credit hasn't created demand for credit that otherwise didn't exist. It's simply an alternate capital provider.

Allison Nathan: That said, are the worries about several high-profile defaults warranted, or overblown?

Michael Arougheti: They're overblown. Downturns in credit cycles are a normal part of the business. But the last two cycles—which spanned the GFC and the pandemic—were super cycles dotted with a series of mini cycles around the 2013 Taper Tantrum, 2022 Gilt Crisis, and 2023 SVB collapse. So, even some long-time investors have never experienced a credit cycle downturn and so are reacting strongly to defaults even though fundamentals remain generally solid. Looking at private and bank credit portfolios and credit card charge-off data, nothing indicates a turn in the credit cycle. While defaults will inevitably rise as the cycle matures and some managers will underperform, that doesn't mean the entire credit market will tip over, despite a handful of attention-grabbing defaults last fall that were relatively small exposures and, notably, mainly funded by bank-led asset-based facilities.

Allison Nathan: Could the increased use of payment-in-kind (PIK) features—which allow borrowers to preserve cash by shifting interest payments to the loan's principal balance—be masking deteriorating credit quality?

Michael Arougheti: To answer this, PIK should be thought about in three baskets. First is "PIK at the outset", where the instrument is underwritten with PIK because the lender wants to earn a higher return and the borrower wants more flexibility in deploying their cash. That's a structural choice, not a distress measure. Second is "good PIK", where a performing company wants to divert cash from debt service to growth. This is attractive to institutional investors because the borrower is willing to pay the lender more, driven by a belief that the return from redeploying cash into the business exceeds the cost of capital. Third is "bad PIK", where a company can't service its debt because operations are deteriorating. Bad PIK is rising, which warrants watching. But not all PIK is bad PIK.

Allison Nathan: How concerning is private credit exposure to the software industry amid worries about AI disruption?

Michael Arougheti: Some concern may be warranted, but the fears around software exposure also seem overdone. Software names in private credit portfolios are typically the biggest US private software companies, which are largely institutionally equity-backed. These loans are senior secured, high up in the capital structure, and extended to firms with high free cash flow. For context, at Ares, software is ~9% of our private credit portfolio. These are large businesses, with average EBITDA of ~\$350mn. The average loan-to-value is ~38%, meaning 62% of the capital structure sits below us, owned by brand-name private equity sponsors. So, the bulk of the risk is in the private equity market. These firms are also generally growing at low double-digit rates. And the loans typically have around 3.5 years of remaining maturity, which aligns with the timeframe of expected revenues. These structural mitigants provide meaningful downside protection. That's not to say some software firms won't be disrupted and losses won't occur. But for every firm AI disrupts, another will likely benefit.

Allison Nathan: Fueling these concerns is a perception that private credit lacks transparency. What's your response?

Michael Arougheti: The question is, transparency to *whom*? Institutional investors in private credit funds have perfect transparency into what they own and how it's performing on a monthly basis, along with access to company management and private credit portfolio teams. That's far more transparency than public market investors have given reporting mandates and regulatory limitations on public company disclosures. Third-party firms also evaluate and value institutionally-managed private credit portfolios, and public accounting firms review that. The banks providing leverage to private credit portfolios also frequently review the underlying credit exposures. It's true that the Fed doesn't have perfect transparency into private credit portfolios because they aren't bank-chartered or highly leveraged. Even so, the Fed meets with us regularly and sees our portfolios through the regulation of our bank partners. So, I struggle with the idea that private credit isn't transparent.

Allison Nathan: How concerning is the recent surge in redemption requests, and could they trigger fire sales?

Michael Arougheti: It's not particularly concerning owing to the structure of non-traded BDCs, which is where redemption requests are concentrated. Key aspects of these structures are often misunderstood. First, so-called "redemption gates" aren't actually gates. Before non-traded BDCs existed, investors could either have daily liquidity in a traded BDC or illiquidity in a 10-12 year private commingled fund. The innovation of non-traded BDCs was to allow investors to own illiquid assets while providing a structured path to liquidity of 5% of net asset value per quarter or 20% per year, with the structure self-amortizing so investors could be made whole without forced asset liquidations. The 5%/20% limits weren't pulled out of thin air but rather designed to track the weighted average life of the underlying loan portfolio. So, meeting the contractual 5%/20% is exactly what these structures were designed to do.

Second, no "asset-liability mismatch" or "run on the bank" dynamic exists in non-traded BDCs. Unlike a bank with short-term deposits and long-term assets, these funds align liquidity with the five-year amortization profile of their loans. They also typically hold 20-30% of assets in liquid securities like loans,

bonds, or high-grade investments that can be sold to generate cash, and managers can draw on loan facilities to meet redemption requests. Those features exist precisely to avoid having to liquidate private assets below their intrinsic value.

Third, scale matters. The non-traded BDC market is ~\$300bn, representing less than 10% of the nearly \$4tn private credit market. Even if every non-traded BDC met its maximum redemption limit, that would amount to ~\$5bn of loan sales out of their tradable bucket every quarter, which is small relative to the ~\$85bn of loans that trade quarterly in the syndicated loan market. More broadly, adequate dry powder exists in the market to resolve private credit portfolios in an orderly fashion without any disruption to pricing.

Lastly, the current volatility in non-traded BDCs is a technical, not fundamental, issue driven by narrative and positioning. Case in point: our non-traded BDC fund saw no credit underperformance and zero non-accruals as of 4Q2025, and over 95% of the investors didn't request their money back in the most recent redemption period. That said, some managers have and may continue selling assets to demonstrate confidence in their marks to the retail base, which may benefit their franchise but not necessarily be in stakeholders' best interest. So, unwinds could occur. But that would play out over years and in an orderly fashion.

Allison Nathan: What do you make of comparisons between private credit worries and the lead-up to the GFC?

Michael Arougheti: The setups differ dramatically. First, while the Iran conflict has generated significant uncertainty, the US economy remains relatively strong. Second, credit today is owned by a much broader swath of investors and with much less asset-liability mismatch, leading to a significant decline in volatility in the traded credit markets. If anything, private credit has become a buffer to market volatility and risk as opposed to a conduit for them. Third, banks' exposure to BDCs and private credit today is ~0.8% compared to their nearly 20% exposure to subprime mortgages in the lead-up to the GFC. The assets themselves are also fundamentally less risky. A CLO security rated single-A or higher—which is where banks typically invest—has never defaulted. And as regulators seek more transparency, they will discover that banks have offloaded risk and now hold only the safest loans. Lastly, regulators have much more transparency into the financial system today than in the lead-up to the GFC, when off-balance sheet structures and subprime loan portfolios weren't visible to them.

Allison Nathan: Will recent developments impact the growth outlook for private credit?

Michael Arougheti: The recent developments will likely shift market share within private credit rather than slow growth overall. The current narrative is creating noise, but also opportunity for those able to cut through that noise. In practice, that likely means slower growth in non-traded BDCs, with capital reallocating toward strategies like opportunistic credit, credit secondaries, and direct lending. Over time, as investors see how these structures actually performed, capital will likely flow back. In the meantime, private credit will continue to find ways to fund the markets' needs, creating significant opportunity for managers who are structured the right way.

Private credit, explained

WHAT IS PRIVATE CREDIT?

Private credit refers to debt financing provided by non-bank financial institutions directly to companies. The rise of private credit was largely fueled by stricter regulatory frameworks following the Global Financial Crisis, including Dodd-Frank and Basel III, that forced traditional banks to tighten lending standards and reduce risk. This created a funding gap that non-bank lenders stepped in to fill. Private credit's growth was further accelerated by a prolonged period of low interest rates, which encouraged both institutional and retail investors to seek the higher yields and illiquidity premiums available through private debt.

Borrowers

Borrowers in the private credit space tend to be small and mid-sized businesses, although the borrower profile has evolved from a niche group of small businesses into a broad spectrum of corporates and entities. Borrowers can also be the portfolio companies of private equity sponsors who borrow funds to finance acquisitions or buyouts.

Lenders

The major non-bank lenders in the private credit space include alternative asset managers (many of which also participate in private equity), specialized credit funds, business development companies (BDCs)—which are SEC-regulated investment vehicles—and occasionally large individual investors.

Banks

Banks have increasingly become involved in the private credit ecosystem in recent years, largely as indirect participants, with banks acting as key capital providers to private credit funds. Banks also provide leverage to private credit funds through subscription lines and fund financing. Some banks have also increasingly taken a more proactive role in the market by establishing their own internal private credit capabilities.

Investors

Investors in private credit funds include institutional investors like pension funds, family offices, endowments, insurance companies, as well as retail investors (see pg. 9 for a more detailed ownership breakdown).



What types of private credit strategies exist?



1 Direct Lending

Direct lending is the most common (and 'traditional') form of private credit. Private lenders provide financing directly to corporate borrowers. Such loans are typically secured and involve corporate recourse, offering borrowers flexible solutions tailored to their specific financial needs.



2 Asset-Based Finance (ABF)

Asset-based private credit refers to a type of private credit in which lending is backed by specific collateral such as real estate, equipment, or financial assets. The loan amount depends on the value of the pledged assets, which can reduce credit risk for lenders. ABF is common among businesses with strong asset bases but limited cash flow or short-term financing needs.



3 Mezzanine, Second-Lien Debt, and Preferred Equity

These three forms of credit, known as junior capital, provide borrowers with subordinated debt. These instruments are not secured by assets and rank below senior loans for repayment in the event of a default or bankruptcy. Junior capital often comes with equity "kickers" like warrants. Borrowers use mezzanine financing to fund growth or acquisitions when senior debt alone isn't sufficient, and investors accept higher risk in exchange for higher returns.



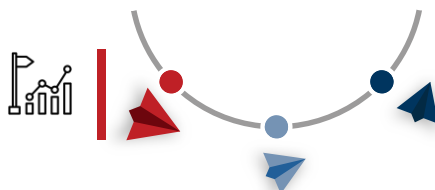
4 Distressed and Opportunistic Credit

These are investments in companies under financial stress or those that target market dislocations. Distressed debt refers to loans or bonds from companies experiencing financial difficulties or facing bankruptcy. Private credit investors in this space seek to profit by purchasing the debt at a discount and working toward a turnaround or restructuring that increases its value.

How can investors access private credit?

BDCs are one way investors, particularly retail investors, can access private credit strategies. They are investment vehicles that invest in small and mid-sized private businesses. They generate income mainly through interest payments and distribute income to shareholders, typically in the form of dividends. The BDC market is comprised of three main structures that offer a different level of accessibility to investors.

Publicly traded BDCs trade daily on major exchanges and are available to any investor with a brokerage account.



Non-traded BDCs (perpetual/evergreen funds) are often sold through RIAs/broker-dealers and don't trade on an exchange. Instead, they provide liquidity through periodic share repurchase programs. Most non-traded BDCs limit total repurchases to 5% of net asset value (NAV) per quarter, though the cap is discretionary. If redemption requests exceed 5%, the fund has the ability to repurchase shares on a pro-rata basis.

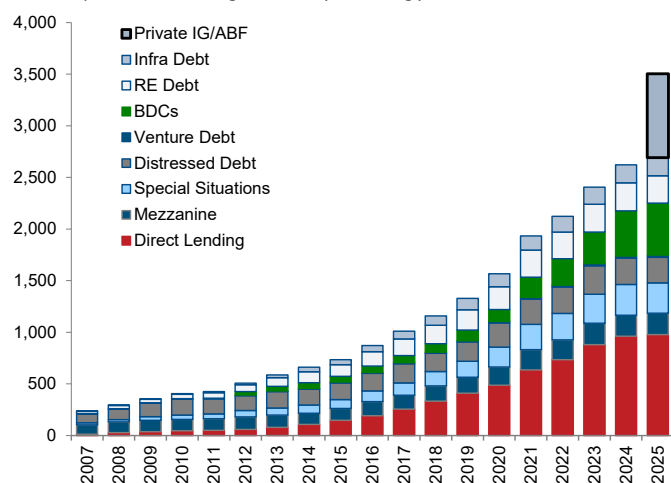
Private BDCs (drawdown funds) are closed-end investment vehicles where investors commit a capital amount upfront but provide the cash only when the fund manager calls for it to fund specific loans. These funds generate returns through income and capital appreciation, with profits distributed to investors as the fund matures. Capital is locked up for the life of the fund, and access is restricted to accredited investors and institutions.

Source: Federal Reserve, Blackstone, Carlyle, KKR, various news sources, Goldman Sachs GIR.

Sizing the private credit market

The private credit market has grown significantly since 2007 to around \$3.5tn in assets under management (AUM) today, with direct lending accounting for the largest slice of the market

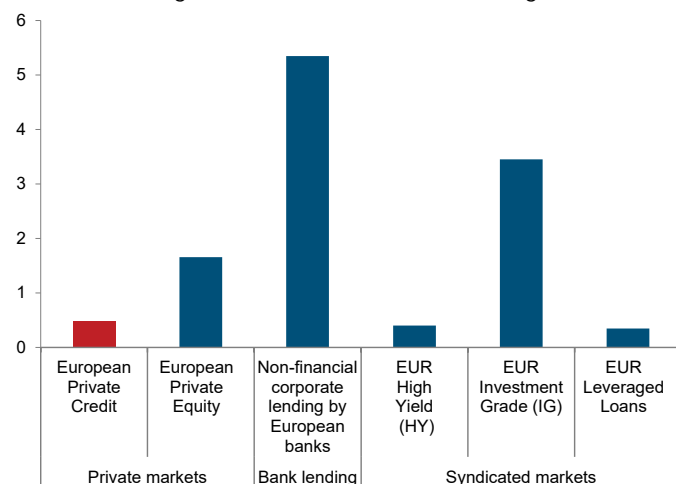
Global private lending AUM by strategy, \$bn



Source: PitchBook LCD, Preqin, LSEG Data and Analytics, BDC Collateral, Goldman Sachs GIR.

...and the European private credit market is relatively tiny

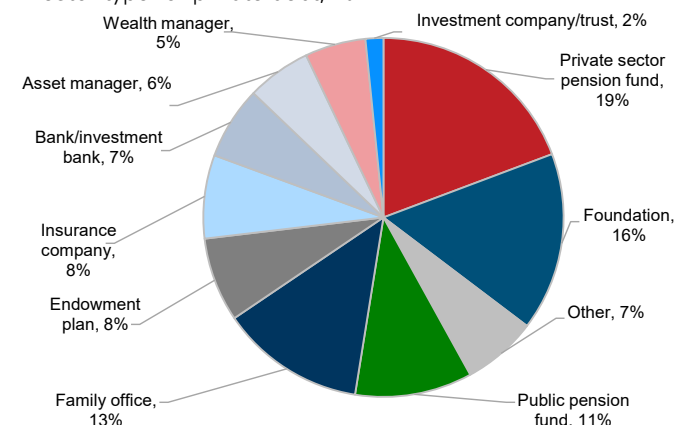
Amt outstanding across various Euro area lending markets, €tn



Source: Preqin, ECB, Haver, Bloomberg, PitchBook LCD, Morningstar, GS GIR.

...with the ownership structure of private credit funds largely skewed toward liquidity-agnostic investors

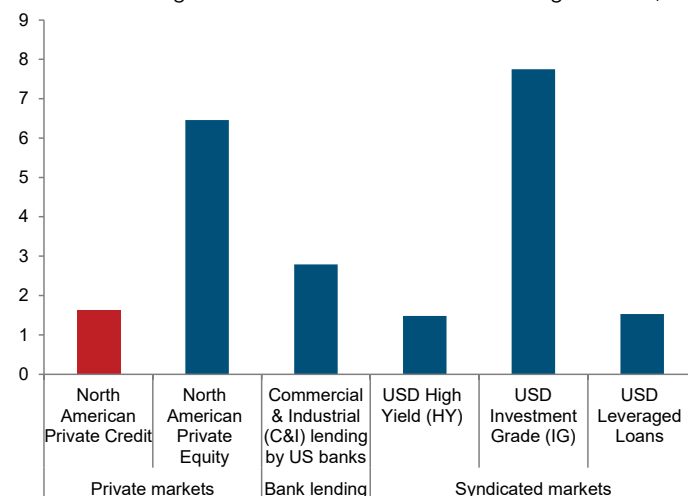
Investor type for private debt, %



Source: Preqin, Goldman Sachs GIR.

Despite this growth, the private credit market is modest in size relative to other North American corporate financing markets...

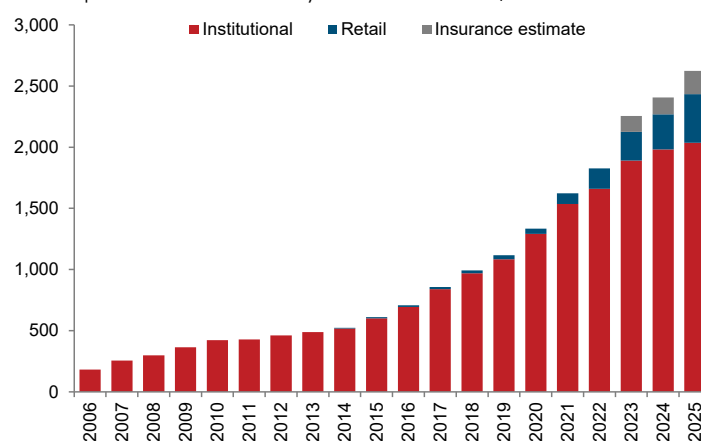
Amt outstanding across various N. American lending markets, \$tn



Source: Preqin, Cliffwater, Federal Reserve Board, Haver, Bloomberg, PitchBook LCD, Morningstar/LSTA, Goldman Sachs GIR.

While retail investors are playing a growing role in the private credit market, institutional investors continue to dominate...

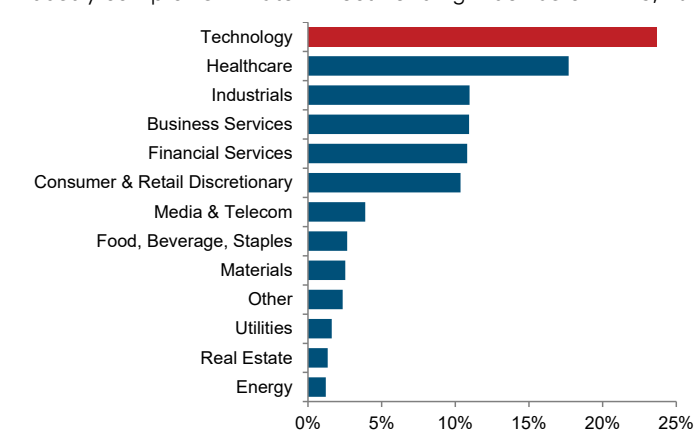
Global private credit AUM by investor channel, \$bn



Source: PitchBook LCD, Goldman Sachs GIR.

At nearly a quarter of loans, the technology sector accounts for the largest part of the US direct lending market

Industry comp of Cliffwater Direct Lending Index as of YE25, %



Source: Cliffwater, Goldman Sachs GIR.

Special thanks to GS Credit Research team (Amanda Lynam and Shamshad Ali) and Asset Managers Equity Research team (Alexander Blostein and Anthony Corbin) for data. Exhibit 1 originally appeared in Alexander Blostein's October 19, 2025 Americas Asset Managers note. Exhibits 2-6 appeared in the Credit team's April 13, 2026 and July 29, 2025 Private Credit Monitors (see latest Monitor for more details on chart calculations).

Interview with Amanda Lynam

Amanda Lynam is Chief Credit Strategist at Goldman Sachs. Below, she explains the role of private credit in the financial system and why many concerns about the space are misplaced.



Jenny Grimberg: What role does private credit play in the financial system, and how has that evolved?

Amanda Lynam: Private credit is one of the three core pillars of the financing continuum alongside the bank lending and public debt markets. While the private credit market was initially reserved for smaller companies

that lacked access to the public debt markets, it has significantly expanded since the Global Financial Crisis (GFC), and especially since the pandemic, to include borrowers who could otherwise access public debt markets or, in many cases, have already done so and are refinancing their debt into the private credit market. So, over time, the boundaries between the three markets have become increasingly fluid.

Jenny Grimberg: So, it's not the case that companies who use the private credit market do so simply because they lack alternatives?

Amanda Lynam: No; that's a common misperception. Firms with access to public markets could still opt for private credit for several reasons. First and foremost is certainty of execution, especially during periods of market volatility when public markets may be less accessible. For example, amid recession fears in 2022-2023, the leveraged loan and high yield (HY) bond markets weren't widely available for lower-rated borrowers due to lower risk appetite and structural constraints. In particular, the predominant buyers in the leveraged loan market are Collateralized Loan Obligations (CLOs), rating-sensitive vehicles that can only hold a small amount of triple-C rated debt, which restricts their lending to lower-rated firms in times of economic uncertainty. Private credit doesn't face such constraints, making it a more reliable source of capital in choppy markets.

Second is the availability of customized and bespoke financing solutions. Public markets tend to work best for borrowers that fit neatly within standardized credit frameworks. For unique companies—like those in early growth stages, with high customer concentration risk, or with binary success outcomes—private credit can offer tailored solutions the public markets may not be comfortable providing. And third is higher barriers to entry in the public markets. As companies stay private for longer, public market deal sizes have become extraordinarily large. The average deal size in the HY market today is well above \$700mn. This makes public issuance less practical for firms seeking relatively modest amounts of capital.

Jenny Grimberg: Is private credit riskier than public credit?

Amanda Lynam: Not necessarily. The additional spread that private market financing commands over public market financing—known as the illiquidity premium—is often mischaracterized as purely a reflection of higher risk. In many instances, it just reflects compensation for the fact that private credit lenders commit to owning a loan for the entirety of its life. Borrowers are also paying extra for the customization and

greater certainty of execution that private credit offers. And while private credit tends to serve smaller or less mature companies than public debt markets, lenders manage this risk through thorough due diligence, the use of covenants, and active oversight, including the ability to renegotiate deal terms early if a borrower's financial health deteriorates.

Jenny Grimberg: The current worries about private credit are focused on non-traded business development companies (BDCs). What are these structures, and how important is their role within private credit?

Amanda Lynam: Non-traded, or evergreen, BDCs are vehicles for retail investors to access private credit strategies. They offer a faster path to investment with lower minimum denominations and more simplified tax reporting than institutional private credit vehicles. Non-traded BDCs represent ~15% of "traditional" private credit AUM (per PitchBook LCD), which doesn't include asset-based finance or investment grade (IG) private credit. So, despite the significant focus on them, non-traded BDCs comprise only a small fraction of the private credit universe. The vast majority of private credit AUM is institutional capital that is locked up in long-term vehicles with fundamentally different structures.

Jenny Grimberg: So, are the concerns around the surge in non-traded BDC redemption requests overdone?

Amanda Lynam: Let me first say that despite the attention-grabbing headlines about major private credit firms capping redemptions at 5% of net asset value (NAV) per quarter, these redemption "gates" aren't new, even if some investors seemed previously unaware of them, and are a feature, not a bug, of these structures. Such features are designed to safeguard investors' capital by preventing managers from being forced to sell assets at depressed prices to meet short-term liquidity demands. The current stresses in these retail vehicles don't exist in the institutional market, because institutional credit vehicles were never structured to offer redemptions in the first place. So, concerns about these redemptions, and potential for spillover to other vehicles, seems misplaced.

Jenny Grimberg: Are recent defaults in the space cause for concern?

Amanda Lynam: While private credit default rates have risen somewhat, these statistics can paint a misleading picture of fundamental health. Some people compare defaults in private credit to defaults in public credit, but they are not actually comparable. Maintenance covenants—periodic financial metrics borrowers must meet—are much more common in private credit than in the broadly syndicated leveraged loan market. Because of that, a covenant breach in private credit can be classified as a covenant default, which counts in the default statistics. By contrast, roughly 92% of the broadly syndicated leveraged loan market is "covenant-lite", meaning the loans don't have regular maintenance covenants. Consequently, the first sign of default is often a distressed exchange or a missed payment, which can quickly translate into a monetary loss.

For that reason, realized losses are a better gauge of the health of the market than default rates. As of end-2025—the latest available data—realized losses in private credit, which we track through the Cliffwater Direct Lending Index that comprises over 20k US middle-market loans representing \$550bn in AUM, were around 65bp, below the long-term average of 100bp and generally in line with the public credit markets. And the yield on that index is 9.9%, so the income is more than offsetting the losses. So, underlying market fundamentals appear relatively healthy, and we expect that to remain the case as long as the economic backdrop remains resilient.

Jenny Grimberg: Concerns have also grown about the use of payment-in-kind (PIK), which allows borrowers to defer cash interest payments, and non-accrual loans, which are delinquent in payment. What are you actually observing, and how worrying is it?

Amanda Lynam: PIK is another metric that requires nuance because not all PIK is created equal. “Good” PIK is PIK embedded at loan origination, often for high-growth borrowers seeking to preserve liquidity for investment or acquisitions and comes with guardrails around duration and magnitude. “Bad” PIK is PIK introduced after origination when a borrower is experiencing financial stress. “Bad” PIK modestly rose after the Fed embarked on its hiking cycle, but not to outsized levels. And total PIK has remained around 7–8% of BDC total income over the past several quarters, well below the 2020 peak. Non-accrual rates have also remained in a tight range over that period. So, directional indicators of potential credit deterioration aren’t sending concerning signals right now.

Jenny Grimberg: How concerned should investors really be about private credit’s software exposure?

Amanda Lynam: It’s something to watch, but software exposures aren’t uniquely a private credit problem—while software is the largest sector in private credit, it’s also the largest in the broadly syndicated leveraged loan market. So, if AI does disintermediate a significant share of software firms, both the public and private markets would be impacted. In such an environment, private credit could actually be well-positioned to deploy dry power because the public markets would likely be choppy, and opportunistic credit strategies could see capital inflows. AI disruption risks have also long been on many private credit managers’ radars even if the pace of disruption may be occurring faster than they anticipated. So, their investments have focused on software firms with moats like proprietary data or deeply embedded customer relationships. I largely view the software exposure in both private and liquid markets as a catalyst for additional dispersion, not widespread market disruption.

Jenny Grimberg: Given all that, how likely is private credit to pose a systemic risk?

Amanda Lynam: Very unlikely, in my view. While there is absolutely potential for performance dispersion and losses at the manager and fund levels, the tentacles through which those losses could amplify stress across the broader financial system seem largely contained. A key feature of private credit is asset-liability matching. Most capital is locked up, and even in retail-focused structures, liquidity is limited by design. That

significantly reduces the risk of a “run on the bank.” Leverage at the fund level is also modest, particularly compared to the banking system, and the private credit market is small relative to other parts of the financial ecosystem. In the US, the Bloomberg USD IG Corporate Bond Index is roughly \$8tn in size while private equity is six times larger than private credit. More broadly, unlike during the GFC—when hidden correlations and excessive leverage in structured securities triggered a domino effect—losses in private credit would largely be borne by the investors in those funds. While that would undoubtedly be painful, it wouldn’t be systemic.

It’s also worth mentioning that rather than being a source of systemic risk, private credit has the potential to help stabilize markets in times of stress. For example, during the 2023 US regional banking crisis, private lenders stepped in to fill some of the financing void when banks and public debt markets stepped back. As a result, corporate default rates didn’t rise nearly as much as they otherwise could have during that period.

Jenny Grimberg: But couldn’t the interconnectedness between private credit firms and banks act as a transmission channel for systemic shocks?

Amanda Lynam: Loans to non-depository financial institutions comprise 15% of overall bank lending, much less than other types of loans, and private credit accounts for only around a quarter of that, with the loans largely held by large banks with over \$10bn in assets. A recent Federal Reserve report found that bank loans to private credit vehicles have low loss rates. So, the interconnectedness isn’t a particular concern.

Jenny Grimberg: Conversely, could upcoming regulatory changes like Basel III Endgame boost lending capacity in the banking system and slow the growth of private credit?

Amanda Lynam: A regulatory rollback is unlikely to materially dampen private credit’s growth given the size of the addressable market. And for banks with excess capital, it is not obvious to us that it would be redeployed into middle-market lending given the many competing uses for capital, including shareholder returns, M&A, market-making activities, or lending to private credit firms themselves. So, banks and private credit firms will likely remain strategic partners, with banks providing the origination networks and leverage, while private credit firms serve as the ultimate holders of the loans and associated risk.

Jenny Grimberg: More broadly, will the current stresses in private credit dent its growth?

Amanda Lynam: Noise will likely persist over the near term, which may slow growth in some areas of the market, particularly retail-focused strategies. However, over the longer term, these challenges will likely prompt private credit to deploy capital in areas beyond traditional direct lending, such as opportunistic credit, mezzanine financing, and private asset-based finance. And the North American private credit market is far from saturated, especially given the ongoing AI-driven capex buildout. Private credit also has significant room to capture a larger share of overall lending in Europe and Asia, where financing remains relatively bank-dependent. So, while growth is unlikely to be linear, the scope for private credit to continue expanding over the longer term remains very much intact.

Interview with Bruce Richards

Bruce Richards is CEO, Chairman, and Co-Founder of Marathon Asset Management. Below, he argues that private credit is meaningfully overexposed to the software sector, which he expects will result in significant defaults and losses, though not a systemic crisis.

The views stated herein are those of the interviewee and do not necessarily reflect those of Goldman Sachs.



Allison Nathan: Are the concerns around software exposure in private credit portfolios justified, or overblown?

Bruce Richards: Let's put perspective on software exposure within private credit by focusing on three key numbers: 3, 13, and 23. 3% of the high yield bond market is software

loans versus 13% of the broadly syndicated loan market and a whopping 23% of the private credit market. For the top 10 private lenders, that number climbs to 26%. Given the roughly 20 investable industry sectors that exist, exposing 26% of a portfolio to software is excessive, especially considering that software is just a subset of the Technology, Telecom, and Media (TMT) sector. So, it's hard to argue that there isn't too much software risk in direct lending.

Allison Nathan: But concerns about technology concentration risk are nothing new and are as much a feature of the public equity market as the private market. So, why the particular concern here?

Bruce Richards: Comparing portfolio exposure to market cap weight is a mischaracterization of how portfolio managers manage risk, whether it be in a bank lending, asset-based or direct lending program. The cardinal rules of risk management for any lender running a diversified portfolio are to limit the maximum amount of exposure to any one company and industry. And software companies with anywhere from \$10 to \$100mn of revenues represent only around 1% of all private companies in the US. That number is around 7% across the public markets, whether in the NASDAQ, S&P 500, or Russell 2000. So, software exposure is outsized any way you slice it.

Allison Nathan: But many of these software companies are well-established and profitable. And while agentic AI technology may disrupt some of them, it may enhance others. Does that give you comfort that some wins will offset some losses?

Bruce Richards: There will be tremendous wins. I'm somewhat constructive on software in the public markets because public companies have, net of cash, half of one turn of leverage or debt to EBITDA of 0.5. Many companies have strong margins, solid earnings, and substantial free cash flow, which will allow them to make the operating investment required to reposition themselves as AI-first software firms.

But many companies are not in that position, with leverage of 8 to 10 times debt to EBITDA. Those elevated leverage levels are my primary concern because they leave companies with very little to no free cash flow after debt service. In fact, many companies are actually burning cash on an operating basis after debt service. Without adequate free cashflow, these highly

leveraged software companies will have little ability to make the upfront investment necessary to transition into AI-first firms.

Allison Nathan: Some have argued that private credit concerns are overdone because these funds aren't highly leveraged. What is that argument missing?

Bruce Richards: Let me explain with a simple example. \$1bn of capital with one turn of leverage on the balance sheet amounts to \$2bn of assets. \$2bn of assets with 26% of the portfolio in software means that the software exposure on \$1bn of investor capital is \$520mn—or 52% of the capital. So, the only one turn of leverage that private credit funds or business development companies (BDCs) have still amounts to exceptionally high software exposure. At the same time, the underlying software firms typically have upwards of 8 to 10 turns of leverage in the direct lending business and a minimum of 4 to 5 turns of leverage in the broadly syndicated loan market. So, it's important to distinguish between the leverage of the fund versus the amount of debt on the specific company the fund is lending to. Both metrics are important to fully understand the risk.



The only one turn of leverage that private credit funds or BDCs have still amounts to exceptionally high software exposure. At the same time, the underlying software firms typically have upwards of 8 to 10 turns of leverage in the direct lending business."

Allison Nathan: Some have also argued that even if software companies are disrupted, private lenders' position in the capital structure protects them relatively well. What's your view?

Bruce Richards: It's true that private—and public—equity investors have more downside exposure. But they also have all of the upside exposure. There will be winners and losers, and even the best software investors in the private equity space will undoubtedly see some zeros in the 2020-2025 investment vintages. But they'll also see some companies with higher multiples than their original underwriting assumed as AI adoption drives valuations higher. So, it's possible that some equity investors will end up doing quite well overall across those vintages, even in the event of elevated defaults. Conversely, lenders can't afford the zeros because they only make par back; they don't have the upside that the private and public equity markets do.

At the same time, a substantial part of the cushion that sits below private credit in the capital structure has already been eaten through given the collapse in software valuations across

public and private markets. Valuations for mid-sized software companies are generally down 50% or more from peak levels.

And when a software company goes bankrupt, even senior loans will take a sizable hit. It's important to understand that software bankruptcies look different than bankruptcies in other industries because enterprise customers won't tolerate uncertainty. A CTO or CIO needs to know that their firm's software is reliable and can't gamble with operational or reputational issues that arise in a bankruptcy process. So, keeping a software business intact through a bankruptcy process is exceptionally challenging. As a result, while defaults in other industries typically result in recovery rates for high yield bonds, broadly syndicated loans, and direct lending of 60 to 80 cents on the dollar, recovery rates will likely prove considerably worse in software bankruptcies. Given the collapse in valuations, heavy debt loads, and substantial uncertainty about which software companies will be successful amid the AI transition—as well as likely limited capital availability until that becomes clearer—software default recovery rates will likely be in the 0 to 30 cent on the dollar range for direct lenders.

Allison Nathan: So, what do you expect for software default and loss rates?

Bruce Richards: I expect incredibly high software sector default rates in 2027, 2028, and 2029, with peak defaults of 15% and double-digit default rates in each of those years given the valuation reset this year and likely reluctance of the broadly syndicated loan and direct lending markets to extend new credit to highly leveraged, below-investment grade software companies. Direct lending recovery rates of 0 to 30 cents would amount to 70-100% losses on those loans.

“I expect incredibly high software sector default rates in 2027, 2028, and 2029, with peak defaults of 15%... Direct lending recovery rates of 0 to 30 cents would amount to 70-100% losses on those loans.”

Allison Nathan: Is there a historical analogue for such an industry-specific default cycle?

Bruce Richards: The energy sector's huge technological transformation in 2013-2014 owing to the rise of horizontal drilling and hydraulic fracking, which caused a significant shift in the price structure for energy, comes to mind. Energy had been the number one sector for the high yield bond and broadly syndicated loan markets, with capital rushing in. But the technological shift led to the collapse of several companies, leaving creditors with substantial losses. Debtor-in-possession (DIP) financing came in senior to broadly syndicated loans in the restructuring process, and default rates for these loans hit 15 to 20% at their peak and remained at double digits for three successive years.

While Oil & Gas and Software are very different industries, the common thread between the two periods is a technological transformation driving a change in pricing structure that highly

leveraged companies are unable to sustain, leading to exceptionally high default rates and low recovery rates. Very little capital was available in the Oil & Gas sector in the ensuing years to refinance debt other than the new capital that came in senior to existing debt during the bankruptcy process. I believe the software sector will see the same lack of capital in the coming years, especially given the elevated leverage levels already in the sector today.

Allison Nathan: Could the losses in the software sector that you expect pose systemic risk?

Bruce Richards: No. During the Global Financial Crisis (GFC), banks became ensnared given their exposure to collapsing home prices. And consumers were highly exposed both at the high end, given that the majority of high-income households' net worth resided in housing, and at the low end, given the loss of over 10mn jobs. So, banks failed, businesses were wiped out, and consumers were punished.

Today, banks are stronger than they've ever been. So, there is no systemic risk in the financial system. And there is no systemic risk stemming from consumers because consumers don't pay for software, companies do. Consumer net worth today is \$180tn—an over-threefold increase from the GFC—and consumers are benefitting from near-maximum employment, high housing prices, and record-high equity levels and valuations. And, while the AI transformation will undoubtedly result in software winners and losers, as we discussed, the sector represents only around \$300bn of capital on a \$31tn economy. So, I am not at all concerned that the troubles I foresee for the software sector will lead to a broader crisis.

“Ultimately, while I am very concerned about the coming wave of software defaults and losses, I am very constructive on direct lending.”

Allison Nathan: Even if you don't expect a broader crisis, could software losses dent private credit's growth outlook?

Bruce Richards: No. Direct lending is a critical business for capital formation that will emerge from this correction bigger and stronger than ever. It's worth noting that private credit has grown 18-fold over 18 years to \$1.8tn today and has never experienced a correction until now. This reset will result in the return of manager discipline with respect to maximum exposure levels to any one company or industry, tighter documentation, and stricter covenants. So, the next vintage will likely be very strong. And European direct lending carries materially less of the software exposure I am concerned about than its US counterpart, given that software constitutes a far larger share of both the US economy and US private equity activity than it does in Europe. Ultimately, while I am very concerned about the coming wave of software defaults and losses, I am very constructive on direct lending, which has become an essential financing solution for many companies and an integral and invaluable part of our financial system that will only continue to grow.

BDCs: NAVs vs. prices

Shamshad Ali finds that BDC portfolio fundamentals and signposts of future portfolio health look less concerning than public BDC equity prices suggest

Since the fall of 2025, business development companies (BDCs) have become a key barometer of private credit health and sentiment as retail participation in middle-market lending has grown. Concerns about software exposure in private credit funds and a surge in non-traded BDC redemption requests on the back of worries about forward returns, underwriting standards, and broader secular disruption from AI drove a significant selloff in public BDC equities since June 2025. While they have bounced off their March lows, public BDCs still trade at a nearly 20% discount to their net asset value (NAV), raising a key question: is the NAV right, or is the price?

Current BDC portfolios show some signs of deterioration in credit quality at the margin, but little evidence of broad-based distress. As such, the fundamentals don't seem to justify the price. But fundamentals are backward-looking. Prices, which are forward-looking, may be reflecting signs of trouble ahead that aren't yet baked into portfolio loan values, which tend to adjust with a lag going into credit distress. That said, while the macro outlook will hold the key to how BDC loans ultimately perform, we don't think the current signposts validate the extent of concerns embedded in the price.

BDCs are trading at a nearly 20% discount to their NAV

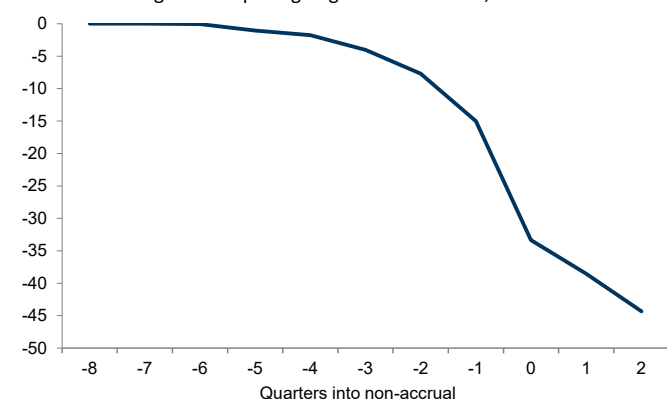
BDC share price/NAV per share, ratio



Source: Bloomberg, Goldman Sachs GIR.

Loan valuations tend to adjust with a lag going into credit distress

Cumulative change in loan price going into non-accrual, %



Source: PitchBook LCD, company filings, Goldman Sachs GIR.

Little sign of broad-based stress in portfolio fundamentals...

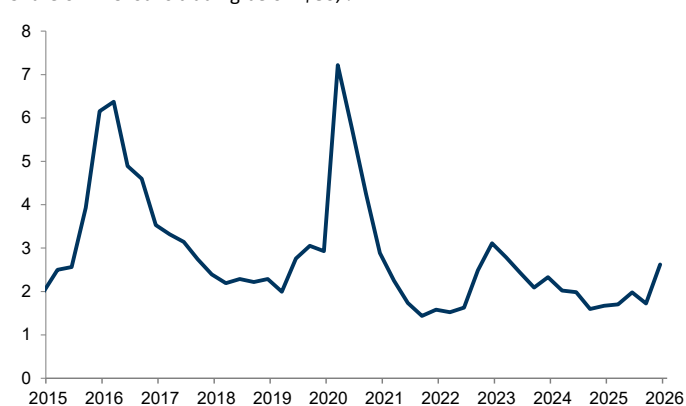
BDC portfolio fundamentals generally remained healthy through 4Q25. The share of first- and second-lien loans trading below \$80, an important indicator of credit portfolio health as a senior secured loan trading below \$80 signals credit distress, has remained relatively flat over the past few quarters, while other common measures of delinquency have moved modestly higher.

The use of payment-in-kind (PIK), which allows borrowers to defer interest payments by adding them onto the loan's principal balance rather than paying them in cash, is one such measure. PIK can be baked in at origination ("PIK-at-origination"), often when lending to growing businesses with long ramps to profitability, or structured into an existing loan if a borrower has difficulty making interest payments in cash ("PIK-at-restructuring"). While the latter is more concerning, both types of PIK have risen in recent quarters, suggesting borrowers are seeking greater flexibility.

Non-accrual loans—loans where a lender has stopped recording interest income as it's earned because it's doubtful that the interest will be collected—is another such measure. The share of such loans has risen to just below 1% of fair value, still comfortably below historical averages. Taken together, these metrics suggest incremental deterioration in BDC fundamentals, not broad-based impairment.

The share of BDC loans trading <\$80 has remained relatively flat

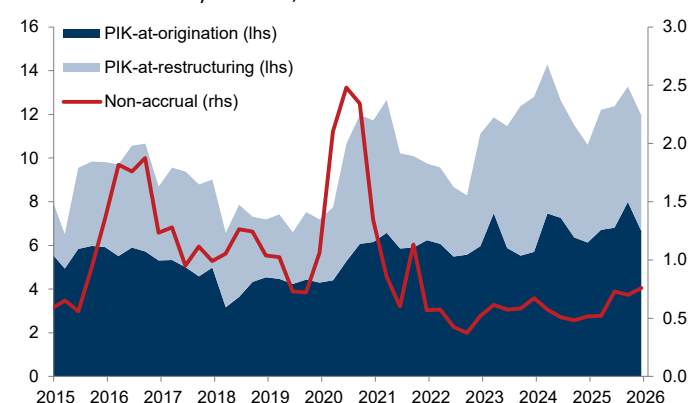
Share of BDC loans trading below \$80, %



Source: PitchBook LCD, company filings, Goldman Sachs GIR.

While the share of PIK has risen, over half is at origination, and non-accruals remain below historical averages

PIK and non-accrual by fair value, %



Source: PitchBook LCD, company filings, Goldman Sachs GIR.

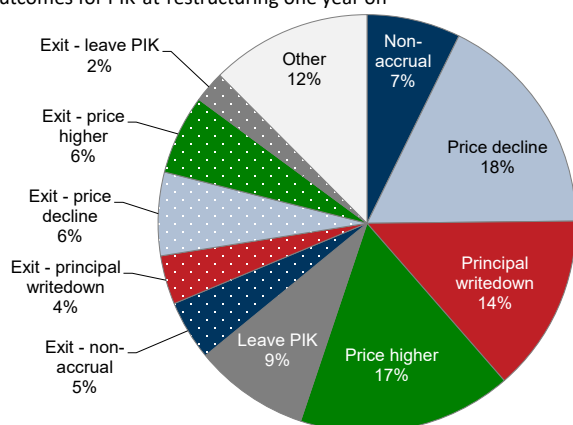
...and the current signposts aren't particularly concerning

But even if fundamentals don't look concerning today, could they in the future? While the macro outlook ultimately remains the main driver of loan health, two signposts leave us generally constructive.

Although investors often treat loans that are flagged as PIK-at-restructuring or non-accrual as reliable markers of future credit impairment, in practice the outcomes are more mixed than that characterization implies. Looking at over 3,000 loans tagged as PIK-at-restructuring, we find that 53% experienced negative outcomes, including principal writedowns, lower prices, or transition into non-accrual status, four quarters later. But over a third *improved*, either by exiting PIK status or increasing in price. So, while less favorable outcomes are more likely, negative outcomes aren't a certainty.

Loans tagged as PIK-at-restructuring tend to see negative outcomes, but not always

Outcomes for PIK-at-restructuring one year on*



*"Exit" refers to positions that left the portfolio before the four quarters and prior to their maturity month. We use the values prior to exit to categorize these loans. "Other" refers to loans which left portfolios immediately after the PIK toggle.

Source: PitchBook LCD, Goldman Sachs GIR.

Given the prospect of non-performing loans, we also consider the size and profile of loans coming due, known as the maturity wall. In our view, refinancing risk is relatively benign. Near-term maturities are relatively manageable, with loans maturing by 2027 accounting for only 10% of the total loan balance, and the loans coming due generally having higher weighted average coupons, which improves refinancing incentives given lower prevailing yields—in the absolute or relative to later maturities.

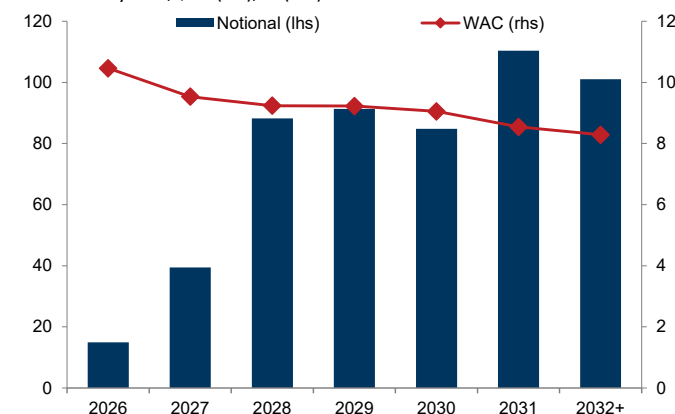
Another concern is redeployment risk—the risk that managers may not be able to reinvest the proceeds from maturing loans into new loans with comparable risk-return profiles. However, we see no indication that near-term loans are healthier than longer-maturity loans when looking at prices or delinquency status, with upcoming maturities also fairly well diversified by origination vintage.

So, while a prolonged slowdown in investment activity remains a key concern for the viability of BDCs and capital access for middle-market companies, the maturity wall does not strike us as an immediate source of stress. We also see potential for expanded deployment opportunities over the medium- to long-term, especially outside of traditional direct lending.

Taken together, we think that NAVs could be optimistic, but not overly so. We would continue to flag that dispersion across BDCs will likely increase as underwriting remains in focus.

The maturity wall is not elevated over the near term, and loans coming due generally have higher weighted average coupons

BDC maturity wall, \$bn (lhs), % (rhs)*



*Includes first- and second-lien and subordinated loans.

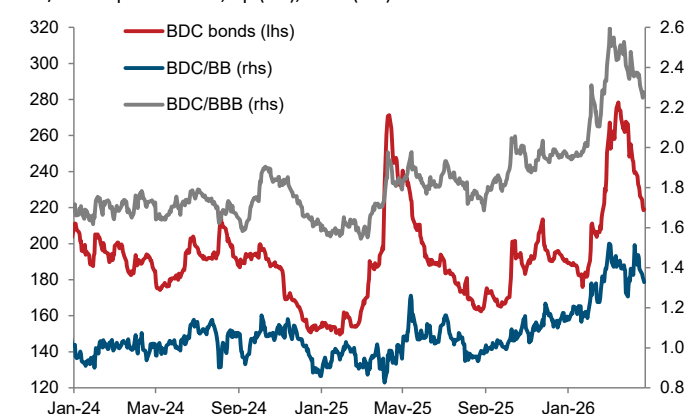
Source: PitchBook LCD, Goldman Sachs GIR.

Look to credit spreads

Looking ahead, we view BDC credit spreads as a better barometer for NAV risk than BDC equity prices, as equity prices will likely continue to reflect an elevated liquidity premium following the recent redemption surge and lower future dividend prospects for floating rate lending. BDC credit spreads appear to have turned a corner. Despite continued headline risk and a recent credit rating downgrade, secondary spreads have tightened by 60bp from their local wides in March and primary market activity has resumed. While we think the fundamentals should provide a level of support for senior capital in BDCs, we would be cautious of some of the recent exuberance. Since mid-March, the most distressed BDCs have seen their credit spreads narrow the most, by 93bp vs. only 44bp for the median bond spread. As a result, while we see value in BDC spreads at current levels, especially for investors comparing spreads to BB bonds, we think manager selection will be critical.

BDC credit spreads have tightened by 60bp since mid-March

BDC/index spread ratios, bp (lhs), ratio (rhs)



Source: iBoxx, Bloomberg, Goldman Sachs GIR.

Shamshad Ali, Senior Global Credit Strategist

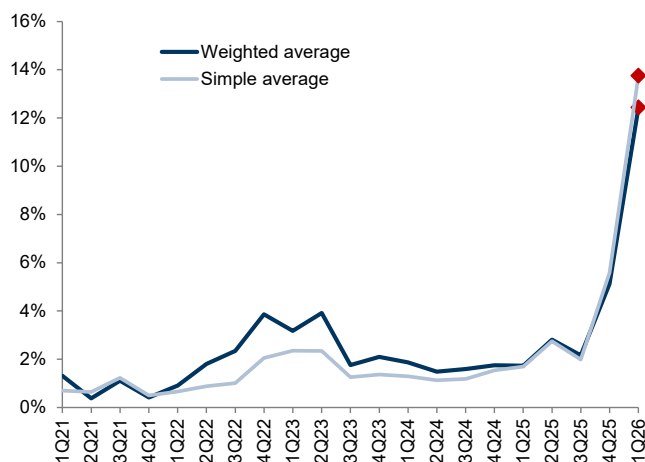
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Goldman Sachs & Co. LLC

Private credit concerns, in pics

Recent private credit concerns have prompted a rise in retail investor redemption requests...

Evergreen/non-traded BDC private credit fund redemption request rates, %



Source: Company data, Goldman Sachs GIR.

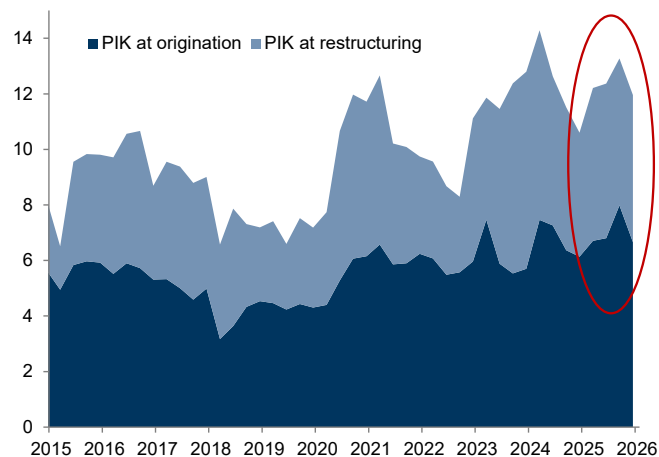
BDCs are trading at a substantial discount to their net asset value (NAV)...

BDC share price/NAV per share, ratio



Source: Bloomberg, Goldman Sachs GIR.

Payment-in-kind (PIK) usage, which can be linked to financial stress of a borrower, rose last year but not to outsized levels
Share of PIK by fair value, %

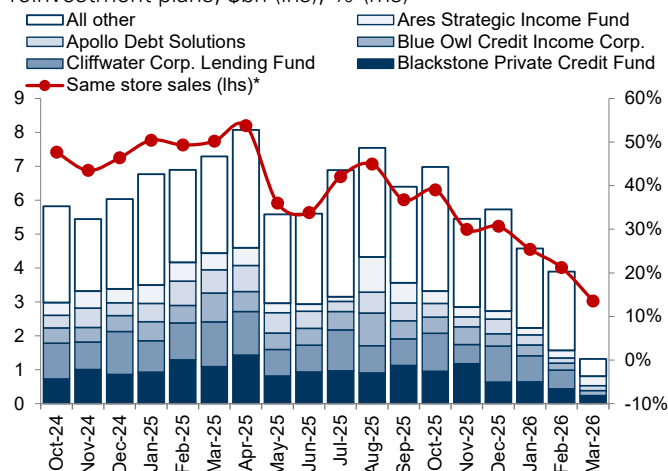


Note: Includes first- and second-lien loans in BDC portfolios. Captures data through December 31, 2025.

Source: PitchBook LCD, Goldman Sachs GIR.

...as well as a sharp reduction in gross flows into retail private credit products in recent months

Semi-liquid private credit monthly gross flows ex. dividend reinvestment plans, \$bn (lhs), % (rhs)



*Percent of prior NAV, annualized.

Source: Company data, Goldman Sachs GIR.

...and the share of underperforming private credit loans increased slightly in 2H2025

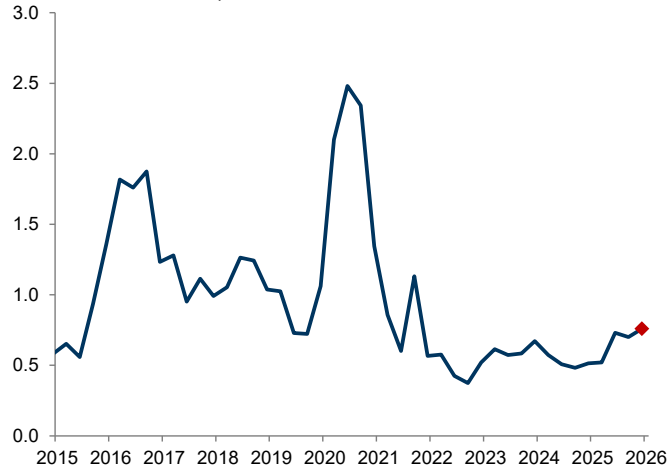
Share of underperforming private credit loans, %



Source: Company data, LSEG Data & Analytics, BDC Collateral, GS GIR.

Non-accrual rates increased only slightly and are not signaling broad-based impairment in the BDC complex

Non-accrual rates by fair value, %



Note: Includes first- and second-lien loans in BDC portfolios. Captures data through December 31, 2025.

Source: PitchBook LCD, Goldman Sachs GIR.

Special thanks to GS Global Credit Strategist Shamshad Ali and US Financials Equity Research Analyst Anthony Corbin for charts.

Private credit: limited macro spillovers

Manuel Abecasis argues that private credit stress will generate limited macro spillovers

Recent private credit concerns have driven a pickup in retail investor redemptions as well as growing worries about broader risks to financial stability and the economy given private credit's increased interconnectedness with other financial institutions (see pg. 19) as well as its significant software exposure.

Despite the worries, we think that private credit stress is unlikely to generate large macroeconomic spillovers, for a few reasons. First, private credit remains a relatively small part of the financial system, with direct lending accounting for only around 4% of all credit to the private non-financial sector (or roughly \$1.7tn) compared to the 45% and 30% that residential and securitized mortgages comprised in the run-up to the Global Financial Crisis (GFC), respectively. While the modest size of direct lending doesn't imply that broader contagion is impossible, it raises the bar for private credit stress to become macroeconomically significant.

Second, private credit faces less of a funding mismatch than other financial institutions like banks. Third, leverage at private credit funds remains modest. And fourth, while banks' exposure to private credit has risen in recent years, it remains modest overall, is senior to private credit fund investors' stakes, and the banking system is well-capitalized. As a result, we estimate that private credit stress will likely lead to only small GDP losses, even in relatively extreme default scenarios.

Limited economic losses from private credit

To gauge how higher losses in the private credit sector might affect economywide lending, we estimate the economic impacts under several potential loss scenarios¹. For each loss scenario, we consider three cases: (1) losses are proportional across all funds but don't spill over into banks; (2) losses are concentrated in a small number of funds and spill over into the banks that lend to them; and (3) these concentrated losses see lower recovery rates than even our conservative baseline assumption (30% vs. our assumption of 40%), to capture the potential risks from lower recoveries on software loans.

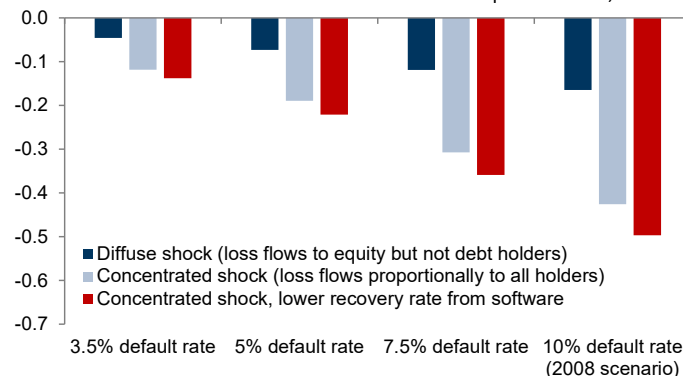
A scenario in which private credit default rates rise from around 1% in 2025 to 3-4% (the lower end of the range for leveraged loans in prior credit cycles) would result in around \$45bn in additional defaults, which would generate a small drag of 0.2% or less on the stock of loans, or roughly 1.5% or less of the gross flow of new lending, even after accounting for a pullback in lending from other financial institutions. That, in turn, would result in a small GDP drag of around 0.1%.

A more severe scenario in which private credit default rates rise to 10% (the top of the historical range for leveraged loans) in a concentrated shock would result in around \$150bn in additional defaults, leading to a 5-6% pullback in the gross flow of new lending to the private sector. That would translate to a GDP drag of 0.4-0.5%. So, even if defaults rose sharply, the macroeconomic impact would be moderate and much smaller than the pullback in new lending in both the 1990 and 2008 recessions (of around 30% and 55%, respectively).

These scenarios focus only on spillovers from stress that is specific to the private credit sector. A broader tightening in financial conditions or a deterioration in corporate balance sheets and the macroeconomic backdrop would pose larger risks than those we estimate here.

Private credit stress will likely lead to only small GDP losses

Estimated GDP effects of downside stress scenarios for private credit, %



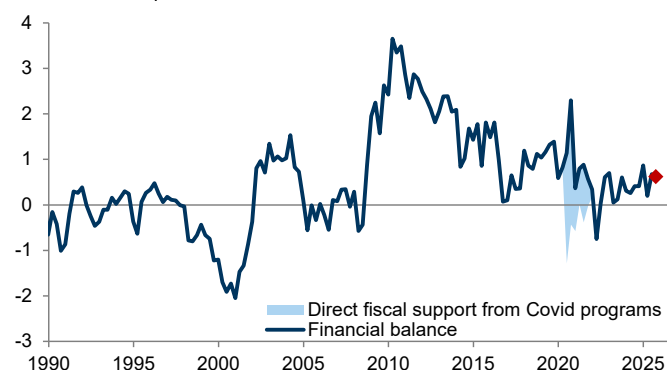
Source: Goldman Sachs GIR.

A supportive macro backdrop

While private credit lending will likely tighten in coming months, corporate balance sheets are very healthy, bank lending to businesses has picked up significantly over the last year, and increased AI-related investment demand will likely act as a tailwind to credit growth over the next several quarters. Taken together, this should further limit the impact of a pullback in private credit on the broader economy.

Corporate balance sheets are healthy...

Nonfinancial corporate financial balance, % of GDP



...and bank lending has accelerated recently

Bank commercial and industrial loans, % chg., year ago



Source: Federal Reserve, Goldman Sachs GIR.

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¹ These estimates leverage our representation of the financial system's architecture based on Fed data and insights from our equity analysts.

Alts managers: private credit opportunities

Alexander Blostein and Anthony Corbin assess the implications of private credit concerns for Alternative Asset Managers

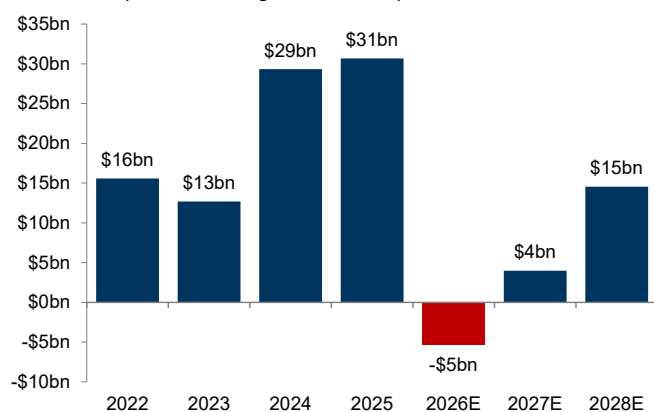
Private credit has been a significant driver of management fee growth for Alternative Asset Managers, accounting for roughly 50% of the growth from 2023-2025. Accordingly, mounting concerns about private credit have weighed heavily on the group, with stocks down over 25% YTD and valuations now 20% below historical averages at 18x NTM P/E. In particular, headlines around redemption requests in retail vehicles have sparked worries about the durability of the wealth channel, which contributed nearly 30% of the group's management fee growth over 2023-25. While we think concerns about the wealth channel are largely overblown, we expect its growth to moderate and a widening dispersion across firms based on their exposure to the wealth channel. However, beyond wealth, private credit opportunities remain compelling, and we expect accelerated institutional deployment to support healthy management fee growth for the group.

Headwinds in the evergreen credit channel

While investors largely expected slower growth from evergreen credit funds—also known as non-traded business development companies (BDCs)—heading into this year given lower base rates/returns, intense media scrutiny has amplified investor anxiety and driven a surge in redemption requests from the retail investor base. These requests reached 12% in 1Q26 (~50% annualized), while gross flows into BDCs in March were over 50% lower than their 2025 monthly run-rate, resulting in a 4% (16% annualized) gross inflow rate (compared to ~50% last year). In response, most asset managers have capped redemptions at 5% of net asset value (NAV) per quarter.

We expect net outflows in retail credit products in 2026

Net flows into public alt manager retail credit products, \$bn



Source: Company data, Goldman Sachs GIR.

We expect these dynamics to persist, with outflows from evergreen funds likely to continue for some time as the industry works through redemption queues. Based on March's gross inflow run-rate and a 20% maximum annual redemption rate, we estimate a 4% annualized pace of net outflows, or approximately \$5bn of outflows across the public alt managers in 2026 before flows turn marginally positive in 2027. We expect a wide dispersion among managers based on underlying

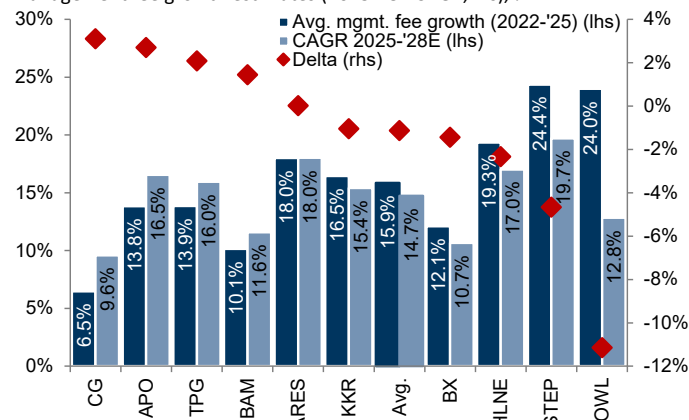
credit quality and vintage, with weaker-performing or later-vintage vehicles facing more acute outflow pressure. This will likely remain a drag on growth for several firms, but we ultimately do not expect retail redemption requests to trigger a "run on the bank" dynamic that would lead to fire sales.

Other compelling private credit growth opportunities

While retail private credit vehicles face near-term headwinds, the institutional channel offers significant growth opportunities. Indeed, despite the recent focus on the wealth channel, over 80% of alts managers' private credit-related management fees are derived from institutional sources compared to the roughly 7% of fees-related revenues from evergreen credit vehicles (though BX, OWL, and HLNE are relatively more skewed toward the wealth channel). And though institutional fundraising has declined in recent years largely owing to tighter credit spreads and faster deployment from non-traded BDCs, we expect faster institutional deployment supported by significant institutional available capital (\$419bn in direct lending/opportunistic unlevered available capital) as wealth pulls back and spreads widen. Substantial loan maturities through 2028 should further boost institutional fundraising and investment activity, particularly for opportunistic, mezzanine, and special situations strategies, which have been relatively dormant in recent years.

We expect Alt managers' management fees to grow at a healthy 15% CAGR through 2028 on average

Historical average management fee growth (organic, ex. catch-up fees, lhs) vs. management fee growth estimates (2025-28E CAGR, lhs), %



Source: Company data, Goldman Sachs GIR.

The path forward: wider dispersion

Looking ahead, we expect Alt managers' management fees to grow at a healthy 15% compound annual growth rate (CAGR) through 2028 (vs. 16% from 2024-25) on average. However, we expect wider dispersion among managers depending on their exposure to the wealth channel—a dynamic that has not yet been reflected in relative stock performance and valuation multiples, presenting potential opportunities.

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Banks and private credit: the intersection

Richard Ramsden discusses the evolving intersection between banks and private credit

The rapid expansion of the private credit industry over the past two decades has reshaped the lending landscape. The post-Global Financial Crisis (GFC) regulatory environment made it increasingly difficult for banks to compete in certain lending categories, and regulatory constraints around lending to corporates above specified leverage thresholds have pushed some borrowers toward non-bank financing providers that can offer greater flexibility on structure and terms, including covenant-lite and payment-in-kind (PIK) features (see pgs. 10-11).

But while banks have ceded share to the private credit industry, particularly in commercial lending, they have also benefitted from the industry's growth and have become increasingly interconnected with it, providing direct financing opportunities to private credit funds and partnering with private credit firms to serve customers whose financing needs are beyond banks' risk or balance sheet constraints. Amid the current stresses in private credit, this interconnectedness has raised concerns about systemic risk, which we view as overdone.

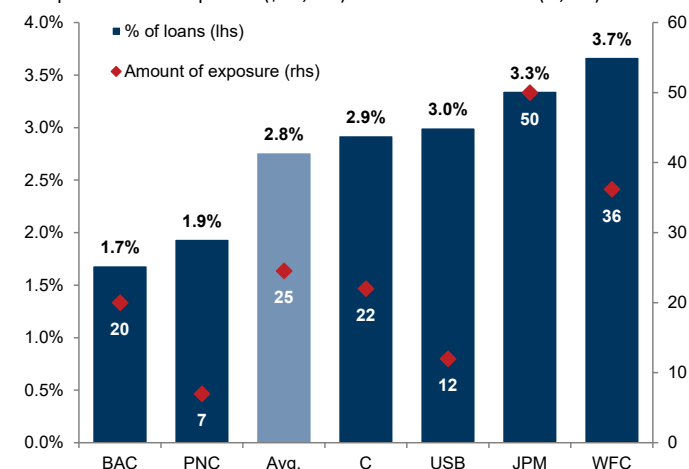
Meaningful linkages...

Currently, banks provide financing to private credit managers in the form of working capital facilities and fund-level leverage. These facilities are typically secured by diversified loan portfolios, which generally have relatively conservative advance rates or loan-to-value ratios (LTVs). Based on disclosures from large banks, advance rates are generally in the 60-70% range, suggesting that defaults would need to rise materially before banks would incur losses. Banks are typically insulated from first-loss risk within their private credit exposure through both underlying operating company equity and a meaningful layer of fund investor capital. Banks also actively monitor collateral values and, in some cases, have reduced the leverage they provide private credit funds over the last six months. As a result, loss rates on banks' private credit portfolios haven't materially risen, and we expect risk-adjusted returns to remain healthy in the near term.

Lending to private credit firms has also become an important source of lending growth for the banking industry, though it remains a relatively small portion of bank balance sheets. Loans to private credit currently account for 3% of bank loan portfolios, having grown at double-digit compound annual growth rates (CAGRs) over the last five years. This exposure sits within the broader non-depository financial institution (NDFI) lending category that represents around 15% of total bank lending. However, as private credit firms experience net outflows, the demand for these working capital loans will likely decline, leading to slower growth—or even some contraction—in bank lending to private credit.

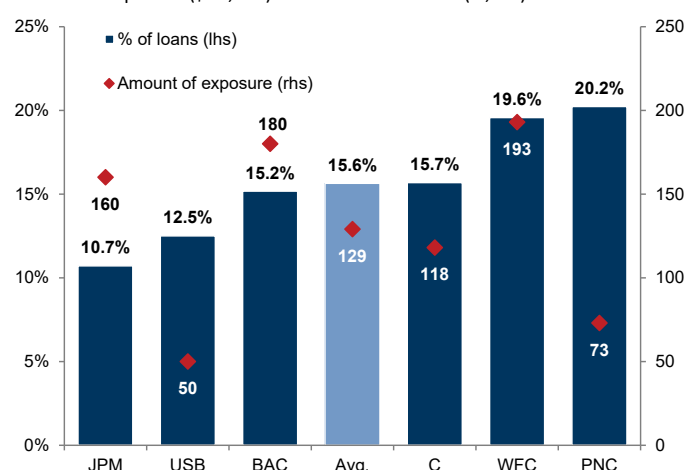
Loans to private credit account for 3% of total bank loans...

Bank private credit exposure (\$bn, rhs) and as share of loans (%), lhs



...with broader NDFI lending accounting for 15%

Bank NDFI exposure (\$bn, rhs) and as share of loans (%), lhs



Note: WFC, USB, and PNC as of 1Q26, others as of 4Q25; chart excludes MS.

Source: Company data, Goldman Sachs GIR.

Banks have also increasingly partnered with private credit firms to serve customers whose financing needs may fall outside a bank's risk appetite or balance sheet constraints. These partnerships enable banks to distribute risk, helping diversify portfolio exposure while allowing banks to offer clients a broader range of lending options than they could provide on a standalone basis, as they can now sell and offload loans that do not fit their underwriting criteria. So, while a pullback in private credit may allow banks to regain some share in loan origination, it could also reduce the range of lending options banks can offer clients, particularly in situations where banks are unwilling to hold the loans on their own balance sheet.

...but limited systemic risk

Taken together, banks' exposure to private credit is real but structurally buffered, with direct exposures generally secured, conservatively leveraged, and protected by substantial first-loss capital. As a result, while ongoing monitoring is warranted, particularly if private credit stresses continue, private credit seems unlikely to pose a risk to the traditional financial system.

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Summary of our key forecasts

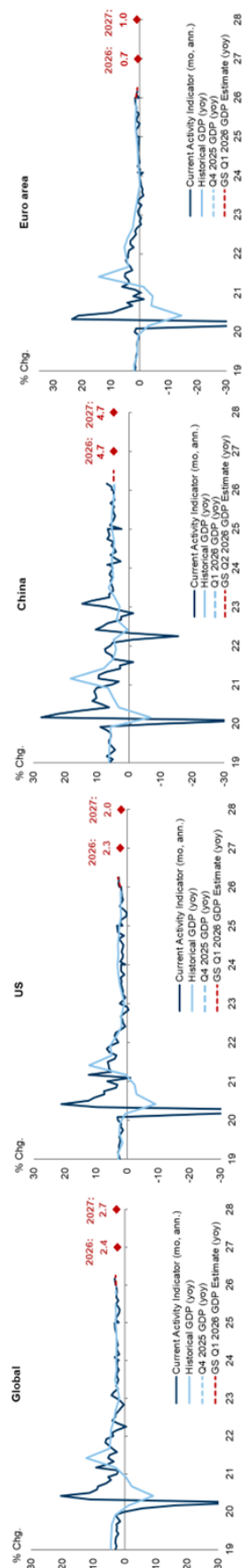
GS GIR: Macro at a glance

Watching

- **WATCH THE IRAN WAR.** While the US and Iran have agreed to extend the ceasefire, the situation remains fluid and oil flows through the Strait of Hormuz remain very low, with risks to economies and markets growing the longer these supply disruptions continue.
- **Globally,** we expect real GDP growth of 2.4% yoy in 2026 amid growing headwinds from higher energy prices owing to the Iran war. We expect global core inflation to decline to 2.3% by the end of 2026, reflecting a fading tariff boost and further normalization in shelter and wage inflation but a boost from higher energy prices.
- **In the US,** we expect real GDP growth of 2.1% on a Q4/Q4 basis in 2026, reflecting a drag from higher oil prices, tighter financial conditions, and elevated geopolitical risk. We expect core PCE inflation to fall to 2.6% yoy by December 2026, as the tariff boost fades and wage and shelter inflation continue to cool, though higher oil prices provide some offset. We expect the unemployment rate to end 2026 at 4.6%.
- **We expect the Fed** to deliver two 25bp cuts this year in September and December for a terminal rate range of 3-3.25%.
- **In the Euro area,** we expect below-potential real GDP growth of 0.6% on a Q4/Q4 basis in 2026, reflecting higher energy prices and tighter financial conditions owing to the Iran conflict. We expect core inflation to rise to a peak of 2.5% yoy in 3Q26 amid higher energy prices.
- **We expect the ECB** to deliver two 25bp hikes this year in June and September to a peak policy rate of 2.5% before cutting back to 2% in 2027, although we view whether the ECB hikes twice or holds rates unchanged this year as a close call.
- **In China,** we expect real GDP growth of 4.7% yoy in 2026 as resilient export growth, continued government policy easing, and a decreasing drag from the ongoing property market downturn offset headwinds from higher energy prices and sluggish domestic demand. We expect CPI/PPI inflation to rise to 1.0%/1.2% yoy this year owing to higher global commodity prices, government policy efforts, and a low base.

Goldman Sachs Global Investment Research.

Growth



Source: Haver Analytics, Goldman Sachs Global Investment Research.

Note: GS CAI is a measure of current growth. For more information on the methodology of the CAI please see "Technical Updates to Our Global CAIs," Global Economics Comment, Sep. 01, 2025.

Forecasts

Economics	Markets										Equities									
	GDP growth (%)					Interest rates 10Yr (%)					FX					Commodities				
	2026	2027	2028	2029	2030	2026	2027	2028	2029	2030	2026	2027	2028	2029	2030	2026	2027	2028	2029	2030
Global	2.2	2.4	2.7	2.7	2.6	US	4.31	4.10	4.15	EUR/US	1.17	1.14	1.20	Price	7.600	US	2.7	4.6	2.1	4.4
US	2.1	2.0	2.3	2.2	2.2	Germany	3.00	3.00	3.00	GBP/US	1.35	1.33	1.33	EPS	\$309	US	2.9	6.3	2.1	6.2
China	4.6	4.5	4.7	5.0	4.6	Japan	2.44	2.00	2.00	\$/JPY	159	160	155	Growth	12%	China	1.0	--	1.0	--
Euro area	0.6	0.9	0.7	1.4	1.0	UK	4.80	4.40	4.25	\$/GBP	6.82	6.85	6.70	Credit	4026	China	1.0	--	1.0	--
Policy rates (%)	2026					2027					2028					2029				
	GS	Mkt.	GS	Mkt.	GS	GS	Mkt.	GS	Mkt.	GS	GS	Mkt.	GS	Mkt.	GS	GS	Mkt.	GS	Mkt.	GS
	3.13	3.54	3.13	3.37	3.13	Crude Oil, Brent (\$/bbl)	105	100	90	USD	IG	95	90	US	2.7	4.6	2.1	4.4	3.7	--
	2.50	2.51	2.00	2.46	2.00	Nat Gas, NYMEX (\$/mmbtu)	2.52	3.50	3.50	EUR/USD	HY	272	335	315	Euro area	2.9	6.3	2.1	6.2	--
	1.40	1.50	1.30	--	1.30	Nat Gas, TTF (EUR/MWh)	44.89	50	40	EUR	IG	92	120	114	China	1.0	--	1.0	--	--
	1.00	1.20	1.50	1.73	1.50	Copper (\$/lb)	13.247	12.700	12.500	EUR	IG	92	120	114	China	1.0	--	1.0	--	--
	1.00	1.20	1.50	1.73	1.50	Gold (\$/troy oz)	4,712	4,865	5,308	EUR	HY	286	375	360	China	1.0	--	1.0	--	--

Source: Bloomberg, Goldman Sachs Global Investment Research. For important disclosures, see the Disclosure Appendix or go to www.gs.com/research/hedge.html.

Market pricing as of April 24, 2026

Glossary of GS proprietary indices

Current Activity Indicator (CAI)

GS CAIs measure the growth signal in a broad range of weekly and monthly indicators, offering an alternative to Gross Domestic Product (GDP). GDP is an imperfect guide to current activity: In most countries, it is only available quarterly and is released with a substantial delay, and its initial estimates are often heavily revised. GDP also ignores important measures of real activity, such as employment and the purchasing managers' indexes (PMIs). All of these problems reduce the effectiveness of GDP for investment and policy decisions. Our CAIs aim to address GDP's shortcomings and provide a timelier read on the pace of growth.

For more, see our CAI page and Global Economics Comment: Technical Updates to Our Global CAIs.

Dynamic Equilibrium Exchange Rates (DEER)

The GSDEER framework establishes an equilibrium (or "fair") value of the real exchange rate based on relative productivity and terms-of-trade differentials.

For more, see our GSDEER page, Global Economics Paper No. 227: Finding Fair Value in EM FX, 26 January 2016, and Global Markets Analyst: A Look at Valuation Across G10 FX, 29 June 2017.

Financial Conditions Index (FCI)

GS FCIs gauge the "looseness" or "tightness" of financial conditions across the world's major economies, incorporating variables that directly affect spending on domestically produced goods and services. FCIs can provide valuable information about the economic growth outlook and the direct and indirect effects of monetary policy on real economic activity.

FCIs for the G10 economies are calculated as a weighted average of a policy rate, a long-term risk-free bond yield, a corporate credit spread, an equity price variable, and a trade-weighted exchange rate; the Euro area FCI also includes a sovereign credit spread. The weights mirror the effects of the financial variables on real GDP growth in our models over a one-year horizon. FCIs for emerging markets are calculated as a weighted average of a short-term interest rate, a long-term swap rate, a CDS spread, an equity price variable, a trade-weighted exchange rate, and—in economies with large foreign-currency-denominated debt stocks—a debt-weighted exchange rate index.

For more, see our FCI page, Global Economics Analyst: Our New G10 Financial Conditions Indices, 20 April 2017, and Global Economics Analyst: Tracking EM Financial Conditions – Our New FCIs, 6 October 2017.

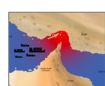
Goldman Sachs Analyst Index (GSAI)

The US GSAI is based on a monthly survey of GS equity analysts to obtain their assessments of business conditions in the industries they follow. The results provide timely "bottom-up" information about US economic activity to supplement and cross-check our analysis of "top-down" data. Based on analysts' responses, we create a diffusion index for economic activity comparable to the ISM's indexes for activity in the manufacturing and nonmanufacturing sectors.

Macro-Data Assessment Platform (MAP)

GS MAP scores facilitate rapid interpretation of new data releases for economic indicators worldwide. MAP summarizes the importance of a specific data release (i.e., its historical correlation with GDP) and the degree of surprise relative to the consensus forecast. The sign on the degree of surprise characterizes underperformance with a negative number and outperformance with a positive number. Each of these two components is ranked on a scale from 0 to 5, with the MAP score being the product of the two, i.e., from -25 to +25. For example, a MAP score of +20 (5;+4) would indicate that the data has a very high correlation to GDP (5) and that it came out well above consensus expectations (+4), for a total MAP value of +20.

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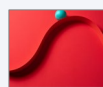
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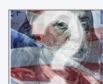
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Reg AC

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We, Richard Ramsden, Alexander Blostein, CFA and Anthony Corbin, hereby certify that all of the views expressed in this report accurately reflect our personal views about the subject company or companies and its or their securities. We also certify that no part of our compensation was, is or will be, directly or indirectly, related to the specific recommendations or views expressed in this report.

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Rating and pricing information

Ares Management Corp. (Buy, \$115.23) and StepStone Group (Buy, \$51.82).

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Growth is based on a stock's forward-looking sales growth, EBITDA growth and EPS growth (for financial stocks, only EPS and sales growth), with a higher percentile indicating a higher growth company. **Financial Returns** is based on a stock's forward-looking ROE, ROCE and CROCI (for financial stocks, only ROE), with a higher percentile indicating a company with higher financial returns. **Multiple** is based on a stock's forward-looking P/E, P/B, price/dividend (P/D), EV/EBITDA, EV/FCF and EV/Debt Adjusted Cash Flow (DACF) (for financial stocks, only P/E, P/B and P/D), with a higher percentile indicating a stock trading at a higher multiple. The **Integrated** percentile is calculated as the average of the Growth percentile, Financial Returns percentile and (100% - Multiple percentile).

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	Rating Distribution				Investment Banking Relationships		
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